

The Macroeconomics of Automation: Sorting Frictions and the Asymmetry Between Robots and AI*

Carlos Chávez[†]

February 2026

Abstract

Does automation rebuild work around those it displaces, and does the answer differ for robots and AI? I model both frontiers advancing together, each reinstating tasks it destroys, workers sorting across the rest. Displacement scales come from commuting-zone moments under design-based inference, with the cognitive moments net of a college-wage trend that predates AI and runs out by 2012. Physical task creation is calibrated from firm-level evidence and carried across its full range. Within zones, AI-exposed occupations lose relative wage in both skill groups after 2010. Robots deliver about twice the productivity per unit of advance that AI does, and the ranking holds across the whole physical calibration range and in every bootstrap draw. The case for tilting research toward robots is distributional, and a planner averse to inequality supports robot research about 70 percent more. Sorting disruption is consistent with the evidence and not separately identified.

JEL: E24, J23, J31, O33, O41

Keywords: Automation, reinstatement, productivity, labor share, robots, artificial intelligence, directed technical change, optimal policy

*I am grateful to James Heckman for guidance and support. All errors are my own.

[†]Center for Economics of Human Development, University of Chicago. Email: cchavezpadilla@uchicago.edu

1 Introduction

When a technology automates human work, does it rebuild new work around the people it displaces? For robots and AI the answers differ. In the model estimated here, robots deliver about twice the productivity per unit of frontier advance that AI does, and the difference is large enough to reorder productivity, the labor share, and optimal innovation policy. That automation both displaces workers and generates new work is established ([Acemoglu and Restrepo, 2019](#); [Autor et al., 2024](#)), and so is the fact that robots and AI reach different tasks and different workers ([Webb, 2020](#); [Bloom et al., 2025](#)). The two frontiers have not been put on a common metric, however, and so nothing in that work says whether a marginal advance of the physical frontier or of the cognitive frontier delivers more aggregate value once two things are priced on the same scale: how fast each creates new work, and how badly each disturbs the matching of workers to the tasks that survive.

The two frontiers differ in what displacement does to the match between workers and the tasks that get rebuilt. When a manufacturing firm installs a robot, the surrounding production process is reorganized: quality-control positions, maintenance roles, and programming tasks emerge alongside the displacement, and displaced production workers slot into them. When an AI system automates cognitive work, document review, data classification, routine analysis, the evidence over the horizon this paper measures shows the displacement and not the rebuilding, since exposed occupations lose relative wage in both skill groups and no new cognitive work appears at the rate the physical frontier creates it. The model also allows the displacement to scramble the sorting of skilled workers across the cognitive occupations that remain. How large the gap in task creation is, how much of the cognitive drag the sorting channel can account for, and what both imply for aggregate outcomes, is the question this paper answers.

The model has physical and cognitive automation advancing at the same time, each creating new work at its own rate, inside a labor market where workers sort across sectors by comparative advantage. Productivity, labor shares, skill premia, and optimal policy all follow from one growth decomposition that separates what each frontier contributes from what general equilibrium adds back.

Estimating it requires matching each parameter to evidence that can actually measure it, and the two frontiers are not symmetric in that respect. The cognitive side is measured here, from

how wages and the college employment share respond across U.S. commuting zones to an industry-level robot shock and an occupation-level measure of AI exposure. Those responses pin down how much new cognitive work is created and how badly the assignment of workers to the surviving cognitive tasks is disturbed. Whether they measure an effect of AI is tested three ways before they enter the estimator. Backward, against the 1990–2000 census, zones that would later measure as AI-exposed were already on a steeper college-wage trend. Forward, across windows through 2022–2024, that college gradient runs out to zero, while the non-college gradient reverses sign after 2005 and persists. Within zones, exposed occupations lose relative wage with every zone-level trend absorbed. The moments enter net of the pre-period trend, and the cognitive reinstatement estimate is the one object that depends on how much of that trend is subtracted. The physical side cannot be measured the same way. New work created around robots appears economy-wide, not only in the places where robots are installed, so a comparison across commuting zones cannot see it. It is calibrated instead from firm-level evidence, at the neutral value where task creation just offsets task displacement, and every result is reported across the whole range that evidence brackets under the accounting that maps wage-bill changes into task value. Danish adopters studied by Humlum (2024) cut the production-worker wage bill by a fifth while raising sales and total labor demand. I translate that evidence into task-value units in Appendix D.3, and the resulting range brackets the neutral value. The agreement of Danish, U.S., German, and Spanish evidence suggests the pattern belongs to the technology and not to local labor-market institutions.

Robot automation is the more productive of the two per unit of frontier advance at the estimates, by 119 percent at the neutral calibration, and the ranking holds at every value of physical task creation the firm-level evidence brackets, from a ratio of 1.80 at the narrow bound to 2.93 at the broad one. The cognitive moments carry the sampling variation of the occupation scores that generate them, of which 77.9 are effective, and the ninety-percent interval for the ratio runs from 1.57 to 8.35, above parity in every one of the 200 bootstrap draws. The size of the gap does depend on how much of the pre-AI trend is removed from the cognitive moments: subtracting none of it gives a ratio of 1.21, subtracting half gives 1.48, and the linear continuation gives 2.19. Robot dominance holds at every rate. A wage-only system in related work places cognitive task creation near zero (Chávez, 2026), and the adjusted estimate here, 0.03 tasks created per task displaced, reproduces it from a different moment system.

What carries the gap is the difference in task creation. The direct displacement components of the two frontiers are within six percent of each other, and the reinstatement-and-reallocation component is +0.314 for robots against +0.011 for AI. The sorting friction the model allows on the cognitive frontier is consistent with the college-wage evidence and is weakly identified once the pre-trend is removed, so the paper reports it with its interval and rests nothing on its value, because the ranking holds at every sorting disruption the bootstrap draws visit and at every share of it that is transitional. The ranking describes the horizon over which research policy is set, the years to a decade or two ahead and not the far steady state.

Three macro results follow.

Supply-side dominance. The distributional consequences of asymmetric automation come almost entirely from the supply-side productivity difference and not from demand. Solving the model with non-tradable services, an endogenous participation margin, and households that differ in their propensity to consume, and computing each demand channel directly, the total general equilibrium feedback is more than two orders of magnitude below the level at which it would change any conclusion, on both frontiers. The demand side changes the feedback on the robot frontier by a tenth of a percent of itself and is negligible on the cognitive one.

Labor share dynamics. Both frontiers lower the labor share, and robots do about 41 percent of the damage AI does per unit of advance, a ratio that holds across the whole physical calibration range. The gap traces to task creation: the work robots rebuild is paid to labor, and the cognitive frontier rebuilds almost none, so an AI advance lowers the wage bill outright. Since both technologies lower the labor share, the relevant comparison is which damages it less.

Optimal frontier-specific policy. Over the transition horizon a planner supports robot research more than AI research, by a gross subsidy factor of 1.72 at the estimates, about 70 percent higher for robots, and above one in 93.5 percent of the bootstrap draws. The reason is distributional. If investors capture the output a frontier delivers, relative productivity gives a planner no reason to tilt research at all, and what remains is that a robot advance raises the non-college wage while an AI advance lowers it, and a planner with any aversion to inequality weights that group more heavily. The subsidy ranking survives across the entire physical calibration range, including the edge where the productivity motive vanishes and only the distributional one remains. This is a statement about the transition. As AI matures its productivity content rises, and along every development

path considered the ranking turns at or before full maturity. What differs is how much of the path robots lead, from all of it under the slowest path the current evidence favors to about half under the scenario [Autor \(2024\)](#) describes. A uniform policy leaves the two frontiers misallocated relative to each other by the whole of the distributional wedge, because a common rate cancels from the condition that governs how investment splits between them.

The estimates describe the automation environment of 2005 to 2019, before generative AI, with the 2022–2024 window used as a test of timing and not as a moment, and the physical side is calibrated from outside the sample because local variation cannot measure it. The cognitive reinstatement estimate is the one object the identifying assumption moves. It runs from 0.57 tasks per task displaced when half the pre-AI trend is removed to 0.03 under the linear continuation.

This paper extends the task-based automation framework of [Acemoglu and Restrepo \(2018, 2022\)](#) in three directions: two frontiers with asymmetric reinstatement, a frontier-specific productivity decomposition that the single-frontier model cannot deliver, and frontier-specific optimal policy. Relative to [Hémous and Olsen \(2022\)](#), who set automation against horizontal innovation in new products, the contribution is the empirical mapping from measured reinstatement to macro objects. [Bloom et al. \(2025\)](#) already separate traditional capital, industrial robots and AI within a nested CES and derive their different consequences for the skill premium. I measure the reinstatement and sorting margins separately for each frontier and carry them into a common productivity and policy comparison. Closest to the quantitative exercise here is [Acemoglu \(2025\)](#), which prices the aggregate productivity gain from cognitive automation within a single-frontier task framework. The two-frontier structure allows a question that framework cannot pose, how the gain from artificial intelligence compares, per unit of advance, with the gain from the physical frontier. The three-sector extension with non-tradable services relates to the structural change literature ([Herrendorf et al., 2014](#); [Baumol, 1967](#)), and the dynamic model with heterogeneous savings to the macro-inequality literature ([Karabarbounis and Neiman, 2014](#); [Piketty, 2014](#)). The evidence on the physical frontier builds on [Graetz and Michaels \(2018\)](#) and [Acemoglu and Restrepo \(2020a\)](#), and the empirical motivation draws on [Webb \(2020\)](#); [Eloundou et al. \(2024\)](#); [Babina et al. \(2024\)](#); [Acemoglu et al. \(2022\)](#). That AI may either automate cognitive tasks or create new human ones, depending on the direction it is given, is the distinction [Acemoglu and Restrepo \(2020b\)](#) draws. The cognitive frontier is where this paper measures it.

Section 2 presents the model and its demand-side extensions as theory. Section 3 documents the reduced-form evidence, the inference each research design warrants, the identification tests, the estimation of the cognitive frontier, and the external calibration of the physical one. Section 4 computes the general equilibrium feedback and establishes supply-side dominance. Section 5 extends the model to continuous time. Section 6 presents the productivity decomposition, the labor share, and the threshold at which the frontier ranking would turn. Section 7 develops welfare and optimal policy. Section 8 concludes.

2 Model

The model extends the Acemoglu–Restrepo task framework to two automation frontiers with frontier-specific reinstatement. The production structure keeps the frontiers separate throughout, since a scalar automation measure can mask heterogeneous frontier effects (Chávez, 2026). Three extensions follow the core: non-tradable services, endogenous labor supply, and endogenous capital income distribution. The dynamic extension follows in Section 5.

2.1 Production

A final good Y combines physical and cognitive bundles via CES:

$$Y = \left[\mu Y_P^{\frac{\rho-1}{\rho}} + (1-\mu) Y_C^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}}, \quad (1)$$

where $\rho > 0$ is the elasticity of substitution between physical and cognitive output. The cognitive weight $(1 - I_A)^\varphi$ below is taken on $I_A \in [0, 1)$ for $\varphi = 0$ and on $I_A \in [0, 1]$ for $\varphi > 0$, which excludes the corner $(I_A, \varphi) = (1, 0)$ where it has no real value. Equation (1) is displayed for $\rho \neq 1$. The quantitative core sets $\rho = 1$, where the exponents are undefined and the aggregator is the Cobb–Douglas limit $Y = Y_P^\mu Y_C^{1-\mu}$, which is how $\rho = 1$ is to be read throughout.

Disaggregated task structure.

Each sector $j \in \{P, C\}$ produces output from a continuum of tasks. The task space comprises three segments:

- (i) *Automated tasks* $z \in [0, I_j]$: produced by capital with task-specific productivity $g_{Kj}(z)$.
- (ii) *Surviving labor tasks* $z \in (I_j, 1]$: produced by labor with task-specific productivity $g_{Lj}(z)$.
- (iii) *Reinstated tasks* $z \in (1, 1 + \delta_j I_j]$: new complementary tasks created at rate δ_j per unit of frontier advance, produced by labor.

Output in sector j aggregates all tasks via CES with elasticity $\sigma > 1$ (exponent $\varsigma \equiv (\sigma - 1)/\sigma$):

$$Y_j = \left[\int_0^{I_j} y_{Kj}(z)^\varsigma dz + \int_{I_j}^1 y_{Lj}(z)^\varsigma dz + \int_1^{1+\delta_j I_j} y_{Rj}(z)^\varsigma dz \right]^{1/\varsigma}, \quad (2)$$

where $y_{Kj}(z) = g_{Kj}(z) k_j(z)$ is output of automated task z using capital $k_j(z)$, $y_{Lj}(z) = g_{Lj}(z) \ell_j(z)$ is output of surviving task z using labor $\ell_j(z)$, and $y_{Rj}(z) = g_{Rj}(z) \ell_j^R(z)$ is output of reinstated task z using labor $\ell_j^R(z)$. Following [Acemoglu and Restrepo \(2018\)](#), the task productivities g_{Kj} , g_{Lj} , and g_{Rj} are defined in efficiency units, so that raising them to the task exponent yields the task content coefficient in the CES aggregator. The exact aggregator below carries the exponent $\sigma - 1$. Some expository passages use a distinct additive shorthand in ς . Appendix [A](#) shows that it is generically unequal to the exact CES aggregate, and it enters no quantitative computation.

Reinstated task productivities.

Reinstated tasks are created by entrepreneurs who draw a potential complementary task at frontier j with productivity γ from a sector-specific distribution G_j on $[\underline{\gamma}_j, \bar{\gamma}_j]$. An entrepreneur creates the task if and only if the value $v_j(\gamma)$ of operating it exceeds the creation cost $c(\gamma)$. Creating a mass N of tasks at frontier j costs $C_j(N) = c_0 N^{1+\eta}$ in the aggregate, with $\eta > 0$, so entry runs to the point where the marginal creation cost $c_0(1+\eta)N^\eta$ equals the marginal surplus $\theta_j(v_j - W_{k(j)}^d)$, and to the corner $N = 0$ when that surplus is not positive. Entry requires an existing frontier to build around, so $N_j(0) = 0$: at $I_j = 0$ no complementary task is drawn, whatever the surplus. In equilibrium, a measure $\delta_j I_j$ of tasks is created. The following lemma establishes that the heterogeneous productivities aggregate exactly under CES.

Lemma 1 (CES Aggregation of Reinstated Tasks). *Let reinstated tasks at frontier j have productivities $\{\gamma\}$ drawn from equilibrium distribution G_j^* (the distribution of γ conditional on entry).*

Let the reinstated measure be positive, $\delta_j I_j > 0$. Under CES aggregation with exponent ς and optimal labor allocation across reinstated tasks, the reinstated-task contribution to the CES aggregator satisfies:

$$\int_1^{1+\delta_j I_j} (g_{Rj}(z) \ell_j^R(z))^\varsigma dz = \delta_j I_j \cdot (\gamma_j B_j)^\varsigma \cdot \left(\frac{L_j^R}{\delta_j I_j} \right)^\varsigma, \quad (3)$$

where:

$$\gamma_j \equiv \left(\mathbb{E}_{G_j^*} [\gamma^{\sigma-1}] \right)^{1/(\sigma-1)} \quad (4)$$

is the $(\sigma - 1)$ -power mean of reinstated-task productivities, B_j is a sector-specific efficiency scalar, and L_j^R is total labor allocated to reinstated tasks. At $\delta_j I_j = 0$ the reinstated interval is empty, $L_j^R = 0$, and both sides are zero by continuous extension.

The power mean γ_j in equation (4), with exponent $\sigma - 1$ per Lemma 2 of Appendix A, is an exact aggregation object, with no approximation involved in this step. When reinstated tasks are homogeneous (G_j^* degenerate at a single γ_j), the power mean reduces to γ_j itself. The proof is in Appendix A.

Compressed production function.

The firm in sector j allocates capital across automated tasks and labor across surviving and reinstated tasks to maximize Y_j . The following proposition derives the compressed form.

Proposition 1 (Task Aggregation). *Under optimal within-sector factor allocation:*

$$Y_j = \left[Z_j(I_j) + \tilde{\Gamma}_j(I_j, \delta_j) \cdot (L_j^{\text{eff}})^\varsigma \right]^{1/\varsigma}, \quad j \in \{P, C\}, \quad (5)$$

where:

$$Z_j(I_j) = \int_0^{I_j} g_{Kj}(z)^\varsigma \bar{k}_j(z)^\varsigma dz, \quad (6)$$

$$\tilde{\Gamma}_j(I_j, \delta_j) = \left[\int_{I_j}^1 g_{Lj}(z)^{\sigma-1} dz + \delta_j \cdot I_j \cdot (\gamma_j B_j)^{\sigma-1} \right]^{1/\sigma}, \quad (7)$$

with $\bar{k}_j(z)$ denoting optimal capital per automated task. Both Z_j and $\tilde{\Gamma}_j \cdot (L_j^{\text{eff}})^\varsigma$ carry units $[\text{output}]^\varsigma$, since Z_j absorbs optimal capital quantities and $\tilde{\Gamma}_j$ collects the task-content terms that multiply

aggregate effective labor raised to ς .

Proof. See Appendix A.

$Z_j(I_j)$ is not a pure technology parameter, because it contains optimal capital deployment $\bar{k}_j(z)$ at each automated task, which in general equilibrium depends on the rental rate, so Z_j is endogenous to factor prices. The notation suppresses this dependence, and the productivity and policy comparative statics solve the factor-price response at each perturbed frontier position, with the reinstatement rates δ_j held at their baseline values (Appendix A, Step 5).¹

The surviving-task integral and the reinstatement term enter the bracket of $\tilde{\Gamma}_j$ additively because both represent task-content terms that multiply $(L_j^{\text{eff}})^\varsigma$ after optimal labor allocation. The equal-effective-labor-share result under CES (Appendix A, Step 2) is what permits this additive separation inside the aggregate.

The reinstatement term is proportional to $\delta_j \cdot I_j$, the total measure of reinstated tasks, not to δ_j alone. The entry condition (11) makes this product invariant to I_j except through the equilibrium objects $(v_j, W_{k(j)}^d)$ (Appendix A, Step 5), so the stock of reinstated tasks responds to frontier advance only through equilibrium task values, and the interaction between δ_j and I_j generates the nonlinearities exploited in the productivity decomposition.

Effective task content.

Production depends on reinstatement only through the product that enters the bracket of (7): the task quantity δ_j and the reinstated-task productivity $\gamma_j B_j$ enter jointly and are not separately identified by any moment this paper uses. The estimated primitive is therefore the *effective task content* of reinstatement,

$$\chi_j \equiv \delta_j (\gamma_j B_j)^{\sigma-1}, \quad (8)$$

evaluated at the baseline frontier position, so that $\tilde{\Gamma}_j = [\int_{I_j}^1 g_{Lj}(z)^{\sigma-1} dz + \chi_j I_j]^{1/\sigma}$. All estimation and every quantitative statement in the paper are in terms of χ_j . The parameter χ_j is not a number of tasks. A task-count translation $\delta_j = \chi_j / (\gamma_j B_j)^{\sigma-1}$ is conditional on a normalization of the

¹Equation (7) is the exact CES aggregate derived in Appendix A, equation (A.9), and is what enters every quantitative computation. Where the text refers to the two terms separately, it uses the additive shorthand $\int_{I_j}^1 g_{Lj}(z)^\varsigma dz + \delta_j I_j (\gamma_j B_j)^\varsigma$ introduced above. Appendix G.10 reports the productivity decomposition under an alternative wage-only calibration from related work (Chávez, 2026), with π_R/π_A between 1.71 and 2.60 across its configurations.

reinstated-task productivity $\gamma_j B_j$, and task counts are reported only as an illustration under that normalization. The theory derivation of Appendix A retains δ_j and $\gamma_j B_j$ as separate primitives, because the entry margin that determines δ_j is a distinct economic object there, and the mapping (8) carries its results into the estimated system.

Effective labor aggregates non-college (S) and college (H) workers via CES with elasticity $\varepsilon_w > 1$:

$$L_j^{\text{eff}} = \left[(a_S^j \cdot \ell_S^j)^{\frac{\varepsilon_w - 1}{\varepsilon_w}} + (a_H^j \cdot \ell_H^j)^{\frac{\varepsilon_w - 1}{\varepsilon_w}} \right]^{\frac{\varepsilon_w}{\varepsilon_w - 1}}, \quad (9)$$

with skill–sector productivities satisfying comparative advantage: non-college workers have comparative advantage in physical production, college workers in cognitive production ($\kappa \equiv (a_S^P/a_H^P)/(a_S^C/a_H^C) > 1$). Sector prices satisfy $p_P/p_C = [\mu/(1 - \mu)](Y_C/Y_P)^{1/\rho}$.

Sorting disruption and augmentation on the cognitive frontier.

Cognitive automation moves the skill weights of the cognitive sector’s labor aggregator through two channels. First, displacement scrambles the sorting of non-college workers across the remaining cognitive tasks: the non-college weight becomes $a_S^C (1 - I_A)^\varphi$, with $\varphi \geq 0$ the sorting disruption parameter. Second, AI that complements surviving high-skill cognitive work raises the productivity of the tasks college workers retain: the college weight becomes $a_H^C (1 + \alpha_A I_A)$, with $\alpha_A \geq 0$ the augmentation parameter. The two channels are structural mirrors, and both enter production separately from the reinstatement channel χ_A , so the model admits high augmentation with zero reinstatement, the reverse, and both at once. The physical frontier has neither channel, a modeling assumption discussed with the estimates.

The qualitative comparative statics of α_A follow from its position in the aggregator. A higher α_A raises the effective productivity of college labor in the cognitive sector, so it raises the college wage W_H and widens the skill premium W_H/W_S , by raising the return to college employment in the cognitive sector it raises the college employment share, and by raising the value the cognitive sector extracts from its surviving tasks it raises the productivity content of a unit of cognitive frontier advance, π_A . Augmentation therefore opposes sorting disruption on the productivity margin while reinforcing it on the premium margin. In principle a college-complementary wage pattern could identify α_A separately from (χ_A, φ) . In practice commuting-zone data cannot deliver that separa-

tion, because a four-block design that attempts it is rank-deficient, with α_A at its zero boundary in every bootstrap draw because the college-complementary exposure measure's wage response is no larger than the displacement-loaded measure's (Appendix G.9). The estimated system therefore sets $\alpha_A = 0$, and the augmentation channel remains in the model as the structural interpretation a different research design could revive.

Frontier and sector indices correspond one-to-one: $R \equiv P$ for the robot frontier operating on physical tasks, and $A \equiv C$ for the AI frontier operating on cognitive tasks. The endogenous automation mix $\omega \equiv I_R/(I_R + I_A)$, defined for $I_R + I_A > 0$, the robot share of total automation investment, and the reinstatement rates (δ_R, δ_A) are determined in the two-frontier equilibrium:

$$\omega = \omega\left(\frac{W_S}{W_H}, \frac{r_R}{r_A}\right), \quad \frac{\partial \omega}{\partial (W_S/W_H)} > 0, \quad (10)$$

$$\delta_j = \frac{1}{I_j} \left[\frac{\theta_j}{c_0(1+\eta)} (v_j - W_{k(j)}^d) \right]^{1/\eta}, \quad (11)$$

for $I_j > 0$ and $v_j > W_{k(j)}^d$, with $\delta_j = 0$ at the zero-entry corner otherwise. This is the marginal-cost condition of the previous paragraph divided by I_j . Here θ_j is the task complementarity parameter at frontier j , v_j is the value of a reinstated task, and $W_{k(j)}^d$ is the post-displacement wage of the skill group $k(j)$ that frontier j 's automation displaces, following the comparative-advantage assignment of equation (9): $k(R) = S$ and $k(A) = H$. Here r_R and r_A are the rental rates on robot and AI capital, and $c_0 > 0$ and $\eta > 0$ are the scale and the curvature of the task-creation cost function $c(\cdot)$ introduced above. The two frontiers can differ both in the number of feasible complementary tasks per unit of advance and in the value margin $v_j - W_{k(j)}^d$ that governs whether those tasks are created. What enters the macro model is the *realized* effective task content χ_j that this margin produces in equilibrium. The cognitive content χ_A is estimated from the commuting-zone moments, and the physical channel is calibrated externally from firm-level evidence, with results reported over its full evidence range.

2.2 Three-Sector Extension: Non-Tradable Services

The quantitative results come from the two-sector core. This extension is solved separately, in Appendix B.1, and it produces the computed demand-side feedback channels. I extend the production

structure to three sectors: physical (P), cognitive (C), and non-tradable services (N). The final good becomes:

$$Y = \left[\mu_P Y_P^{\frac{\rho-1}{\rho}} + \mu_C Y_C^{\frac{\rho-1}{\rho}} + \mu_N Y_N^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}}, \quad \mu_P + \mu_C + \mu_N = 1. \quad (12)$$

Physical and cognitive sectors retain the two-frontier task structure. Non-tradable services are produced with labor only ($Y_N = A_N \cdot L_N^{\text{eff}}$) and are not automated: $I_N = 0$, $\delta_N = 0$. This is the Baumol sector. Accordingly the three-sector weights (μ_P, μ_C, μ_N) are not separately calibrated. The two-sector core uses the physical share $\mu = 0.45$, and of the three-sector weights only $\mu_N = 0.35$ enters, as the scale of the demand-side channels.

With non-homothetic preferences, non-college workers spend a larger share of income on local services ($\alpha_S^N > \alpha_H^N$). AI displacement that reduces $W_S L_S$ directly reduces service demand, releasing non-college workers to the tradable sectors and depressing W_S further. The channel operates through a local demand multiplier and enters the feedback through labor reallocation, bypassing the CES price dampening that limits the composition channel.

A natural concern is that AI could automate services ($I_N > 0$), which would change the sign of this channel. If AI automates customer service, food ordering, or personal care scheduling, the non-tradable sector shrinks through automation itself and the labor reallocation channel weakens. Appendix E evaluates a parametric extension in which the service sector is partially automated. The supply-side ranking survives partial service automation.

2.3 Endogenous Labor Supply

Worker i of skill k participates if $W_k \geq \underline{w}_k(i)$, where reservation wages are drawn from $F_k(\cdot)$. The participation elasticity is $\varepsilon_k^{LS} \equiv W_k f_k(W_k) / F_k(W_k)$, calibrated for each skill group from Chetty et al. (2013) (Table 6).

The participation margin *dampens* the wage response (some workers exit when wages fall) but *amplifies* the income response. Which force wins is a quantitative question. Solving the margin at the estimated parameters, the dampening prevails and participation reduces the total feedback.

2.4 Endogenous Capital Income Distribution

Automation generates capital income from returns to automated tasks (Π^{auto}) and reinstatement rents (Π^{reinst}). With heterogeneous marginal propensities to consume ($m_S > m_H > m_K$), a shift toward AI raises the capital income share and lowers the aggregate MPC, feeding back through the non-tradable channel.

The demand composition channel, which operates through the relative price of the two final goods, completes the set of four. Computed at the estimated parameters, their net contribution is a tenth of a percent of the direct supply-side effect, which is the content of the formal supply-side dominance result.

3 Estimation and Calibration

The structural parameters $\theta = (\Lambda_R, \chi_A, \varphi)$ are recovered by minimum distance from reduced-form commuting-zone moments under design-based inference. The migration and incidence elasticities are calibrated externally, as is the physical reinstatement channel, which is reported over its evidence range, and the commuting-zone estimates are then mapped to national parameters.

3.1 Reduced-Form Moments

A minimum distance estimator (MDE) maps the six reduced-form moments of the paper’s two exposure measures into the model’s parameters, by matching the model’s predicted cross-sectional moments to their empirical counterparts (Section 3.4). The AI moments enter the estimator net of a college-wage trend that predates AI, and Section 3.3 states the identifying assumption that treatment rests on and tests it backward, forward and within zones. The college local labor-supply elasticity ε_H , which governs the incidence of local demand shocks on college wages through the share $\eta_H = \varepsilon_D / (\varepsilon_D + \varepsilon_H)$, is calibrated externally, and it is distinct from the college participation elasticity ε_H^{LS} of Section 2.3. This subsection documents the reduced-form coefficients that serve as inputs.

I regress three commuting-zone-level outcomes, log change in non-college wages ($\Delta \ln W_{NC}$), log change in college wages ($\Delta \ln W_C$), and change in college employment share (Δs_C), on robot and AI exposure, using 722 commuting zones, long differences from pooled 2005–2007 baseline to

pooled 2015–2019 endpoint ACS samples. Robot exposure is the EU5 Bartik instrument, built from 1993–2014 robot adoption quantities in Denmark, Finland, France, Italy, and Sweden (Acemoglu and Restrepo, 2020a). AI exposure is the Webb (2020) AI exposure index in Bartik percentiles. All regressions include 38 baseline demographic and labor-market composition shares, the baseline farm-occupation employment share, and 8 census region dummies, and are weighted by baseline commuting-zone employment.²

Table 1 reports the results. Commuting zones with greater robot exposure experienced lower non-college wage growth ($\hat{\beta}_{R,NC} = -0.0079$, $t = -4.25$, precisely estimated under conventional inference, though Section 3.2 shows the design-based robot-side precision is far lower) and moderately lower college wage growth ($\hat{\beta}_{R,C} = -0.0055$, $t = -1.99$). AI exposure is associated with higher college wage growth ($\hat{\beta}_{AI,C} = +0.0052$, conventional $t = +2.07$, but exposure-robust $t = 1.46$, so it is not significant under the design-based inference this paper prefers), lower non-college wage growth ($\hat{\beta}_{AI,NC} = -0.0069$, $t = -2.88$, which survives exposure-robust inference at $t = -2.24$ but clears neither permutation null, $p = 0.10$ within skill bins and $p = 0.16$ on task residuals), and a rising college employment share ($+0.0030$, conventional $t = +2.29$, though it does not survive the within-bin permutation null). The AI moment with the strongest support against a null is the skill-premium response in column (4): $+0.0121$, with $t = +4.51$ conventionally and $+3.60$ under exposure-robust inference. It is the only moment that clears a permutation null at conventional levels ($p = 0.014$ within skill bins), though it is marginal against the task-residual null ($p = 0.068$). Differencing the two wage responses removes the commuting-zone-level shocks common to both, which cuts the variance of the contrast below the sum of the two variances but not below either one, so at 0.0027 its standard error exceeds both 0.0024 and 0.0025 . What the contrast delivers is a larger coefficient against a comparable standard error, and so the largest t . This pattern is what the minimum-distance system maps into Λ_R , χ_A and φ , after the AI gradients are netted of the pre-AI college-wage trend (Section 3.3).

The college-share moments carry the expected signs (robots reduce, AI increases). The AI college-share coefficient survives exposure-robust inference ($t = 2.11$) but not the within-skill-bin

²The correlation between the robot and AI exposure instruments is $r = -0.067$, so the two exposure measures vary nearly independently across commuting zones, which is what allows the robot and AI gradients to be separated. That differential variation identifies the physical displacement scale and the cognitive parameters. The physical reinstatement channel is calibrated externally because local variation cannot measure it (Section 3.5).

permutation null ($p = 0.38$), so it is the weakest of the three AI moments on design-based inference. It nonetheless remains part of the joint minimum-distance system: the wage and share moments jointly inform χ_A and φ through the full 6×6 cross-equation variance-covariance matrix. In the unadjusted system, leave-one-moment-out estimates keep the translated δ_A near one when either AI wage coefficient is removed, while removing the college-share block increases uncertainty and boundary incidence. The share moments therefore matter for precision without uniquely carrying the point estimate.

Figure 1 visualizes the two headline relationships. The robot–non-college wage gradient is steep and approximately linear across the support. The AI–college wage relationship is positive but noisier, consistent with the larger standard errors in column (2).

Appendix I reports three robustness checks. The first two are estimated on the completed control set of Table 1, and the third drops the farm share. Adding the Autor et al. (2013) China import shock to the baseline specification leaves all six moments virtually unchanged (Panel A), indicating that robot and AI exposure capture channels distinct from trade competition. With augmented controls that include log population and baseline tech employment share (Panel B), the AI coefficient on college wages attenuates from 0.0052 to 0.0022 and loses even its conventional significance. It was in any case not significant under the design-based inference this paper prefers. The attenuation reflects the endogeneity of these controls, because commuting zones with high AI-task concentration are the large, tech-intensive metropolitan areas whose size and sectoral composition are themselves shaped by AI exposure. These controls absorb the cross-sectional variation carried by AI-task concentration, so the baseline specification of Table 1 is the one the MDE uses. Outlier and design-based diagnostics for these moments are in Section 3.2.

The baseline control set comprises 38 pre-determined composition shares plus the baseline farm-occupation employment share. Farm share belongs to the same class as the other 38, baseline, pre-determined, and compositional, and unlike the log-population and tech-share controls rejected above it does not absorb the exposure variation. A Rotemberg decomposition of the AI exposure measure under this set places the top identifying weight on computer systems analysts, the occupation the model’s mechanism says should carry an AI moment. Dropping the farm share leaves the AI college-wage coefficient nearly unchanged (+0.0052 against +0.0056) while weakening the non-college and college-share moments. The farm-dropped specification is reported as Panel C of Table I.1, and

all moments feeding the estimation use the completed set. Appendix [I.1](#) decomposes the exposure gradients into early and late sub-windows, where the AI gradients are not detectable in 2005–2010 and emerge after 2010, and Appendix [I.2](#) runs the census pre-period test on which the identification section builds.

A natural extension would enrich the moment system to four exposure measures, adding the [Webb \(2020\)](#) robot measure’s occupation-level task vulnerability on the physical side and splitting the cognitive side between the displacement-loaded Language Model Occupational Exposure (LMOE) measure and the college-complementary Webb AI measure, together with direct task-creation and task-displacement moments reconstructed inside this paper’s replication package from the raw displacement/reinstatement task kit of [Acemoglu and Restrepo \(2022\)](#) and the Lin new-work title data, placed on a common cardinal scale before structural use. That four-block system was built and estimated for this paper, and the richer system is not separately identified in these data. The Jacobian is rank-deficient, the augmentation parameter sits at its zero boundary, and the direct task-content moments fail a pre-period placebo. Appendix [G.9](#) reports the diagnostic. The estimation below therefore uses the parsimonious system the data can support.

Table 1: Reduced-Form Effects of Robot and AI Exposure

	Dependent Variable			
	$\Delta \ln W_{NC}$	$\Delta \ln W_C$	Δs_C	$\Delta \ln(W_C/W_{NC})$
	(1)	(2)	(3)	(4)
<i>Panel A: Conventional inference</i>				
Robot exposure (EU5)	−0.0079*** (0.0019)	−0.0055** (0.0028)	−0.0016 (0.0011)	—
AI exposure (Webb pctl)	−0.0069*** (0.0024)	0.0052** (0.0025)	0.0030** (0.0013)	0.0121*** (0.0027)
<i>Panel B: Design-based inference</i>				
Robot: permutation p (19 shocks)	0.11	0.27	0.46	—
AI: exposure-robust t (AKM)	−2.24	+1.46	+2.11	+3.60
AI: within-bin permutation p	0.10	0.17	0.38	0.014
AI: task-residual permutation p	0.16	0.31	0.09	0.068
Completed controls + region FEs	Yes	Yes	Yes	Yes
Observations	722	722	722	722

Notes: Each column reports a weighted OLS regression of the indicated outcome on both exposures under the completed control specification of Section 3.1: 38 baseline composition shares, the baseline farm-occupation employment share, and 8 census region dummies, weighted by baseline employment. Robot exposure is the EU5 Bartik over 1993–2014 adoption quantities. AI exposure is the Webb (2020) index in Bartik percentiles. Column (4) is the skill-premium response, a linear combination of columns (1)–(2) whose information the joint estimation already uses. Panel B holds each estimate to its design: permutation over the 19 industry shocks on the robot side, where the coarse permutation distribution limits resolution to roughly the ten-percent level, and exposure-robust standard errors (Adão et al., 2019), abbreviated AKM in the row labels, with skill-bin-constrained permutation nulls over 338 occupation scores on the AI side. Treating those scores as the resampling units carries conditions beyond how concentrated they are: conditional on the exposure shares and the controls, the score-level residual contributions are taken to be mean independent across scores and free of dependence within the skill bins the permutation respects. The effective count answers concentration alone, and these conditions are maintained, not tested. Robust standard errors in parentheses. Coefficients and standard errors are displayed rounded to four decimals. Reported t -statistics are computed from the unrounded estimates and need not equal the ratio of the displayed figures. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Two of these coefficients carry the estimation, the robot non-college wage response and the AI skill-premium contrast, and a shift-share coefficient of either kind can be produced by a few extreme zones. The binned scatters below show that neither is. Both gradients are close to linear across the support of exposure, so the coefficients describe the whole cross-section and not its tails, which is the condition under which the minimum-distance system can read them as moments.

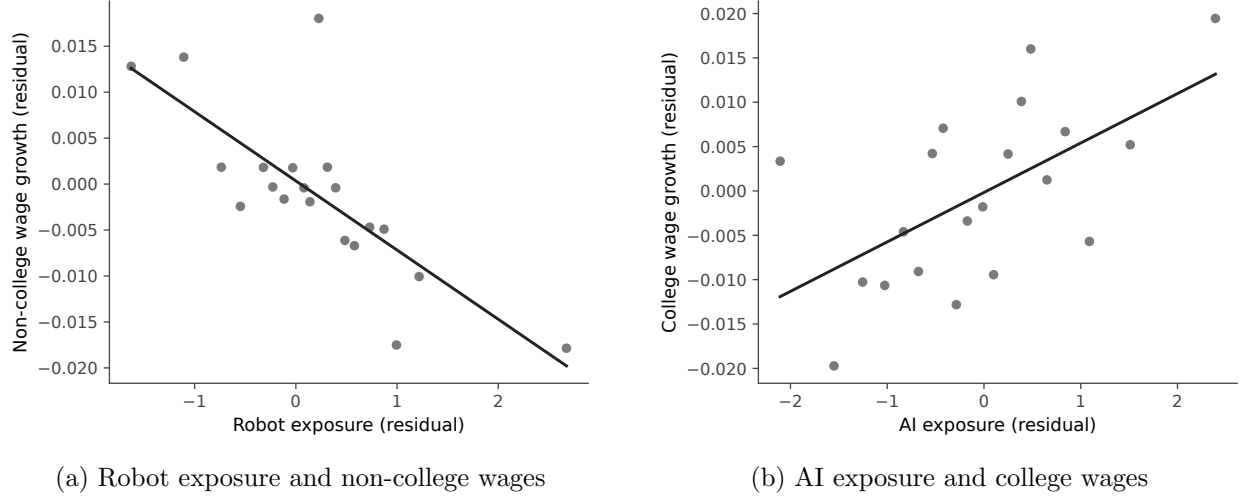


Figure 1: Partial Correlation Plots: Technology Exposure and Wages

Notes: Each panel displays a binscatter with 20 quantile bins, weighted by baseline commuting-zone employment. Panel (a) plots $\Delta \ln W_{NC}$ against EU5 robot exposure, residualized on AI exposure and baseline controls. Panel (b) plots $\Delta \ln W_C$ against Webb AI exposure, residualized on robot exposure and baseline controls.

3.2 What Identifies the Robot Moments

The robot moments are estimated on the employment-weighted cross-section. The coefficient is the employment-weighted partial linear projection, which weights commuting zones by employment times residual exposure, and it equals the wage response of the average worker only under a common slope. The exposure measure is a shift-share object: nineteen industry-level European adoption shocks spread over baseline industry employment shares. The identifying claim is that the shocks, not the shares, are as good as random (Borusyak et al., 2022), which makes the industry count the relevant sample size for inference. The effective number of shocks, the inverse Herfindahl of the exposure weights, is 3.05. The AI exposure measure aggregates 338 occupation scores with far more dispersed weights and gives 77.9 effective occupation shocks, the sample size the exposure-robust

AI inference exploits.

The robot design is concentrated in automotive adoption: automotive accounts for 93.9 percent of the identifying covariance in the non-college wage moment, which is how the International Federation of Robotics data are built. Conventional commuting-zone inference therefore overstates precision, and I use shock-level inference throughout. Under this design-based inference the non-college wage response is the informative robot moment, and the college wage and college employment share moments are imprecisely estimated (permutation p of 0.27 and 0.46). Appendix H reports the Rotemberg decomposition, the specification and state-deletion ladders, and the margins that move.

The two tiers carry into the reported uncertainty. The robot block pins down Λ_R , the scale of the displacement shock. Table 5 reports the two-way bootstrap standard error (0.0092), which resamples at the industry level on the robot side, against 0.0030 under conventional commuting-zone inference. The physical reinstatement channel is not in these moments, since it is calibrated externally (Section 3.5), so no conclusion about reinstatement asymmetry rests on the automotive experiment. The restriction $\psi \equiv 1$ is maintained because an industry-level demand shock is skill-neutral by construction (Section 3.6).

3.3 Identification: Pre-Trends, Timing, and Within-Zone Variation

The AI gradients of Table 1 are cross-sectional long differences. They measure an AI effect only if, absent AI, the skill-specific wage trend is orthogonal to the exposure score. That condition is testable in two directions, backward and forward in time, and within zones, and this subsection reports all three before the moments enter the estimator.

Backward. Appendix I.2 regresses the 1990–2000 census wage changes on the same 2005 exposure scores. Zones that would later measure as AI-exposed already had faster college-wage growth: +0.0195 log points per standard deviation of exposure per decade, three times the 2005–2019 gradient in the same units, and the coefficient survives both constrained permutation nulls. The non-college gradient in 1990–2000 is positive, +0.006, the opposite sign of its post-2005 value.

Forward. Large language models diffused from late 2022. If the exposure score measures an AI effect, the window that starts after 2019 is the first in which it should predict more, and if it measures a trend, it should predict the same gradient as before or less. Table 2 estimates the three AI moments on the same exposures, controls and weights over five windows of the PUMA-vintage-corrected panel.

Table 2: AI Exposure Gradients by Window

Window	$\Delta \ln W_{NC}$	$\Delta \ln W_C$	Δs_C
1990–2000 (census)	+0.0064 [0.16]	+0.0195 [0.03]	+0.0018 [0.98]
2000–2007 (census to ACS)	−0.0002 [0.98]	+0.0177 [0.06]	—
2005–07 to 2010–12	−0.0064 [0.55]	+0.0081 [0.47]	+0.0024 [0.49]
2010–12 to 2015–19	−0.0075 [0.33]	+0.0009 [0.90]	+0.0037 [0.23]
2015–19 to 2022–24	−0.0051 [0.52]	−0.0017 [0.86]	−0.0002 [0.95]

Notes: Coefficient on standardized AI exposure, expressed per decade, from the Table 1 specification (robot exposure, farm share, 38 baseline composition shares, 8 region dummies, baseline employment weights) on 722 commuting zones. Brackets: permutation p -values over the 338 occupation scores. The 1990–2000 and 2000–2007 rows are the census pre-period of Appendix I.2. The post-2005 rows use pooled endpoints from the PUMA-vintage-corrected wage panel, which is why the 2005–07 to 2015–19 window reproduces the signs of Table 1 at somewhat smaller magnitudes.

The college-wage gradient is largest before AI existed and falls monotonically: +0.0195 in the 1990s, +0.008 in 2005–2012, +0.001 in 2012–2019, and −0.002 in the window that contains the diffusion of large language models. That is a pre-trend running out, and no window after 2012 shows the college wage rising with exposure. The non-college gradient moves the other way. It was positive in the 1990s, zero in 2000–2007, and negative in every window from 2005 on, including 2015–19 to 2022–24, at a stable −0.005 to −0.008 per decade. A sign reversal after 2005 that persists through 2024 is not the continuation of a trend. The college employment share follows the college wage: its gradient is present through 2019 and absent after.

Within zones. The same exposure score varies across occupations within a zone, and a design with zone-by-skill fixed effects absorbs every zone-level trend, including the one the backward test finds. Table 3 regresses the change in the mean log wage of zone-by-occupation-by-skill cells from the ACS microdata on the occupation’s standardized AI score, with the robot score as a control,

baseline cell employment as weights, and standard errors clustered by occupation.

Table 3: Within-Zone Wage Growth of AI-Exposed Occupations

Window	Fixed effects	Pooled	Non-college	College	Cells
2005–2019	none	−0.014 (0.008)	−0.021 (0.009)	+0.011 (0.011)	13,666
2005–2019	zone × skill	−0.014 (0.008) [0.09]	−0.021 (0.008)	+0.010 (0.010)	13,666
2005–2010	zone × skill	+0.008 (0.004) [0.09]	+0.004 (0.004)	+0.020 (0.006)	17,598
2010–2019	zone × skill	−0.025 (0.007) [0.00]	−0.027 (0.008)	−0.019 (0.010)	15,144

Notes: Coefficient on the occupation’s AI exposure score, standardized across the 338 occupations, in a regression of the change in the cell’s employment-weighted mean log wage. Cells are commuting zone by occupation by skill group with at least ten observations at both endpoints. Standard errors in parentheses are clustered by occupation. Brackets: permutation p -values over the occupation scores. The skill-group columns re-estimate on that group’s cells alone.

Zone-by-skill fixed effects leave the coefficients where they were, so zone trends do not drive the occupation-level pattern. In 2010–2019 the more exposed occupations lose relative wage in both skill groups, the non-college loss at $t = -3.4$ with a permutation p -value below 0.001, and the college gain is confined to 2005–2010. Within zones, AI exposure predicts displacement without reinstatement over the sample horizon. This is reduced-form evidence on sign and timing and not a moment for the structural system, whose skill-group wages equalize across tasks within a zone.

What the three tests establish. The non-college wage response to AI exposure is identified: it reverses sign after 2005, persists through 2024, and appears within zones with zone trends absorbed. The college wage response is not: the gradient that the unadjusted moments carry is the tail of a trend that predates AI and is gone by 2012. The estimator therefore uses the moments adjusted for the pre-period gradient, $m^{\text{adj}} = m^{\text{on}} - 1.4 g^{\text{pre}}$, the linear continuation of the 1990–2000 per-decade gradient across the 14-year window, with the pre-period covariance added to the moment covariance. The maintained identifying assumption is that the pre-2005 skill-specific trend in exposed zones would have continued at its 1990s rate. That is the parallel-trends assumption of a difference-in-gradients design, and Table 2 is the evidence on it. The college gradient continues at a declining rate, so the linear continuation subtracts more than the trend delivered over the window. That pushes the adjusted college moment down, cognitive reinstatement toward zero and

the productivity ratio up, which is why Table 4 reports the estimates at smaller continuation rates as well. The unadjusted system is reported in Appendix I.3 and every headline object is shown under both.

Table 4 re-estimates the system as the assumed continuation of the pre-trend varies from none, the unadjusted moments, to the full linear continuation. Cognitive reinstatement falls and the productivity ratio rises monotonically with the rate, and robot dominance holds at every rate. The rate is the one assumption the cognitive estimates rest on, and the table is the range of results it spans.

Table 4: Estimates by Assumed Continuation of the Pre-Trend

Continuation (decades subtracted)	$\hat{\delta}_A$	$\hat{\varphi}$	J	π_R/π_A	$(1 + \tau_R^*)/(1 + \tau_A^*)$
0 (unadjusted moments)	0.99	1.99	0.01	1.21	1.33
0.5	0.57	1.55	0.71	1.48	1.43
0.7	0.43	1.38	1.75	1.61	1.48
1.0	0.24	1.13	2.00	1.83	1.57
1.4 (linear continuation, baseline)	0.03	0.80	2.86	2.19	1.72

Notes: Each row subtracts the stated multiple of the 1990–2000 per-decade gradient from all six on-treatment moments, adds the pre-period covariance scaled accordingly, and re-estimates $(\Lambda_R, \chi_A, \varphi)$ by two-step minimum distance. Macro objects at the neutral physical calibration $\delta_R = 1.00$. At 0.7 the global minimum of the objective lies in a second basin with $\varphi = 6.9$ and negative cognitive productivity content; the row reports the regular basin, whose objective is 0.5 higher. The linear continuation over the 14-year window is 1.4 decades.

3.4 Minimum-Distance Estimation

The parameter vector is $\theta = (\Lambda_R, \chi_A, \varphi)$: the physical displacement scale, the cognitive effective task content, and sorting disruption. The moment vector is the six reduced-form coefficients of Table 1, the change in non-college log wages, the change in college log wages, and the change in the college employment share, each regressed on the EU5 robot shift-share and the Webb (2020) AI exposure measure, each net of the linear continuation of its 1990–2000 pre-period gradient (Section 3.3), with the pre-period covariance added to the moment covariance. The three robot moments identify Λ_R , the three AI moments identify (χ_A, φ) , with the wage-level contrast carrying χ_A and the premium contrast carrying φ . The adjustment removes most of the premium contrast,

so φ is weakly identified here and is reported with its profile interval. The system has six moments and three parameters, three overidentifying restrictions.

Two parameters that a richer system would estimate are fixed outside the estimator, each for a stated reason. Migration and local labor-supply elasticities are calibrated externally: $\varepsilon_D = 3.0$ and $\varepsilon_S^{\text{mig}} = 0.78$ from [Notowidigdo \(2020\)](#)-class estimates, and the college incidence elasticity is fixed at $\varepsilon_H = 1.38$, the value the same incidence literature supports. Estimating ε_H jointly returns a 90 percent bootstrap interval of $[-0.60, 6.26]$, avoidable underidentification that calibration plus a reported sensitivity grid over $\varepsilon_H \in [0, 3]$ replaces (Table [G.5](#)). The physical effective task content is not estimated at all, because CZ variation cannot measure it, and it enters as the external calibration of Section [3.5](#).

Two-step minimum distance with the inverse of the adjusted covariance, the two-way on-treatment matrix (robot block over the 19 industry shocks, AI block over the 338 occupation scores) plus the pre-period covariance, gives

$$\hat{\Lambda}_R = 0.0090, \quad \hat{\chi}_A = 0.013, \quad \hat{\varphi} = 0.804,$$

with $J = 2.86$ on three overidentifying restrictions, reported as a fit criterion without a probability, since the chi-square reference is unavailable under the robot design's shock concentration. Under the reinstated-task efficiency normalization $(\gamma_A B_A)^{\sigma-1} = \bar{g}_A = 0.5$ used for illustration throughout, $\hat{\chi}_A$ translates to a task-count reinstatement rate $\hat{\delta}_A = \hat{\chi}_A / \bar{g}_A = 0.026$, and the translation, not the estimate, is conditional on that normalization. Cognitive reinstatement over the sample horizon is close to zero, and its profile interval is $[0, 0.24]$ for χ_A , $[0, 0.48]$ in translated units, censored at zero. The 6×3 Jacobian of the moment conditions has rank three, standard-error-scaled singular values $(2.51, 0.44, 0.135)$, and condition number 18.6, with the weakest direction loading on χ_A , so the data separate the parameters, and the one they separate least is the one the adjustment pushes to zero. φ is the parameter the adjustment leaves least determined. Its profile interval is $[0.64, 7.2]$, and a one-step estimator with a diagonal weight matrix returns 6.9, a second basin of the objective where the cognitive productivity content is negative. The two-step estimate 0.80 is the regular basin, and nothing in Sections [6](#) and [7](#) rests on its value, since robot dominance holds at every φ the draws visit. Λ_R maps exposure units into the model's shock size for moment-fitting

and does not enter the macro objects of Sections 6–7, so its imprecision does not propagate into the headline results. The (χ_A, φ) pair is weakly correlated across bootstrap draws, and Appendix G.7 documents that correlation and what the macro objects do across it. Table 5 reports the estimates with their uncertainty, and Table 4 how they move with the assumed continuation of the pre-trend.

Table 5: Estimated Parameters (Minimum Distance)

Parameter	Value	Source	Identification
<i>Estimated here (moments of Section 3.1 net of the pre-period gradient, Section 3.3)</i>			
$\hat{\chi}_A$	0.013 (0.081)	MDE, 6 moments	AI $\rightarrow (\Delta W_{NC}, \Delta W_C, \Delta s_C)$
translated $\hat{\delta}_A$	0.03 (0.16)	$\chi_A/\bar{g}_A$ at $\bar{g}_A = 0.5$	Normalization-conditional task count
$\hat{\varphi}$	0.80 (1.51)	MDE, 6 moments	AI skill-premium (wage–wage) contrast
$\hat{\Lambda}_R$	0.0090 (0.0092)	MDE, 6 moments	Robot $\rightarrow (\Delta W_{NC}, \Delta W_C, \Delta s_C)$
<i>Fixed outside the estimator</i>			
ε_H	1.38	Incidence literature	Calibrated, grid [0, 3] in Table G.5

Notes: Standard errors in parentheses are the standard deviations of 200 bootstrap draws of this estimator: the six adjusted moments are drawn from a normal distribution with the adjusted covariance, the two-way on-treatment covariance (robot block over the 19 industry shocks, AI block over the 338 occupation scores) plus the covariance of the pre-period gradients, and each draw is re-estimated with $\chi_A \geq 0$ and $\varphi \geq 0$ imposed. The lower bound binds in 51 draws for Λ_R and 83 for χ_A , so those two standard deviations understate the spread on the unconstrained side. Every set statement in the paper comes from these draws. The moments are the six coefficients of Table 1, 722 U.S. CZs, pooled 2005–2007 to 2015–2019 ACS long differences net of the 1990–2000 gradient continued linearly over the 14-year window, $\psi \equiv 1$, ε_H fixed at 1.38. The entry for $\hat{\Lambda}_R$ is a dispersion measure and not a confidence statement, because the robot design’s shock concentration (Section 3.2) leaves no normal reference for it, and Appendix H reports an exposure-robust standard error of 0.0193 for the same parameter, about twice the resampling figure. Hansen $J = 2.86$ on 3 overidentifying restrictions is reported without a probability for the same reason. A one-step estimator with a diagonal weight matrix agrees on Λ_R and χ_A and returns $\varphi = 6.9$, the degenerate basin discussed in the text. Appendix I.3 reports the same system on the unadjusted moments. Values are from the baseline model with Cobb–Douglas final-good aggregation, and Table G.3 reports sensitivity to CES aggregation at the calibrated $\rho = 0.80$.

3.5 The Physical Reinstatement Channel: An External Calibration

The physical reinstatement channel is not estimated but externally calibrated. The reason is structural, not a data limitation: reinstated physical tasks are created economy-wide, so they are absorbed by the constant of any cross-CZ regression, and no moment built from local exposure

variation measures them (Section 3.6). The calibrated object is the effective reinstatement offset ratio δ_R , the value of new labor tasks created per unit of task value displaced at the physical frontier. The baseline sets $\delta_R = 1$: task creation offsets displacement, the neutral point of the firm-level evidence. At the estimates the productivity ranking holds at every value of this object in the range the evidence brackets, including the narrow bound at which reinstatement counts only robot-complementary tech tasks, and a bisection finds no parity crossing above $\delta_R = 0.30$. Humlum (2024)’s event studies of robot adoption in Danish manufacturing find that adopters cut the production-worker wage bill by a fifth while sales and total labor demand rise. He does not report a reinstatement rate, so I convert those responses into task-value units, and the conversion is what places adopters in the reinstatement-dominant regime. That step is a decomposition and not a reading of the signs, because total labour demand can rise with weak task creation and a strong scale response, so the observed pattern is consistent with a lower rate as well and the upper end of the range conflates creation with scale by construction. The translation brackets the ratio between 0.60, counting only robot-complementary tech tasks as reinstatement, and 2.00, attributing adopters’ entire net labor-demand expansion to task creation (Appendix D.3 derives the bracket). Because δ_R is a calibration it carries no standard error, and every macro and policy result is reported across the full range [0.60, 2.00] instead (Table 7). Danish institutions, stronger employment protection, union co-determination, and active labor market policies, create incentives for firms to redeploy their workers, so the Danish range may sit toward the high side of the U.S. rate. The lower half of the range is what carries that concern, and the results are reported there too. U.S. firms adopting industrial robots exhibit comparable patterns of internal task reallocation (Acemoglu and Restrepo, 2020a), and European evidence corroborates reinstatement across institutional settings (Koch et al., 2021; Dauth et al., 2021). The wage-only displacement calibration of Chávez (2026) with $\delta_R = 1.14$ is carried through Appendix G.10 as a point inside the range.

Table 6: Calibrated Parameters

Parameter	Value	Source	Identification
<i>Physical frontier (external calibration, no standard error)</i>			
δ_R	1.00 [0.60, 2.00]	Humlum (2024)	Reinstatement offset ratio, neutral baseline
<i>Production structure</i>			
σ	2.00	Acemoglu and Restrepo (2020a)	Task elasticity
ε_w	1.50	CZ wage differentials	Skill elasticity
κ	1.80	Sectoral employment	Comparative advantage
μ, ρ	0.45, 0.80	Output shares, structural change lit.	Physical share (two-good aggregator), goods subst.
ξ_R, ξ_A	0.45, 0.50	Assigned (no external estimate)	Results insensitive, App. G.4
$\varepsilon_D, \varepsilon_S^{\text{mig}}$	3.0, 0.78	Notowidigdo (2020)	Incidence share $\eta_S = 0.794$
<i>Three-sector extension and demand-side</i>			
μ_N	0.35 [†]	Service employment	Non-tradable share
s_N^S	0.45 [†]	CPS service employment	Non-college employment in services
α_S^N, α_H^N	0.40, 0.25	CEX data	Service expenditure shares
$\varepsilon_S^{LS}, \varepsilon_H^{LS}$	0.25, 0.10	Chetty et al. (2013)	Participation elasticities
m_S, m_H, m_K	0.75, 0.45, 0.20	Dynan et al. (2004)	Marginal propensities
<i>Dynamic model</i>			
σ_c	0.50	Standard macro	Intertemporal elasticity
ρ_S, ρ_H	0.06, 0.03	Wealth data	Time preference
d_A, d_R	0.10, 0.10	Standard	Capital depreciation, symmetric baseline

[†] These two service-sector anchors are recorded from the data, but the extended model cannot reproduce both at once: at its equilibrium the service share of value added is 0.15 and the share of non-college workers employed in services is 0.27, and no service-sector scale satisfies both targets simultaneously. Appendix B.1 reports the tension and shows the general equilibrium feedback is negligible under either.

Notes: The physical δ_R is the external calibration of Section 3.5, the neutral baseline 1.00 with the range [0.60, 2.00] derived from [Humlum \(2024\)](#)’s within-firm evidence, and it carries no standard error. The wage-only calibration $\delta_R = 1.14$ ([Chávez, 2026](#)) is a point inside this range (Appendix G.10). Remaining parameters are calibrated from the sources listed. CPS is the Current Population Survey and CEX the Consumer Expenditure Survey. The local incidence relation that maps $(\varepsilon_D, \varepsilon_S^{\text{mig}})$ into the incidence share $\eta_S = \varepsilon_D / (\varepsilon_D + \varepsilon_S^{\text{mig}})$ follows [Moretti \(2011\)](#) and [Notowidigdo \(2020\)](#), whose estimates of differential mobility place non-college workers at the lower migration elasticity. The baseline quantitative model aggregates the two final goods by Cobb–Douglas. The calibrated elasticity $\rho = 0.80$ enters the general-equilibrium channels computed in Section 4.1 and the CES sensitivity in Table G.3. The entry-margin symbols (η, c_0) of equation (11) and the dynamic allocation symbols (ζ_j, ν, α_r) require no calibration: the cognitive task content χ_A is estimated directly and the physical channel enters through the externally calibrated δ_R , so neither requires the entry-margin primitives, and the amplification factor is written in $(\sigma_c, \rho_S, \rho_H, d_A)$ under the parameterization Appendix C adopts, which is an assumption on the composite elasticity and not a consequence of the dynamic primitives.

3.6 Why the Physical Channel Is Calibrated and the Cognitive One Estimated

The identification design assigns each parameter to the variation that can measure it, and the division between estimation and calibration reflects differences in how the two frontiers transmit.

Robot deployment, as measured by the Bartik instrument constructed from European adoption rates, is an *industry-level* demand shock. When robots are adopted in the auto industry, all workers in auto-intensive commuting zones experience wage and employment effects, regardless of their individual occupation or task content. The restriction $\psi \equiv 1$, robots hitting college and non-college workers equally through the demand channel, is maintained on these construction grounds. At the CZ level, the robot Bartik identifies Λ_R , the total industry demand effect inclusive of local multipliers, not task content.

The occupation-level AI exposure measure, by contrast, is constructed from task content and varies across commuting zones with baseline occupational composition. Its coefficients are estimated separately for the non-college wage, college wage, and college-employment-share outcomes, and those three moments inform (χ_A, φ) through the Acemoglu–Restrepo task structure: cognitive displacement is local to exposed occupations, and the wage-level and premium contrasts separate reinstatement from sorting disruption (Appendix G.7), a separation the pre-trend adjustment weakens, which is why φ is reported with its profile interval (Section 3.4).

CZ-level data can speak to task content only after two margins are separated. Λ_R is the local displacement rate: it measures how much *worse off* a high-robot-exposure CZ is relative to a low-exposure CZ, holding national aggregates constant. Reinstatement, by contrast, is largely absorbed by the intercept in a wage regression, because new tasks do not disproportionately appear in robot-heavy CZs. They are created economy-wide.³ The physical reinstatement channel is therefore calibrated. No moment built from CZ exposure variation measures economy-wide physical task creation, and the one attempt to force one, the four-block system’s direct task-content moments, failed its pre-period placebo (Appendix G.9). The cognitive side does not share the problem to the same degree, because cognitive displacement and its sorting consequences are local to exposed occupations, and the wage and share responses that identify (χ_A, φ) are differential across CZs

³This is a standard feature of shift-share designs applied to automation: the local exposure captures the *differential* displacement across regions, while the aggregate productivity and reinstatement effects are absorbed by time effects or the constant. See Acemoglu and Restrepo (2020a) and Adão et al. (2019) for discussion.

(Appendix D develops the aggregation argument).

Table 7 reports how the headline results vary across the range implied by Humlum (2024)’s firm-level estimates, from the narrow bound that counts only robot-complementary tech tasks as reinstatement to the broad bound that attributes the entire net labor-demand expansion of adopting firms to task creation.

Table 7: The Physical Calibration Range: Macro Objects over the Externally Calibrated Offset Ratio $\delta_R \in [0.60, 2.00]$ (fixing $\hat{\chi}_A = 0.013$, translated $\delta_A = 0.03$ under the $\bar{g}_A$ normalization, and $\hat{\varphi} = 0.80$)

δ_R	π_R	π_A	π_R/π_A	$(1 + \tau_R^*)/(1 + \tau_A^*)$	Source
0.60	+0.477	+0.264	1.80	1.64	Humlum narrow bound
1.00	+0.587	+0.268	2.19	1.72	Neutral baseline (creation offsets displacement)
1.14	+0.622	+0.269	2.32	1.73	Alternative wage-only calibration
1.40	+0.682	+0.270	2.52	1.75	Illustrative high value
2.00	+0.804	+0.274	2.93	1.77	Humlum broad bound

Notes: All results computed from the macro model of Section 6, holding the estimated cognitive parameters at $\hat{\chi}_A = 0.013$ (translated $\delta_A = 0.03$) and $\hat{\varphi} = 0.80$. π_j denotes the productivity content of frontier j , and τ_R^*/τ_A^* is the optimal subsidy ratio from the Bergson–Samuelson welfare function with $\lambda = 1.5$. The bounds $[0.60, 2.00]$ are the creation-to-displacement value ratios implied by Humlum (2024)’s firm-level event studies, derived in Appendix D.3, and the baseline row is the neutral point $\delta_R = 1$ at which task creation offsets displacement. The $\delta_R = 1.14$ row is the alternative wage-only displacement calibration (Chávez, 2026), a point inside the range. No parity crossing exists inside the range, and the subsidy ratio stays above one throughout because a robot advance raises the non-college wage, which carries the higher welfare weight.

Across the range the productivity ratio runs from 1.80 at the narrow bound to 2.93 at the broad one, and the relative subsidy factor from 1.64 to 1.77. The ratio rises between every pair of adjacent nodes on the 1401-point grid, refining the grid tenfold changes no sign, and a bisection finds no parity crossing above $\delta_R = 0.30$, so there is no value in the range at which the ranking turns (Proposition 5 states the certificate). At the narrow bound, where reinstatement is restricted to robot-complementary tech tasks, robots still deliver 1.80 times the output per unit of advance that AI does, and the relative subsidy factor stays above one everywhere, carried by the distributional motive.

The CZ-level displacement rate in this paper is consistent with prior estimates from different countries and time periods. [Acemoglu and Restrepo \(2020a\)](#) find that one additional robot per thousand workers reduces local wages by about 0.42 percent in U.S. CZs over 1990–2007. This paper’s non-college coefficient ($\hat{\beta}_{R,NC} = -0.79$ percent per unit exposure) has the same negative sign despite covering a later period (2005–2019), its magnitude is not directly comparable because the exposure units differ. [Dauth et al. \(2021\)](#) find a small and statistically insignificant overall wage effect in German *Kreise* over 1994–2014, despite Germany’s much higher robot density (7.6 versus 1.6 robots per thousand workers). The contrast is informative because in the model higher robot density means more reinstatement has already occurred, so the *net* displacement effect (what the CZ regression captures) is smaller. The pattern, large negative CZ coefficients in the U.S. and small or zero coefficients in Germany, is qualitatively consistent with $\delta_R > 0$. [Humlum \(2024\)](#) directly observes within-firm task reallocation in Denmark, bracketing δ_R between 0.60 and 2.00. The three identification strategies, between-CZ displacement in the U.S., net effects in Germany, and within-firm reinstatement in Denmark, provide convergent evidence that both displacement and reinstatement are economically important, and that their relative magnitudes are consistent with the calibration used in this paper.

The estimated $\hat{\varphi} = 0.80$ is what the sorting margin maps to from the AI exposure gradients over 2005–2019, net of the pre-period trend, and part of it may be transitional reallocation friction, workers temporarily mismatched to tasks. If so, the long-run φ is smaller than the estimate and the productivity asymmetry narrows. The adjustment half-lives reported below make a long-run value near half the estimate the central long-run case, so [Table 8](#) reports the ranking at every transitional share, at the point estimate and across the bootstrap set. Robot dominance survives all of them. At a 25 percent transitional share the productivity ratio is 2.03, at 50 percent it is 1.88, at 75 percent it is 1.74, and at $\varphi_{LR} = 0.05$, the smallest value the solver evaluates, it is 1.64, which is the reinstatement gap almost on its own. Every bootstrap draw stays robot-dominant at every share. The long-run asymmetry is therefore less sensitive to the transitional decomposition than the contemporaneous one is to the level of cognitive reinstatement, because at $\delta_A = 0.03$ the cognitive frontier has almost no reinstated tasks for sorting disruption to degrade.

Two caveats bound what the exercise can show. The set-level rows scale φ uniformly across draws, which is an assumption about which component of the composite is transitional, and the

scaled draws are themselves outside the bootstrap set. And the decomposition treats only φ as transitional when all four structural parameters are medium-run objects. Reinstatement operates at decadal lags in the evidence, so the measured reinstatement rates may understate their long-run values too, and discounting φ alone is evaluated because it is the margin with direct half-life evidence attached.

Table 8: Sensitivity to Long-Run Sorting Disruption ($\delta_R = 1.00$, $\delta_A = 0.03$)

φ_{LR}	π_R	π_A	π_R/π_A	Set dominant	Interpretation
0.05	+0.589	+0.358	1.64	—	Sorting friction removed
0.20	+0.588	+0.337	1.74	100.0%	75% transitional
0.40	+0.588	+0.312	1.88	100.0%	50% transitional
0.60	+0.587	+0.289	2.03	100.0%	25% transitional
0.80	+0.587	+0.268	2.19	100.0%	Baseline estimate

Notes: φ_{LR} is the permanent component of sorting disruption, $\varphi_{LR} = (1 - q)\hat{\varphi}$ for transitional share q (a symbol distinct from the R&D curvature parameters ξ_j) and $\hat{\varphi} = 0.80$, with other parameters at the baseline estimates and the physical channel at its neutral calibration. “Set dominant” is the share of the 200 bootstrap draws with $\pi_R/\pi_A > 1$ when each draw’s φ is scaled by $(1 - q)$, under the assumption that transition scales φ uniformly across the set. No value of φ_{LR} on the grid brings the two frontiers to parity at the estimated reinstatement rates.

The table evaluates five transitional shares, and the question the ranking turns on is whether any share at all brings the two frontiers together. The continuous profile answers it. Over the whole range of long-run sorting disruption the two productivity contents never meet, and the vertical distance between them at $\varphi_{LR} = 0$ is the reinstatement gap on its own.

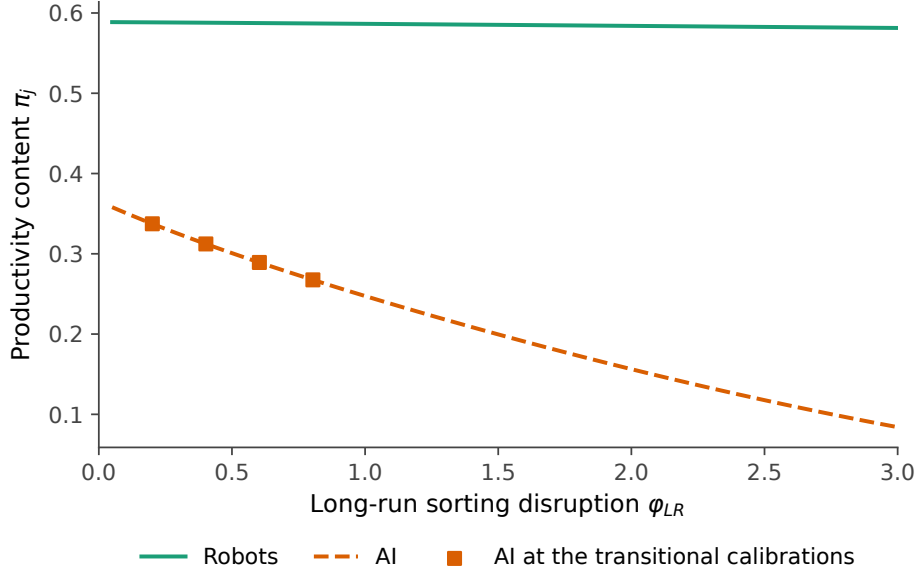


Figure 2: Sensitivity to long-run sorting disruption. π_R (solid) is nearly flat because sorting disruption operates only through the cognitive sector, and π_A (dashed) declines with φ_{LR} . Markers indicate the transitional-share calibrations of Table 8. The two curves do not meet on the range.

φ enters the cognitive block directly and the physical block not at all, so the robot productivity content is nearly flat across the range. It is not exactly invariant, because general equilibrium links the frontiers, and π_R moves from 0.589 to 0.581 between $\varphi_{LR} = 0.05$ and 3.0. The asymmetry between frontiers therefore has two sources, the reinstatement channel ($\delta_R > \delta_A$, which raises π_R relative to π_A) and the sorting disruption channel ($\varphi > 0$, which lowers π_A), and at the estimates the first carries most of it. With the sorting penalty removed π_A rises only to 0.358, against a π_R of 0.589.

The speed-of-adjustment literature provides further discipline. [Dix-Carneiro \(2014\)](#) estimates occupational adjustment half-lives of 4–7 years, and [Traiberman \(2019\)](#) finds similar magnitudes for Denmark. If AI exposure accumulated gradually over 2010–2019 instead of arriving as a single shock in 2005, workers at the end of the sample have had only 5–9 years of adjustment, closer to one half-life than three, and at one half-life roughly half of the transitional friction remains embedded in the regression estimates, which places φ_{LR} near $0.5\hat{\varphi} \approx 0.40$. At that value the ratio is 1.88 and every bootstrap draw remains robot-dominant, so the half-life central case leaves the steady-state ranking where the contemporaneous one is. What the moments pin down within the model is the contemporaneous asymmetry, and the steady-state ranking rests on the transitional decomposition,

which the table reports in full.

3.7 Mapping CZ-Level Estimates to the National Model

The argument for treating CZ-level estimates as national production-function parameters follows three steps. First, the reinstatement rates enter the production function through the task content of labor $\tilde{\Gamma}_j$, which has the same functional form at both the CZ and national levels. Migration and commuting affect local labor supply but not the task production function. Second, the estimation adjusts for migration on the non-college margin, scaling the local labor supply response by the calibrated elasticity $\varepsilon_S^{\text{mig}} = 0.78$ (Table 6), so the fitted χ_A nets out the mobility the wage moments would otherwise absorb. Third, the external firm-level bracket for the physical reinstatement channel, 0.60 to 2.00, is consistent with CZ-level patterns across the U.S., Germany, and Spain despite institutional differences, which supports treating reinstatement rates as properties of the technology, not of local institutions.

The residual concern is spatial spillovers: if robot adoption in one CZ generates task creation in neighboring CZs through supply chain linkages, CZ-level estimates understate the national effect (or overstate it, depending on the direction of spillovers). The concern applies to the displacement scale and the effective cognitive task content, which are estimated from CZ-level variation, and not to the physical channel, which is calibrated from firm-level evidence.

4 Static General Equilibrium

4.1 The General Equilibrium Feedback

A frontier advance moves wages, wages move the automation incentive, and that feedback loop is what could in principle overturn a partial-equilibrium ranking. Write $\mathcal{F}$ for the coefficient of that loop: the response of the automation incentive to the wage change a unit of frontier advance induces. The partial-equilibrium ranking survives, and the fixed point is unique wherever a bound below one holds throughout, when $\mathcal{F}$ is small.

The quantitative core of this paper is two-sector. It contains no service sector, no participation margin and no heterogeneity in marginal propensities to consume, so it produces only the wage-feedback component $\mathcal{F}^{PE}$. At the core's Cobb–Douglas aggregation the relative-price channel is

exactly zero, so the composition channel $\mathcal{F}^{\text{demand}}$ is reported as the increment from evaluating the top-level aggregator at the calibrated $\rho = 0.80$. At the estimated parameters,

$$\mathcal{F}^{PE} = 0.00184, \quad \mathcal{F}^{\text{demand}} = -0.00004. \quad (13)$$

The composition channel is negative. Its sign and magnitude come from the extended model of Appendix B.1, where the skill-specific expenditure shares that drive it are defined: there, non-college households spend more on services and therefore less on physical goods, so the relative-price movement induced by a frontier advance dampens the feedback. Equation (13) reports the channel in isolation (-0.00004), Table 9 reports the sequential increment inside the extended model (-0.00003), whose column sums to $\mathcal{F}^{GE}$, and the two differ in the last displayed digit.

The three demand-side blocks of Sections 2.2–2.4 add three further channels, and they are not present in the two-sector core. Appendix B.1 specifies them as an equilibrium, a demand system for services, a reservation-wage distribution delivering the calibrated participation elasticities, and an explicit allocation of automation and reinstatement rents across households, and solves it. With services switched off, the extended model reproduces the two-sector core. Table 9 reports the budget-consistent service calibration, and Appendix B.1 also reports variants that impose each of the two service anchors of Table 6 in turn.

Table 9: General equilibrium feedback channels, computed

Channel		Robot frontier	Cognitive frontier
$\mathcal{F}^{PE}$	wage feedback	0.00184	0.00128
$\mathcal{F}^{\text{demand}}$	composition	-0.00003	
$\mathcal{F}^{NT}$	non-tradable services	+0.00013	-0.00032
$\mathcal{F}^m$	MPC concentration	-0.00002	
$\mathcal{F}^{\text{part}}$	participation	-0.00009	
$\mathcal{F}^{GE}$	total	0.00184	0.00098

Notes: Computed in the extended equilibrium of Appendix B.1 at the budget-consistent service calibration. With services switched off the model reproduces the two-sector core to machine precision. The composition, MPC and participation channels are reported on the robot frontier, which carries the headline. The MPC channel is *negative* at the baseline calibration, so it dampens there. Components are rounded to five decimals and need not sum to the displayed total.

The net demand-side correction on the robot frontier is -0.2 percent of $\mathcal{F}^{PE}$, computed from the unrounded channels. At the neutral physical calibration the dampening channels, composition, MPC concentration, and participation, almost exactly offset the non-tradable channel, so the demand side, computed directly, nets out to a correction of a fifth of a percent.

Proposition 2 (Supply-Side Dominance). *Across both frontiers and every specification of the demand block considered in Appendix B.1, the total feedback satisfies $|\mathcal{F}^{GE}| \leq 0.0021$. The demand-side channels are more than two orders of magnitude too small to move the automation mix materially, and the distributional consequences of asymmetric automation are therefore supply-side determined.*

Because $\mathcal{F}^{PE}$ nearly cancels on the cognitive frontier, relative ratios are uninformative: the ratio there is large while both quantities are negligible. The absolute feedback is negligible on both frontiers.⁴

⁴The channels are solved, not bounded. An analytic chain that scales each channel to $\mathcal{F}^{PE}$ is not available: the non-tradable channel's scale is set by the service sector, independently of the wage-composition index, and on the cognitive frontier such a bound is violated.

4.2 Contraction and Uniqueness

The feedback loop of Section 4.1 induces a mapping Φ^{GE} on the frontier position: a candidate position x generates wages in the extended equilibrium of Appendix B.1, and the automation incentive at those wages returns the implied position $\Phi^{GE}(x)$. Task shares lie in $[0, \bar{x}]$ with $\bar{x} = 1$.

Proposition 3 (GE Contraction). *Φ^{GE} maps $[0, 1]$ into itself. Its slope is zero below $x_0 = 0.0201$ on both frontiers. On the physical frontier it is at most 0.70 at x_0 and smaller at every point evaluated between x_0 and 0.949, which is the interval swept. On the cognitive frontier the sampled slope stays below one for positions below 0.87 and exceeds one, with a sampled supremum of 1.116, on thirty nodes in $[0.87, 0.90]$. Where a bound below one holds throughout, the fixed point is unique.*

Proof. The self-map is immediate. $\Phi^{GE}(x)$ is an automation threshold, the share of tasks lying below the frontier the incentive implies, so it takes values in $[0, 1]$ whatever x is fed in. Φ^{GE} therefore maps $[0, 1]$ into itself, and the interval is compact.

For the bound, the slope is $(\Phi^{GE})'(x) = \frac{\sigma-1}{\sigma} \frac{\partial \ln \mathcal{W}}{\partial x}$, where $\mathcal{W}$ is the population-weighted wage composite: a candidate position moves wages, and the automation incentive converts that proportional wage movement into a movement in the threshold at rate $(\sigma-1)/\sigma$. That rate is a maintained specification of the incentive, not a consequence of equation (10), which supplies a response and a sign restriction but no threshold condition from which the elasticity would follow. This is the ratio of the induced movement to the initiating one, and so is distinct from $\mathcal{F}^{GE}$, which Proposition 2 reports as the induced movement itself from an advance of 0.01.

The slope is evaluated by central differences at 0.001 spacing across $[0.021, 0.949]$ on each frontier, with a further sweep at 0.00005 spacing around the point where the wage composite begins to respond. Below $x_0 = 0.0201$ the composite does not move with the frontier and the slope is exactly zero. At x_0 it jumps, to 0.6965 on the physical frontier and 0.6242 on the cognitive one. Above x_0 the magnitude declines on the physical frontier at every one of 928 consecutive comparisons, passing 0.185 at the calibrated position and 0.049 at $x = 0.90$, so on that frontier the largest sampled slope is at the corner. The cognitive frontier behaves differently. Its slope falls to zero near $x = 0.16$, changes sign, passes 0.122 at the calibrated position $I_A = 0.25$, and then rises from 0.06 at $x = 0.2$ to 0.80 at $x = 0.8$ and to a sampled supremum of 1.116 on $[0.87, 0.90]$. On the cognitive frontier, therefore, even the sampled evidence for a contraction is confined to positions

below 0.87. The corner height and location are both computed. What happens between points is the subject of the next paragraph.

That is evidence and not a proof. A grid cannot bound behaviour between its points, and a constant estimated from sampled slope differences does not supply that bound: refining the spacing twentyfold near x_0 raised the observed slope variation from 49 to 3489, because the corner is a jump and not curvature. What makes the evaluation informative on the physical frontier is the monotone decline above x_0 , verified at every sampled step, not an appeal to smoothness. An analytic bound would need a closed form for the equilibrium wage response to the frontier, which this model does not provide. Uniqueness is therefore not established here. What the computation gives is a mapping whose slope is below one at every evaluated point on the physical frontier and at every evaluated cognitive-frontier position below 0.87, over an interval it does not cover to the right endpoint, and the fixed point the solver returns is the one every result uses.

On the physical frontier the margin is about a third, well short of the orders of magnitude the level of $\mathcal{F}^{GE}$ would suggest, because that level carries the size of the advance it responds to. On the cognitive frontier there is no margin above 0.87. What the margin does not affect is the comparison this section is about: moving from the partial-equilibrium block to the full demand block changes the slope by -0.2 percent of itself, the net correction of Table 9. The margin is large because the wage response to a frontier advance is itself small, and every demand-side channel that feeds back through it is smaller still. $\square$

5 Dynamic Extension

The economy is described by automation capital stocks (K_R, K_A) and household assets (A_S, A_H) , with laws of motion driven by R&D investment allocation and heterogeneous savings behavior.

5.1 Setup

Automation capital evolves as $\dot{K}_j = s_j(t) \cdot \mathcal{S}(t) - d_j K_j$, where $\mathcal{S}(t) = Y(t) - C(t)$ is aggregate savings (calligraphic, distinct from the skill subscript S) and s_j is the investment share allocated to frontier j . Frontier thresholds respond to capital: $\dot{I}_j = \zeta_j K_j^{\xi_j}$. Investment shares track relative returns: $s_R/s_A = (\mathcal{R}_R^{\text{auto}}/\mathcal{R}_A^{\text{auto}})^{1/\nu}$.

Households solve standard consumption-savings problems with heterogeneous time preference ($\rho_S > \rho_H$, generating endogenous MPC heterogeneity), yielding Euler equations $\dot{C}_k/C_k = \sigma_c(r(t) - \rho_k)$.

5.2 The Savings Composition Channel

Consider a permanent decline in the user cost of AI capital, r_A . The impact effect is identical to the static counterfactual. The dynamic model generates an additional channel: if non-college workers are hand-to-mouth ($C_S = m_S W_S L_S$, justified by high ρ_S) while college workers are Euler-equation savers, then AI displacement raises college income relative to non-college, increases college savings, and tilts investment toward AI, amplifying the initial shift.

Proposition 4 (Dynamic Amplification). *Under differential time preference ($\rho_S > \rho_H$) and the composite-elasticity parameterization of Appendix C, equation (C.10), with $0 < q < 1$, a permanent decline in r_A generates:*

- (a) *An impact effect on the target allocation identical to the static counterfactual. The mix itself is continuous under $\dot{I}_j = \zeta_j K_j^{\xi_j}$, so what moves at the date of the change is the allocation of savings and hence $\dot{\omega}$, not ω .*
- (b) *A savings composition effect that tilts investment toward AI over time. Its sign is the sign of q , which the patience ordering alone does not fix.*
- (c) *Convergence of the K_A channel, the one Appendix C log-linearizes, to a new rest point with a lower target allocation $\omega^\dagger$ and wider skill premium. The appendix derives a single eigenvalue for that channel and no Jacobian for the four-dimensional block, so convergence of the block as a whole is not established here. The mix itself, $\omega = I_R/(I_R + I_A)$, is a function of the frontier positions alone, so it does not move at fixed I_R and I_A ; what the block determines is the allocation the frontier ratio is heading toward, and ω follows it as the frontiers advance. The full system has no rest point in (I_R, I_A) : under $\dot{I}_j = \zeta_j K_j^{\xi_j}$ with positive advance productivity and positive capital, $\dot{I}_j > 0$ always, so the frontier stocks are never stationary, and the block is used only over a horizon on which both stay interior, since the stocks otherwise leave the unit interval the task model is defined on and the object here is the limit of the reduced block*

over the transition horizon, and not a steady state of the whole model. It holds provided $\sigma_c(\rho_S - \rho_H) < d_A + \sigma_c\rho_H$ and the three restrictions under which Appendix C reduces the system: the college-employment and college-consumption-level responses of savings are zero, and the response runs through the K_A channel alone. The transition eigenvalue is $-d_A(1-q)$ with $q = \sigma_c(\rho_S - \rho_H)/(d_A + \sigma_c\rho_H)$ under the parameterization of Appendix C, which sets the composite elasticity to that expression without deriving it from the primitives above, so the patience ordering alone leaves q unrestricted and admits $q > 1$, under which the same formula gives local divergence. At the calibration $q = 0.13$.

Allowing non-college saving at the calibrated marginal propensity to consume ($m_S = 0.75$, Table 6) dampens the amplification, by an amount Appendix F signs but does not size, because it turns on elasticities this calibration does not pin down. The hand-to-mouth limit $m_S = 1$ makes non-college assets $A_S = 0$ and yields the tractable closed form

$$\mathcal{A}^{dyn} \approx 1 + \frac{\sigma_c(\rho_S - \rho_H)}{d_A + \sigma_c\rho_H} = 1.13, \quad (14)$$

a first-order expansion of the exact linearized value $\mathcal{A}^{dyn} = [1 - \sigma_c(\rho_S - \rho_H)/(d_A + \sigma_c\rho_H)]^{-1} = 1.15$ (Appendix C). Both are knife-edge magnitudes, attained only at $m_S = 1$, and the first-order expansion of 1.13 is the more conservative of the two. The reduced block takes approximately 8 years to close half the gap between the impact response and its new rest point.

Proof. Obtained by log-linearizing the K_A channel of the capital-and-asset block around its initial rest point, with the frontier positions held fixed and the college-employment and college-consumption-level responses of savings set to zero, and solving for the ratio of new-rest-point to impact responses. Those restrictions are what reduce the four state equations to one, and the eigenvalue below is that channel's. The derivation proceeds in seven steps (state simplification, savings characterization, investment log-linearization, capital dynamics, savings response, steady-state comparison, and reduction to primitives) and is valid for small perturbations under four conditions: local approximation, steady-state comparison, log-linearity (second-order terms are of order $(\Delta r_A)^2$), and separability in the patience gap. See Appendix C for the full derivation. $\square$

The hand-to-mouth assumption is what makes $A_S = 0$ and the amplification factor separable.

With positive but low non-college savings (a Bewley-style specification), a dampening cross-term proportional to $(1 - m_S) \cdot (\partial W_S / \partial \omega)$ enters the amplification factor and vanishes only at $m_S = 1$. Its size needs η_K , the non-college wage share of savings and $\partial \ln W_S / \partial \omega$, none of them calibrated here, and Appendix F shows that a large enough non-college savings response removes the amplification altogether.

Figure 3 traces the target allocation after a permanent decline in the user cost of AI capital. The impact effect shifts investment toward AI, and the savings composition channel then drives a further gradual decline as college workers, who benefit disproportionately from cognitive automation and save at higher rates, tilt aggregate investment toward the cognitive frontier. The scale of the impact effect is not a model output, since r_A does not enter the static block and the investment elasticity ν is uncalibrated, so the path is plotted in units of that effect. What the model determines is the shape: the savings channel adds 15 percent on top of the impact response, and it closes half of that residual gap in about eight years. The amplification and the half-life follow from the dynamic primitives $(\sigma_c, \rho_S, \rho_H, d_A)$ alone and are common to any frontier vector. The skill premium is a static function of the mix, so under the log-linearization it follows the identical path and is not plotted separately. The mix itself moves continuously: with $\dot{I}_j = \zeta_j K_j^{\xi_j}$ at finite capital, I_j and therefore ω cannot jump at the date of the change, and what responds on impact is the allocation of savings, hence $\dot{\omega}$ and not ω . The figure traces the algebraic mapping of Appendix C, which writes ω as a static function of the capital and asset state, so its date-zero movement is a property of that approximation and not of the integrated frontier laws.

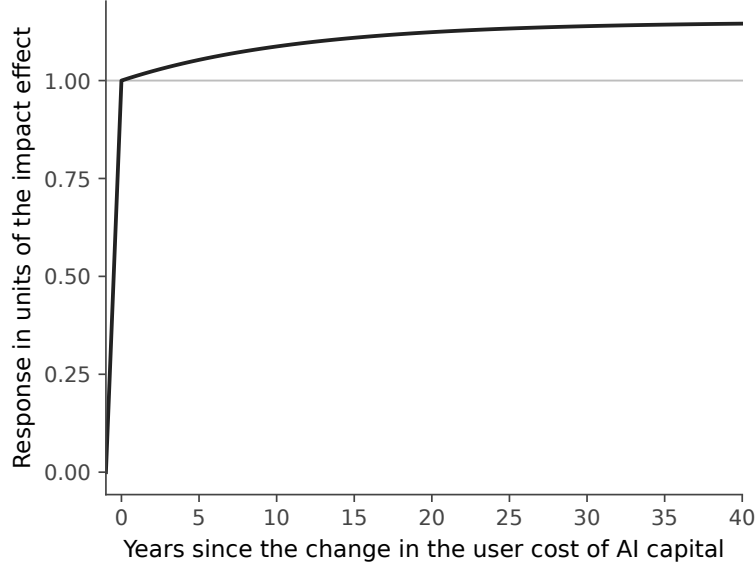


Figure 3: Transition of the target allocation $\omega^\dagger$ after a permanent decline in the user cost of AI capital, at the hand-to-mouth dynamic calibration ($m_S = 1$), in units of the impact response. The impact effect is immediate and the savings composition channel adds a further 15 percent, half of it within about eight years. The impact effect’s own magnitude turns on the uncalibrated investment elasticity ν and is not reported.

6 Results

Everything in this section and the next is computed at the estimates of Section 3, with the cognitive moments net of the linear continuation of the pre-AI trend, and Table 4 reports how the headline objects move under smaller continuations. The reduced-form gradients of Table 1 do not depend on that choice.

6.1 Frontier-Specific Productivity Decomposition

The productivity content of a unit of frontier advance at frontier j splits into a direct component and a residual. Only the direct component has a separate operational definition, obtained by setting the frontier’s reinstatement rate to zero, so the reinstatement and reallocation pieces are reported together and not divided between themselves:

$$\pi_j = \underbrace{\pi_j^{\text{direct}}}_{\text{capital replaces labor}} + \underbrace{\pi_j^{\text{reinst}} + \pi_j^{\text{realloc}}}_{\text{new tasks and the reallocation they induce}}. \quad (15)$$

Table 10 and Figure 4 report the decomposition at the baseline estimates. The direct components are close: shutting down reinstatement ($\delta_j = 0$), a unit of robot advance delivers +0.273 while a unit of AI advance delivers +0.257, a ratio of 1.06. If displacement were the whole story, the two technologies would be nearly interchangeable.

The asymmetry lives in the reinstatement-plus-reallocation component. At $\delta_R = 1.00$ robots rebuild labor’s task content at the pace they displace it, and the component is worth +0.314 of output per unit of frontier advance. At $\delta_A = 0.03$ the cognitive frontier creates almost no new tasks, and the component is +0.011. The gap between the two technologies is therefore a gap in task creation, and the model delivers it through the estimated reinstatement rates alone. Sorting disruption at $\varphi = 0.80$ is the model’s account of the skill-specific wage responses. It degrades the match between non-college workers and the cognitive tasks that remain as others are automated, and it is consistent with the sign pattern of the adjusted wage moments. The physical sector faces no comparable friction, which is a modeling assumption (φ enters only the cognitive frontier) and not something the data impose.

The combined effect is $\pi_R/\pi_A = 2.19$. A marginal unit of robot frontier advance buys 119 percent more output growth than a marginal unit of AI advance, with the sensitivity range reported in Table 11.

The memo rows in Table 10 preview the distributional consequences developed in Section 6.3. Robot advances raise both non-college and college wages (+0.343 and +0.407).⁵ AI advances lower both wages ($\partial \ln W_S/\partial I_A = -0.267$ and $\partial \ln W_H/\partial I_A = -0.238$), so an AI advance lowers the wage bill outright, and the skill premium widens only slightly (+0.029) because the non-college loss is the larger of the two. In the model that difference is sorting disruption, which degrades non-college productivity in the cognitive sector on top of the displacement both groups share. The labor share falls under both frontiers ($\partial s_L/\partial I_A = -0.311$ against -0.127 for robots), and Section 6.3 develops the asymmetry.

⁵These are *national* general-equilibrium semi-elasticities. Advancing the robot frontier raises both wages because reinstatement operates economy-wide. The reduced-form coefficients in Table 1 identify the *local* exposure effect, which is negative, because a commuting zone that absorbs the displacement without the economy-wide reinstatement sees wages fall. Section 3.6 reports the $\delta_R = 0$ local value $\partial \ln W_S/\partial I_R = -0.139$. The positive national sign is a model prediction, not an estimate, and the local–national gap is the standard Acemoglu and Restrepo (2020a) decomposition.

Table 10: Frontier-Specific Productivity Decomposition

Component	Robot (π_R)	AI (π_A)	Ratio
Direct automation	+0.273	+0.257	1.06
Reinstatement + reallocation	+0.314	+0.011	28.71
Total productivity content	+0.587	+0.268	2.19

π_j is the semi-elasticity $\Delta \ln Y / \Delta I_j$ at the step $\Delta I_j = 0.01$ used throughout, not a limit, and the ratio is of two such objects at the same step.

Memo: distributional effects

$\partial \ln W_S / \partial I_j$	+0.343	−0.267
$\partial \ln W_H / \partial I_j$	+0.407	−0.238
$\partial s_L / \partial I_j$	−0.127	−0.311
$\partial \ln(W_H/W_S) / \partial I_j$	+0.064	+0.029

Notes: π_j is the difference quotient $\Delta \ln Y / \Delta I_j$ at $\Delta I_j = 0.01$, centered, and every component is the same quotient. “Direct” sets $\delta_j = 0$ for the shocked frontier. Components are computed unrounded and displayed to three decimals, so they need not sum exactly to the displayed total. Calibration at the estimates with the physical channel at its neutral external calibration: $\delta_R = 1.00$, $\delta_A = 0.03$, $\varphi = 0.80$, $I_R = 0.30$, $I_A = 0.25$, Cobb–Douglas final-good aggregation (Table G.3 varies the top-level elasticity). $\partial \ln W_k / \partial I_j$ is the semi-elasticity of skill- k wages with respect to frontier j . Robot advances raise both groups’ wages and AI advances lower both, with the non-college loss the larger, so the wage bill falls when the cognitive frontier advances.

Figure 4 shows the same decomposition as stacked bars, with the direct segments close in height and the reinstatement segments apart. An out-of-sample check on the sorting interpretation, within-cell residual wage dispersion, is directionally consistent and reported in Appendix G.8. The channel attribution depends on the treatment of the pre-trend. On the unadjusted moments, which the pre-period evidence rejects, the fitted system places the asymmetry on sorting disruption and delivers a smaller ratio (Appendix I.3).

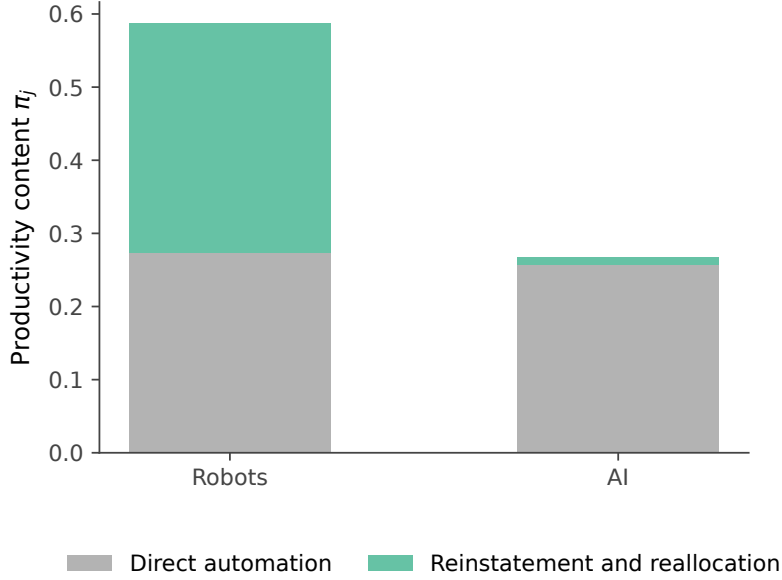


Figure 4: Frontier-specific productivity decomposition at the estimates, with the physical channel at its neutral calibration. Bottom segments: direct automation ($\delta_j = 0$). Top segments: reinstatement and reallocation. The direct components are close across frontiers (+0.273 for robots and +0.257 for AI). The asymmetry is driven by the reinstatement gap: +0.314 for robots against +0.011 for AI.

The quantitative core aggregates the two final goods by Cobb–Douglas ($\rho = 1$). Adopting the calibrated CES elasticity $\rho = 0.80$ moves the headline ratio in the second decimal (Table G.3), so the paper reports the Cobb–Douglas value throughout, and the general equilibrium channels of Section 4.1 are computed on the same core with the composition channel reported as the increment from $\rho = 0.80$.

Proposition 5 (Asymmetric Productivity Content). *The productivity content of robot automation exceeds AI automation by a factor of $\pi_R/\pi_A = 2.19$ at the estimates, with the physical channel at its neutral external calibration. On a grid of 1401 nodes across the calibration’s full range $[0.60, 2.00]$, dominance holds at every node, which is 100.0 percent of the range: the ratio rises from 1.80 at $\delta_R = 0.60$ to 2.93 at $\delta_R = 2.00$, it increases between every adjacent pair of nodes, and refining the grid tenfold to 14001 nodes changes no sign. There is no parity threshold inside the range, and the bisection bracket puts any crossing below $\delta_R = 0.30$. Across a joint grid spanning δ_R over that range and δ_A over the bootstrap percentiles plus the unadjusted estimate as a reference column, $\pi_R/\pi_A > 1$ in 22 of 25 cells: at every node with δ_A at or below the 95th percentile, and in the*

reference column only for δ_R above 1.00. Over the estimated (χ_A, φ) set at the neutral calibration, robot dominance holds in 100.0 percent of the 200 admissible draws, and the ninety-percent interval for the ratio, 1.57 to 8.35, lies above parity.

Table 11: Sensitivity of Productivity Ratio π_R/π_A

δ_R	δ_A				
	0.00	0.04	0.16	0.47	0.99
0.60	1.88	1.76	1.49	1.10	0.79
0.80	2.09	1.96	1.66	1.23	0.88
1.00	2.29	2.15	1.82	1.34	0.96
1.40	2.63	2.47	2.10	1.55	1.12
2.00	3.05	2.87	2.44	1.81	1.31

Notes: Each cell reports π_R/π_A by finite differences ($dI = 0.01$) at the indicated (δ_R, δ_A) pair, all other parameters at the estimates. δ_R spans the external calibration range $[0.60, 2.00]$ with the neutral baseline 1.00 a node. δ_A takes the 5th, 50th, 75th and 95th percentiles of its 200 bootstrap draws (0.00, 0.04, 0.16, 0.47), and the last column evaluates the unadjusted estimate $\delta_A = 0.99$ as a reference. Bold: the neutral δ_R at the bootstrap median. The point estimate $\delta_A = 0.03$ sits between the first two columns, where the ratio at $\delta_R = 1.00$ is 2.19. Twenty-two of 25 cells lie above parity. The three below it are all in the reference column at $\delta_R \leq 1.00$, and the (1.00, 0.99) cell is 0.96 because sorting disruption is held at the estimate $\varphi = 0.80$ and not at the unadjusted value.

6.2 What If AI Learns to Reinstate?

The preceding results condition on the current estimated reinstatement rates. But the cognitive frontier is evolving rapidly, and the CZ-level estimates in Table 5 are identified from 2005–2019 data that largely predate generative AI.

Varying δ_A alone while holding $\hat{\varphi} = 0.80$ fixed and solving $\pi_R = \pi_A$ at the neutral physical calibration yields:

$$\delta_A^{**} \approx 0.92 < \delta_R = 1.00. \quad (16)$$

AI must reinstate 0.92 tasks per task displaced to match the productivity content of robots, against an estimate of $\hat{\delta}_A = 0.03$, so the threshold is roughly thirty times the estimate. The threshold sits a

little below the robot rate because the direct component is slightly larger for AI than for robots, and at equal reinstatement rates the two frontiers would be nearly interchangeable. Sorting disruption ($\varphi = 0.80$) imposes a reallocation cost on the cognitive sector with no physical-sector analogue, but at this estimate it does little to the ranking.

But this one-dimensional exercise is incomplete. As AI capability evolves from routine cognitive automation to generative AI, the sorting disruption parameter φ may also change. Whether it rises or falls depends on the *type* of complementary tasks that generative AI creates and how well they match the skills of displaced workers. I extend the analysis to the full (δ_A, φ) plane, examining three objects: (i) the iso-productivity contour, the set of (δ_A, φ) pairs at which $\pi_R = \pi_A$, (ii) structurally disciplined trajectories through (δ_A, φ) space corresponding to different regimes of AI development, and (iii) bounds from emerging micro evidence on generative AI.

6.2.1 The Generalized Threshold

Define the threshold curve $\varphi^*(\delta_A)$ as the sorting disruption level at which robot and AI productivity equalize:

$$\varphi^*(\delta_A) \equiv \inf \{ \varphi \geq 0 : \pi_R(\delta_R, \delta_A, \varphi) = \pi_A(\delta_R, \delta_A, \varphi) \}, \quad (17)$$

with all remaining parameters held fixed. Both productivity objects carry the full argument list because the equilibrium links the frontiers: π_R moves with δ_A even at fixed (δ_R, φ) . Figure 5 plots π_R/π_A across the (δ_A, φ) plane with the threshold contour.

At the estimates ($\hat{\delta}_A = 0.03, \hat{\varphi} = 0.80$) and the neutral physical calibration, robots are 119% more productive per unit of frontier advance ($\pi_R/\pi_A = 2.19$). The estimate lies far inside the robot-dominant region. Holding $\delta_A = 0.03$, there is no value of sorting disruption on the plotted grid at which AI reaches parity, so φ is not a margin along which the ranking can be overturned at the estimated reinstatement rate. Reinstatement is the binding margin.

The threshold curve rises with reinstatement. At $\delta_A = 1.0$, equalization requires $\varphi^* \approx 1.03$. At $\delta_A = 1.2$, $\varphi^* \approx 1.55$. At $\delta_A = 1.5$, $\varphi^* \approx 2.33$. Higher reinstatement buys AI tolerance for more sorting disruption, and between $\delta_A = 1.0$ and 1.5 the locus climbs at roughly 2.6 units of φ per unit of δ_A . The estimate, at $\delta_A = 0.03$, lies well to the left of the locus at every φ on the grid.

The marginal 90% intervals of the estimates, $\delta_A \in [0.00, 0.47]$ and $\varphi \in [0.42, 6.77]$, lie in the

robot-dominant zone, and under design-consistent inference for the cognitive block 100.0 percent of the 200 admissible draws are above parity.

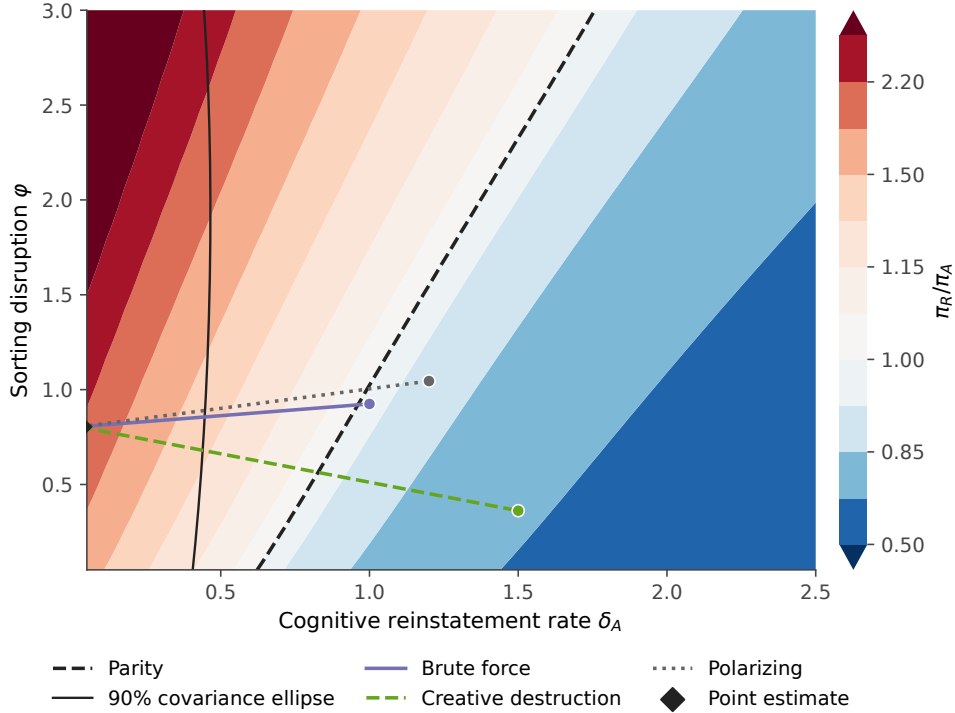


Figure 5: The productivity ratio π_R/π_A over the (δ_A, φ) plane at the baseline estimates. Red shades are robot-dominant, blue shades AI-dominant, and the dashed contour is the parity locus. The diamond marks the estimate $(\hat{\delta}_A = 0.03, \hat{\varphi} = 0.80)$ and the black arc the boundary of the 90% ellipse of the bootstrap covariance, which extends beyond the plotted plane and past the parameter-space boundary the draws respect. Each regime path runs from the estimate to the marker at its full-maturity configuration, in the direction of rising AI capability.

Recent experimental studies on generative AI provide independent information on the plausible region of (δ_A, φ) space. Figure 6 overlays three pieces of micro evidence on the (δ_A, φ) heatmap from Section 6.2.1.

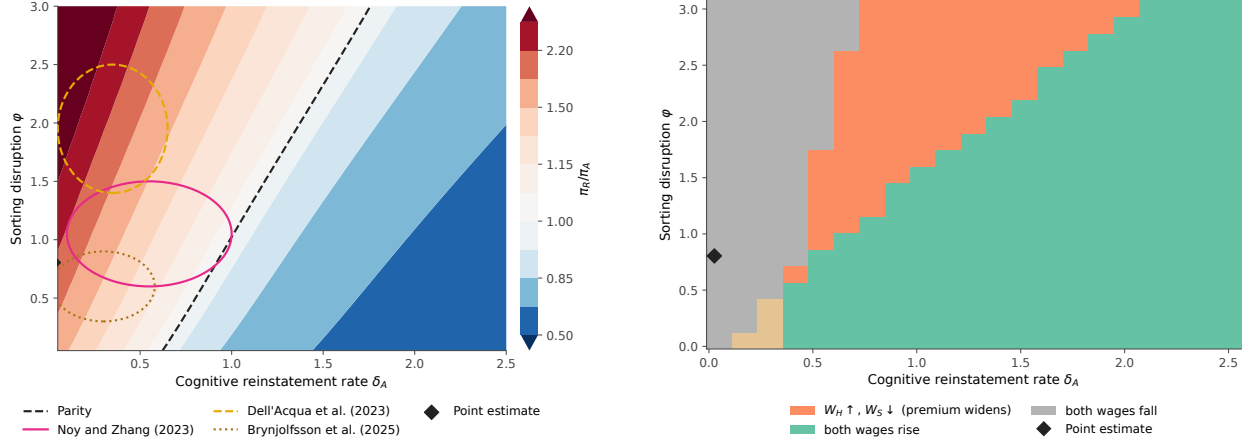
Noy and Zhang (2023) find that access to ChatGPT reduces completion time by 40 percent and raises output quality by 18 percent on professional writing tasks. Two features of their results bear on the model. ChatGPT benefits low-ability workers disproportionately, which is a statement about the direct productivity effect of using the tool and not about φ , which governs the degradation of non-college effective productivity as cognitive tasks are *automated*, and the two margins can move in opposite directions. And treated workers restructure their effort toward idea generation and editing while ChatGPT handles rough drafting, a within-job analog of task reinstatement

that occurs inside existing occupations without creating new ones, which suggests moderate δ_A , below 1.0. The experiment does not decompose the time saving into a direct automation share and a reinstatement share, so the ellipse in Figure 6(a) is a qualitative consistency region on the δ_A axis and does not locate φ .

Dell’Acqua et al. (2026) document a “jagged technological frontier” among Boston Consulting Group (BCG) consultants. Productivity rises by 40 percent on tasks inside AI’s capability frontier, and outside it the likelihood of a correct solution is 19 percentage points *lower* as workers over-rely on AI. The degradation outside the frontier is a misallocation of effort and not the object φ measures, so the study’s marker is schematic in the vertical dimension, and its evidence on task creation suggests limited δ_A .

Brynjolfsson et al. (2025) find a 14 percent productivity gain in customer service, again concentrated among novice workers. The more modest aggregate gain and the absence of task restructuring (agents continue performing the same tasks, just faster) suggest lower δ_A than the writing-task evidence, consistent with the lower-left region of the heatmap.

Panel (b) of Figure 6 maps the sign pattern of AI’s wage effects across skill groups. The identified part of the CZ evidence is the non-college wage loss ($\hat{\beta}_{A,NC} = -0.69\%$, conventional $t = -2.88$ and exposure-robust $t = -2.24$), which reverses sign after 2005 and appears within zones (Section 3.3). The college-wage gradient (+0.52%) is the tail of a trend that predates AI and is netted out of the moments, and the adjusted premium gap is close to zero. At the point estimate the model reproduces that pattern. An AI advance lowers both wages, the non-college wage by more, and the premium moves by +0.029, so the estimate ($\delta_A = 0.03, \varphi = 0.80$) sits in the region of panel (b) where both wages fall.



(a) Productivity ratio π_R/π_A with micro evidence regions. Ellipses indicate approximate (δ_A, φ) ranges consistent with each study's findings. The dashed line marks parity ($\pi_R = \pi_A$).

(b) Sign pattern of AI wage effects. The orange region has W_H rising and W_S falling, the blue region has both rising and the light region both falling. The diamond marks the point estimate, where both wages fall and the non-college wage falls by more.

Figure 6: Micro evidence anchors on the (δ_A, φ) space. The experimental-study markers are *schematic* placements of the task-creation evidence, and their vertical coordinates are not estimates of φ , since the heterogeneous productivity effects those experiments measure concern AI used as a tool and not the degradation of skill–task allocation as cognitive work is automated. The point estimate (star) lies where $\pi_R/\pi_A > 1$ and in the region consistent with the adjusted CZ wage moments.

6.2.2 Structural Co-Movement of δ_A and φ

As AI capability advances, both δ_A and φ respond. I parameterize AI capability by $\alpha \in [0, 1]$, where $\alpha = 0$ represents the current regime (routine cognitive automation, 2005–2019) and $\alpha = 1$ represents fully mature generative AI. From the free-entry condition for reinstated tasks (11), the reinstatement rate depends on the task complementarity parameter θ_A and the match quality between displaced workers' skills and new tasks. I consider three regimes spanning the structural possibilities.

Regime 1: Brute force. AI automates more cognitive tasks without changing the type of complementary tasks it creates. δ_A rises modestly as more automation generates entry opportunities, but φ stays high or rises because the automated tasks are increasingly non-routine, worsening skill-task matching. Parameterization: $\delta_A(\alpha) = 0.026 + \alpha(1.00 - 0.026)$, $\varphi(\alpha) = 0.804(1 + 0.15\alpha)$. Endpoint: $(\delta_A, \varphi) = (1.00, 0.92)$.

Regime 2: Creative destruction. Generative AI creates new task categories (prompt en-

gineering, algorithmic auditing, human-AI collaboration) structurally well-matched to displaced workers’ existing skills. δ_A rises substantially and φ falls. This is the scenario Autor (2024) envisions when arguing that generative AI could “democratize elite expertise.” Parameterization: $\delta_A(\alpha) = 0.026 + \alpha(1.50 - 0.026)$, $\varphi(\alpha) = 0.804(1 - 0.55\alpha)$. Endpoint: $(\delta_A, \varphi) = (1.50, 0.36)$.

Regime 3: Polarizing. AI creates high-productivity complementary tasks requiring new skills (ML engineering, AI safety, complex system design) poorly matched to displaced workers (data analysts, junior lawyers, routine knowledge workers). δ_A rises but φ also rises. Parameterization: $\delta_A(\alpha) = 0.026 + \alpha(1.20 - 0.026)$, $\varphi(\alpha) = 0.804(1 + 0.30\alpha)$. Endpoint: $(\delta_A, \varphi) = (1.20, 1.05)$.

Table 12: Productivity ratio along regime trajectories at full AI maturity ($\alpha = 1$)

Regime	$\delta_A(1)$	$\varphi(1)$	$\pi_R/\pi_A(1)$	Crosses $\pi_R = \pi_A$?
1: Brute Force	1.00	0.92	0.98	At maturity only ($\alpha = 1$)
2: Creative Destruction	1.50	0.36	0.73	Yes ($\alpha \approx 0.55$)
3: Polarizing	1.20	1.05	0.91	Yes ($\alpha \approx 0.85$)

Notes: Each regime parameterizes the evolution of $\delta_A(\alpha)$ and $\varphi(\alpha)$ as AI capability α rises from 0 (current) to 1 (full maturity). $\delta_A(1)$ and $\varphi(1)$ are the terminal values. $\pi_R/\pi_A(1)$ is the productivity ratio at full maturity, computed from the macro model with $\delta_R = 1.00$. “Crosses $\pi_R = \pi_A$ ” reports where the trajectory passes through the parity locus. All three regimes take δ_A from the estimate of 0.03 to at least one task per task displaced, which is why all three eventually cross. The starting point itself is not marginal, since at the estimated sorting disruption parity requires $\delta_A = 0.92$, about thirty times the estimate. Terminal values are computed at the unrounded estimates ($\delta_A = 0.026$, $\varphi = 0.804$), so they can differ in the last digit from arithmetic on the rounded factors. See Figure 7.

Table 12 reports the results. Because cognitive reinstatement starts near zero, the regimes are parameterized by the level of task creation AI reaches at maturity, one task per task displaced under brute force, one and a half under creative destruction, 1.2 under polarization. Regime 2 crosses parity at $\alpha \approx 0.55$. Regime 3 crosses at $\alpha \approx 0.85$, and Regime 1 reaches parity only at full maturity, where the ratio is 0.98. The regimes therefore answer how much of the path to maturity robots dominate. Figure 7 traces the productivity ratio along each trajectory.⁶

⁶The regime parameterizations are stylized. The medium-run conclusion, robot dominance over most of the

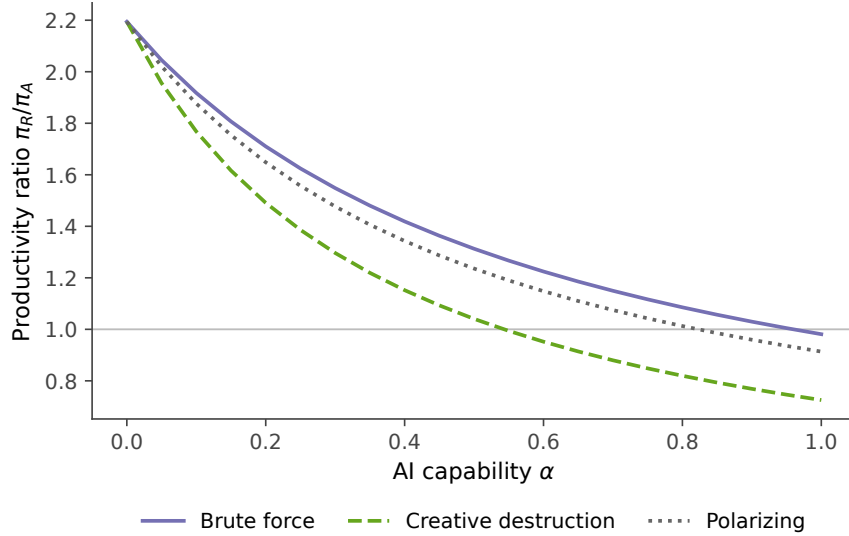


Figure 7: Productivity ratio π_R/π_A along each regime trajectory as AI capability α increases. At the baseline estimates all three trajectories eventually cross parity, Creative Destruction at $\alpha \approx 0.55$, Polarizing at $\alpha \approx 0.85$, and Brute Force only at full maturity.

6.2.3 Disciplining the Trajectories with Micro Evidence

Three recent findings bear on which regime is relevant.

Sorting disruption is real and large for LLM automation. Freund and Mann (2026) build a quantitative task-based model with multi-dimensional skill heterogeneity and show that automation driven by large language models (LLMs) generates *larger* shifts in occupational skill composition than past robot automation. Workers specialized in the most LLM-exposed tasks leave their transformed jobs and experience earnings losses. Two of their other findings point the opposite way. Moderately exposed workers benefit on average, and lower earners gain more than higher earners, a distributional sign opposite to the non-college wage loss the CZ moments carry. The tension is informative about which margin each object measures. The sorting parameter φ is identified by the reallocation cost at the high-exposure margin, the within-occupation skill-composition shifts their model quantifies, while their average-benefit and compression results describe the direct productivity channel at moderate exposure, which in this model raises π_A without loading on φ . Read at the margin the model uses, their evidence is consistent with $\varphi > 1$ and possibly $\varphi \geq \hat{\varphi}$ for generative AI, though the distributional disagreement at moderate exposure remains an open capability path under Regimes 1 and 3, does not depend on the specific values within each regime.

question a validation panel can settle. This evidence is inconsistent with Regime 2’s assumption that φ falls rapidly, and is most consistent with Regimes 1 or 3.

Aggregate labor market effects remain small. Humlum and Vestergaard (2025) link adoption surveys to Danish administrative records and find precisely estimated null effects on earnings and hours through 2024, despite 50%+ adoption rates in exposed occupations. This is consistent with the model’s prediction that the net labor market effect of cognitive automation, displacement minus reinstatement plus sorting disruption, is small when $\delta_A \approx 0.03$ and $\varphi \approx 0.80$. The null’s power is directional. At current adoption scale it cannot distinguish this paper’s configuration from the wage-only alternative of Appendix G.10, since both predict effects inside its band, but a large rise in δ_A at unchanged φ would push predicted effects outside it, so the null disciplines upward departures only.

Task-level productivity gains are concentrated in direct displacement. The experimental literature reports a 14% productivity gain in customer service (Brynjolfsson et al., 2025), a 40% productivity gain for in-frontier consulting tasks (Dell’Acqua et al., 2026), and a 40% reduction in completion time with an 18% quality gain in writing tasks (Noy and Zhang, 2023). These gains operate through the direct automation channel, not through new task creation. The reinstatement rate δ_A governs what happens *after* the task-level gain is realized. The micro evidence shows the direct channel is active but is largely silent on reinstatement.

On balance the evidence favors Regime 1 or Regime 3 over Regime 2 for the current generation of AI. Sorting disruption appears persistent and possibly increasing with LLM capability, not declining. Under the slowest-crossing trajectory, robot automation remains more productive per unit of frontier advance over the whole path to full AI maturity, and under the fastest over about half of it, and the crossing points themselves inherit the transitional-share uncertainty bounded in Table 8.

6.2.4 A Sufficient Condition for Persistent Robot Dominance

The joint analysis yields a sharper version of Proposition 5.

Proposition 6 (Generalized Asymmetric Productivity Content). *On a grid of 25 values of $\varphi \in [0.05, 3.00]$, the productivity content of robot automation exceeds AI automation ($\pi_R/\pi_A > 1$) at*

every evaluated δ_A below $\delta_A^{\min}(\varphi)$, where:

$$\delta_A^{\min}(\varphi) \equiv \inf \{ \delta : \pi_R = \pi_A \text{ at } (\delta, \varphi) \}. \quad (18)$$

At the neutral physical calibration $\delta_R = 1.00$, a bisection certificate over a grid of 25 values of $\varphi \in [0.05, 3.00]$ finds $\pi_R > \pi_A$ below every crossing and puts the smallest crossing at $\delta_A = 0.622$, at the low end $\varphi = 0.05$, rising across the grid to 1.757 at $\varphi = 3.00$. At each of those 25 values of φ , robot dominance holds at every evaluated δ_A below the certified crossing, and in particular at every node with $\delta_A < 0.62$. The statement is a certificate at those points and not a proof over the continuum: the infimum in the definition does not by itself say which side dominates below it, and the sign anchor and the single crossing are verified at each grid value and not established uniformly in φ .

The 90% interval for $\hat{\delta}_A$ from the bootstrap is $[0.00, 0.47]$. At its upper end robots still dominate: the ratio is 1.34 at $\hat{\varphi} = 0.80$ and the neutral physical calibration (Table 11), and the one-dimensional threshold $\delta_A^{**} = 0.92$ lies outside the interval, above the smallest certified crossing of 0.622. That threshold is valid at $\hat{\varphi}$ but understates the difficulty of equalization: in the joint (δ_A, φ) space, AI must simultaneously achieve high reinstatement *and* low sorting disruption, a combination the micro evidence suggests is structurally difficult for cognitive automation.⁷

Outside the estimated parameters, the *level* of the productivity ratio depends on the calibrated skill elasticity of substitution $\varepsilon_w = 1.50$, and because (χ_A, φ) are estimated conditional on it, sensitivity requires re-estimation at each value. That exercise has been run on the unadjusted moment system (Table G.5). Re-estimating across $[1.40, 2.50]$, the range the labor literature spans, π_R/π_A declines from 1.29 to 1.02 and never reverses, and the ratio of social marginal products stays above 1.25 throughout. Holding the estimated pair fixed instead produces a reversal near $\varepsilon_w \approx 1.80$, but the frozen pair no longer satisfies the moment conditions there. The re-estimation is not free either. Across the range $\hat{\delta}_A$ falls from 1.19 to 0.33 while the physical channel is held at its neutral calibration, and the Hansen statistic rises from 0.01 to 1.41, still small on three restrictions and carrying no probability for the reason given in Section 3.4. What moves is the division of labor between the two channels. In that system, setting $\varphi = 0$ leaves the ratio below one at every point

⁷Appendix G, Section G.5 presents a 3×3 joint sensitivity grid of π_R/π_A across (δ_R, φ) pairs.

in the range, from 0.70 at $\varepsilon_w = 1.40$ to 0.92 at 2.50, so on the unadjusted moments the sorting friction is necessary for the ranking, and it supplies 0.45 of the ratio at the calibrated ε_w and only 0.10 at 2.50, where the widened reinstatement gap carries most of the distance. At $\varepsilon_w = 1.40$ the estimated cognitive reinstatement rate even exceeds the neutral physical calibration, 1.19 against 1.00, and sorting disruption delivers the whole asymmetry. At the adjusted baseline the division sits on the reinstatement gap at the calibrated elasticity already, and Appendix L.3 reports the unadjusted system beside it. The ordering of the two frontiers is robust to the skill elasticity, and the division between the channels that produces it is not.

6.3 Labor Share Dynamics

Both frontiers lower the labor share, a generic property of task-based models where capital produces automated tasks, but with asymmetric magnitudes:

$$\frac{\partial s_L}{\partial I_R} = -0.127, \quad (19)$$

$$\frac{\partial s_L}{\partial I_A} = -0.311. \quad (20)$$

Robot automation is approximately 41% as damaging as AI per unit of frontier advance ($|\partial s_L / \partial I_R| / |\partial s_L / \partial I_A| = 0.41$, bootstrap 5th to 95th percentile range 0.29 to 0.44). The intuition runs through reinstatement. Robots at $\delta_R = 1.00$ roughly replace what they automate in labor's task content, so the labor share falls only through the capital income the automated tasks now earn. On the cognitive side, reinstatement at $\delta_A = 0.03$ replaces almost nothing, so labor's task content does not recover after displacement and the labor share falls by the full displacement. Sorting disruption at $\varphi = 0.80$ adds to this by degrading the value of whatever the cognitive frontier does reinstate, but at the estimated reinstatement rate there is little to degrade.

The skill premium tells the same story from the other side. Both frontiers widen it, robots by $\partial \ln(W_H/W_S) / \partial I_R = +0.064$ and AI by $\partial \ln(W_H/W_S) / \partial I_A = +0.029$, but the two movements have opposite sources. Under robots both wages rise and the college wage rises by more. Under AI both wages fall and the non-college wage falls by more, because as cognitive tasks are automated non-college workers in the cognitive sector lose access to tasks where they had a foothold, and the cognitive frontier creates almost nothing for them to move into. That is the sorting-disruption

channel.

Neither technology raises the labor share, so the policy question is which one damages it least.

7 Welfare and Optimal Policy

7.1 Optimal Frontier-Specific Subsidies

A social planner chooses frontier-specific R&D subsidies to maximize a weighted aggregate of group incomes,

$$\mathcal{V} = g_S L_S W_S + g_H (L_H W_H + \Pi), \quad \Pi \equiv Y - L_S W_S - L_H W_H, \quad (21)$$

where non-college workers are hand-to-mouth ($c_S = W_S$), college workers receive the capital income Π , and g_k is the marginal social welfare weight on group k . The weights come from isoelastic preferences with Atkinson inequality aversion λ , $u'(c) = c^{-\lambda}$, evaluated at baseline wage consumption: $g_S = W_S^{-\lambda}$ and $g_H = W_H^{-\lambda}$. At $\lambda = 0$ the weights are equal and $\mathcal{V}$ is output.

Holding the weights and the population masses at their baseline values is what makes the exercise tractable, and it is also what bounds it. Both are held fixed, so differentiating equation (21) returns equation (22) exactly and the policy condition below is the first-order condition of the criterion stated here. The cost is that this is a local comparison around the observed allocation: a planner re-optimizing globally would move the weights as consumption moved, and would count any change in group size. Section 7.3 reports the variants that evaluate the weights at full consumption instead, all of which raise the ratio, so the baseline weighting is the conservative one.⁸ The marginal social return to advancing frontier j , the derivative of equation (21) at fixed weights and masses, is:

$$SMP_j = W_S^{-\lambda} L_S \frac{dW_S}{dI_j} + W_H^{-\lambda} \left(L_H \frac{dW_H}{dI_j} + \frac{d\Pi}{dI_j} \right). \quad (22)$$

At $\lambda = 0$ the weights are equal and $SMP_j = dY/dI_j$, so the two frontiers' social returns stand in the same ratio as their private returns and the relative subsidy is one. At $\lambda > 0$, non-college workers receive higher marginal utility weight ($W_S < W_H \Rightarrow W_S^{-\lambda} > W_H^{-\lambda}$), and the ratio tilts further toward the frontier that benefits them more, robots.

⁸AI advances lower non-college wages ($\partial \ln W_S / \partial I_A = -0.267$) and the labor share ($\partial s_L / \partial I_A = -0.311$). The CZ estimate $\hat{\beta}_{A,NC} = -0.69\%$ is the local partial-equilibrium counterpart, and the general-equilibrium and local estimates agree in sign. The welfare calculations throughout this section use the general-equilibrium derivatives.

7.2 From Social Returns to a Subsidy Rate

A ratio of social marginal products is not yet a policy rate. A subsidy is a wedge in a private first-order condition, so what it has to correct is the gap between the social and the private return, not the level of the social return. The model already contains the margin the wedge acts on. Savings are allocated across frontiers according to relative returns, $s_R/s_A = (\mathcal{R}_R/\mathcal{R}_A)^{1/\nu}$, and a frontier-specific research subsidy at rate τ_j raises the return an investor perceives to $(1 + \tau_j)\mathcal{R}_j$.

Write the private return to a marginal unit of investment as $\mathcal{R}_j = \psi_j T_j$, where $T_j = \partial \dot{I}_j / \partial K_j$ is the frontier-advance technology and ψ_j is the private value of a unit of advance. The planner values that same unit at $\mathcal{S}_j = SMP_j T_j$. The decentralized allocation reproduces the planner's ranking when the subsidized private ratio equals the social one, $(1 + \tau_R)\mathcal{R}_R / [(1 + \tau_A)\mathcal{R}_A] = \mathcal{S}_R / \mathcal{S}_A$, and T_j cancels from both sides:

$$\frac{1 + \tau_R^*}{1 + \tau_A^*} = \frac{SMP_R}{SMP_A} \cdot \frac{\psi_A}{\psi_R}. \quad (23)$$

The cancellation is what makes the object computable. The frontier-advance parameters ζ_j are never calibrated and $\xi_R = 0.45$ differs from $\xi_A = 0.50$, so an expression that retained T_j could not be evaluated here at all.

What remains is the appropriability ratio ψ_A/ψ_R . I take appropriability to be equal across frontiers, $\psi_j \propto \pi_j$, with both returns evaluated at the same finite advance $\Delta I = 0.01$ that defines π_j throughout. The cancellation below is then exact up to the difference between a log difference and a level change over that step, which at $\Delta I = 0.01$ is about five parts in ten thousand and vanishes only in the limit. This is the neutral benchmark, since there is no evidence here that robot research and AI research differ in the share of the output gain an investor captures. Under that benchmark equation (23) becomes

$$\frac{1 + \tau_R^*}{1 + \tau_A^*} = \frac{SMP_R/SMP_A}{\pi_R/\pi_A} = \frac{1 + \mathcal{W}_R^{\text{dist}}}{1 + \mathcal{W}_A^{\text{dist}}} \approx 1.72. \quad (24)$$

The efficiency motive cancels. That is the economics of the wedge and not an artifact. If investors capture the output gain a frontier delivers, a planner with no inequality aversion has no reason to tilt research between two frontiers, whatever their relative productivity, so at $\lambda = 0$ the ratio is one and everything that tilts research toward the physical frontier here is distributional.

Two cautions travel with equation (24). It is a ratio of gross subsidy factors and not of rates, which are different numbers unless subsidies are large. And it inherits the appropriability benchmark: a frontier whose returns are harder to capture privately warrants more support at any λ , and the manuscript has no evidence on that margin.

Here $\mathcal{W}_j^{\text{dist}}$ is the distributional wedge on frontier j , the share of SMP_j that comes from re-weighting each group’s wage gain by its marginal utility instead of by its income, so that $\mathcal{W}_j^{\text{dist}} = 0$ at $\lambda = 0$ and the equity factor collapses to one. Robot reinstatement raises the non-college wage, which an AI advance lowers, and non-college workers carry the higher marginal utility under concave u , so that is the whole of the tilt.

7.3 Robustness Across Social Preferences

Table 13 reports the optimal subsidy ratio across social preferences, from utilitarian ($\lambda = 0$) to high inequality aversion ($\lambda = 3$). The Rawlsian limit is not reported as a ratio because it is not defined as one. AI advances lower non-college wages, so a maxmin planner taxes AI research outright.

Table 13: Relative subsidy factor $(1 + \tau_R^*)/(1 + \tau_A^*)$ across social preferences

λ	Social preference	SMP_R/SMP_A	$(1 + \tau_R^*)/(1 + \tau_A^*)$	Favors robots?
0.0	Utilitarian (no inequality aversion)	2.19	1.00	Indifferent
0.5	Mild inequality aversion	2.50	1.14	Yes
1.0	Log utility	2.97	1.35	Yes
1.5	Baseline	3.77	1.72	Yes
2.0	Moderate inequality aversion	5.37	2.45	Yes
3.0	High inequality aversion	75.3	34.3	Yes (ratio uninformative)

Notes: The planner maximizes a Bergson–Samuelson SWF $\mathcal{W} = \sum_k L_k u(c_k)$ with $u(c) = c^{1-\lambda}/(1-\lambda)$

for $\lambda \neq 1$ and $u(c) = \ln c$ at $\lambda = 1$. SMP_R/SMP_A is the ratio of social marginal products. The relative subsidy factor $(1 + \tau_R^*)/(1 + \tau_A^*)$ is that ratio net of the efficiency ratio π_R/π_A , which cancels out of the wedge under equal appropriability (Section 7.2), so the column is the distributional motive on its own.

At $\lambda = 0$ it equals 1 by construction. At $\lambda = 3$ the AI social marginal product is close to zero, so the ratio in that row is large and informative about the sign of the tilt only. All parameters at baseline:

$$\delta_R = 1.00, \delta_A = 0.03, \varphi = 0.80.$$

The relative subsidy factor is one at $\lambda = 0$, where the planner is indifferent to distribution, and rises with λ from there, reaching 1.72 at the baseline $\lambda = 1.5$, about 70 percent more support for robot research than for AI research. Above $\lambda = 2$ the AI social marginal product approaches zero, because the non-college wage loss from an AI advance receives almost all of the weight, so the level at $\lambda = 3$ carries no information beyond the sign, and in the maxmin limit AI research is taxed outright, which is why that limit is not reported as a ratio.

The direction is conditional on the transition horizon over which the subsidy is computed at today's (δ_A, φ) , and therefore on AI's development path. The cognitive frontier would have to create 0.92 tasks per task displaced to reach productivity parity at the estimated sorting disruption, and the regimes of Section 6.2 describe how far along the capability path that gap closes (Table 12). The subsidy tilt outlives productivity parity, because the equity motive holds the ratio above one even where the two frontiers are equally productive, but on a path where generative AI reinstates the workers it displaces it too eventually reverses. The frontier-specific subsidy is therefore a bet on which development path obtains, and post-deployment data on generative AI's reinstatement and sorting will settle it.

The welfare block also embeds two knife-edge distributional assumptions. The first is that all capital income, including robot reinstatement rents, accrues to college households at the margin. The second is that marginal utilities are evaluated at wage consumption ($c_S = W_S$), which implicitly imposes $m_S = 1$ and so sits in tension with the static calibration's own $m_S = 0.75$. Relaxing either one does not reverse the ranking. Evaluating marginal utility at full per-capita consumption with college households receiving all capital income makes the cognitive social marginal product negative, because the AI advance lowers the wage bill and capital income lowers the college welfare weight, so the planner taxes AI research and the ratio has no reading as a subsidy comparison. Letting non-college households hold 10 or 20 percent of capital income restores a positive cognitive social marginal product and social-return ratios of 9.4 and 3.8, against 3.77 under the paper's convention, and the variants approach the baseline from above as the non-college capital share rises. The baseline is the most conservative of the variants with a defined ratio. Proposition 7 below collects the robustness envelope over the bootstrap set, λ , and the physical calibration range.

7.4 Second-Best Considerations

The preceding analysis assumes the planner can set frontier-specific subsidies (τ_R, τ_A) . In practice, industrial policy instruments are blunt: governments may face political or informational constraints that limit them to a uniform technology subsidy τ or, at best, broad sectoral support. Three second-best considerations are relevant.

If lump-sum redistribution is available, the distributional motive disappears and with it the whole of the tilt: the equity factor is one, so $(1 + \tau_R^*)/(1 + \tau_A^*) = 1$ and research policy is neutral across frontiers. The relative subsidy factor (1.72 at the point estimate, with a median of 1.62 across bootstrap draws) is that Atkinson wedge on its own, since the productivity ratio (2.19) cancels from it: without perfect transfers, technology policy must do double duty as both an efficiency and a redistribution instrument.

The decomposition of social marginal products reveals why. The wage-bill share of a marginal output gain is $s_L + \partial s_L / \partial \ln Y$, where $s_L = 0.612$ is the model's baseline labor share and $\partial s_L / \partial \ln Y = (\partial s_L / \partial I_j) / \pi_j$ from the labor-share derivatives in Table 10. For robots, 39.6 percent of the output gain from a frontier advance accrues as wage income and 60.4 percent as capital income. For AI the wage-bill share is -55.2 percent: the wage bill falls when the cognitive frontier advances, and capital income absorbs the whole output gain and the wage loss on top of it (155.2 percent). Both wages fall when the cognitive frontier advances, so no part of AI's return reaches workers as a group. A planner who cannot redistribute capital income therefore has a strong incentive to tilt technology policy toward the frontier whose gains are paid out as wages.

The second consideration reinforces this conclusion. The productivity ratio π_R / π_A measures output elasticity with respect to frontier *position* and is invariant to capital depreciation rates. The frontier position I_j carries no depreciation law, so what a comparison of costs can hold fixed is the advance flow and not the level. Sustaining a common flow $q = \dot{I}_j$ requires $K_j = (q / \zeta_j)^{1/\xi_j}$ and a replacement flow $d_j(q / \zeta_j)^{1/\xi_j}$, so relative cost turns on ζ_j and ξ_j as well as d_j . It collapses to $(\pi_R / \pi_A) \cdot (d_A / d_R)$ only when the two frontiers share the advance technology, and here $\xi_R = 0.45$ against $\xi_A = 0.50$ with ζ_j uncalibrated. The depreciation-only adjustment below is therefore an illustration under that restriction and not an implication of the dynamic block. If AI capital depreciates faster than physical robots, the cost-adjusted ratio exceeds the output-based ratio,

which would strengthen the case for robot subsidies without changing its direction.⁹

If the planner is restricted to a uniform subsidy $\tau_R = \tau_A = \tau$, the welfare cost depends on the gap in social marginal products at the undifferentiated allocation. At the baseline estimates, SMP_R is 3.77 times SMP_A , a large misallocation. The uniform subsidy over-invests in the AI frontier (relative to the social optimum) and under-invests in robot adoption.

How much a partial tilt would recover, and how costly overshooting would be, are questions this model does not answer: they need the welfare objective as a function of the relative allocation, its feasible constraint and its curvature in that direction, and the model contains no research problem that determines total investment. What the condition delivers is the direction and the size of the wedge, not the shape of the objective around it. Whether frontier-specific subsidies need to be precisely calibrated to deliver most of their benefit is the same unanswered question.

Proposition 7 (Policy Robustness). *Across the bootstrap draws of (δ_A, φ) , the transition-horizon relative subsidy factor $(1 + \tau_R^*)/(1 + \tau_A^*)$ exceeds one in 93.5 percent of draws and has median 1.62 and ninety-fifth percentile 2.60 (computed under the baseline wage weighting of Section 7, the most conservative of the distributional variants). The fifth percentile is -0.00: the 13 draws that fall in the degenerate basin, where π_A is negative, carry a negative factor, and the ratio is not a subsidy ratio there. The factor is at least one at every value of the inequality-aversion parameter λ reported in Table 13, with equality at $\lambda = 0$ where the weights coincide and strict inequality above it, and at every one of 1401 points spaced 0.001 apart across the external physical-calibration range for δ_R , where it rises with δ_R from 1.64 at the lower endpoint to 1.77 at the upper one. As in Proposition 6, that is a certificate at the evaluated points and not over the continuum. In the maxmin limit the ratio is not defined because the planner taxes AI research. On the wider Humlum grid, holding φ at its estimate, it is above one at every percentile of the δ_A sampling distribution and every δ_R node, and it falls below one only in the reference column at the unadjusted estimate $\delta_A = 0.99$, for δ_R of 0.60 and 0.80, two of the three cells that reverse the productivity ranking. A uniform policy, $\tau_R = \tau_A$, leaves the frontiers misallocated by the whole of that wedge, since the common factor*

⁹Any quantification here is illustrative: frontier LLM release cycles of 1–2 years against 10–15 year robot lifespans would put d_A/d_R severalfold above one, but d applies to the frontier capability stock, not to individual model artifacts, so artifact turnover is an upper bound on the economically relevant depreciation rate. The cost-adjusted ratio is $(\pi_R/\pi_A)(d_A/d_R)$, and the ancillary-parameter table in Appendix G covers $d_A \in [0.05, 0.50]$. The cost-adjusted productivity ranking reverses below $d_A/d_R = 1/2.19 = 0.46$. The relative subsidy factor carries no cost adjustment, for the reason given in Appendix G.

cancels from the allocation condition and the relative subsidy stays at one.

I leave the formal dynamic optimal policy problem, where $\delta_A(\alpha)$ and $\varphi(\alpha)$ evolve endogenously with the AI capability frontier, to future work.

8 Conclusion

Robot automation generates about twice the productivity per unit of frontier advance that AI automation does, a ratio of 2.19 at the neutral physical calibration, and it warrants correspondingly stronger research support over the horizon on which such policy is set. The ranking holds at every evaluated value of physical task creation in the firm-level calibration bracket, 100.0 percent of the range, with the ratio rising from 1.80 at its lower end to 2.93 at its upper end, and it holds in 100.0 percent of the bootstrap draws of the cognitive parameters. The reason for it is a reinstatement gap. Robots create new physical tasks around the workers they displace, and that reinstatement accounts for more than half of their productivity return (+0.314 of 0.587). AI displaces without reinstating over the horizon the data cover: its reinstatement contribution is +0.011, against a direct displacement component (+0.257) close to the robot one. The sorting disruption that the model uses to account for the college wage is consistent with the evidence, but the adjusted moments do not pin its magnitude, so the asymmetry rests on the reinstatement gap and not on sorting. Because the estimates describe the automation environment of 2005 to 2019, data from the generative era will show whether the cognitive frontier starts to reinstate.

Demand channels are second order, so the distributional consequences of asymmetric automation are set on the supply side. Both frontiers erode labor’s share of income, and robots do 41 percent of the damage per unit of advance that AI does, the most tightly pinned down of the paper’s magnitudes. Over the transition horizon a planner supports robot research more than AI research, by a gross subsidy factor of 1.72 at the estimates, about 70 percent higher, above one at every node of the physical calibration range and in 93.5 percent of draws. The tilt is distributional in its entirety: the efficiency ratio cancels from the wedge under equal appropriability, and what remains is that a robot advance pays out 39.6 percent of its output gain as wages while an AI advance lowers the wage bill. The empirical result behind that wedge is the non-college wage loss in AI-exposed commuting zones, which reverses sign after 2005, persists through 2024 and appears

within zones, against a college-wage gradient that predates AI and runs out by 2012. Netting that pre-period trend out of the moments is what moves the AI reinstatement estimate toward zero and the productivity ratio toward two.

The economics behind the ranking is simple enough to survive the particular numbers that produced it. A technology that creates new human work around the workers it displaces raises productivity by more than one that displaces and creates nothing in return, and the difference is large enough to matter for growth, for distribution, and for which frontier public policy should push.

References

- Acemoglu, D. (2025). The simple macroeconomics of AI. *Economic Policy*, 40(121):13–58.
- Acemoglu, D., Autor, D., Hazell, J., and Restrepo, P. (2022). Artificial intelligence and jobs: Evidence from online vacancies. *Journal of Labor Economics*, 40(S1):S293–S340.
- Acemoglu, D. and Restrepo, P. (2018). The race between man and machine: Implications of technology for growth, factor shares, and employment. *American Economic Review*, 108(6):1488–1542.
- Acemoglu, D. and Restrepo, P. (2019). Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives*, 33(2):3–30.
- Acemoglu, D. and Restrepo, P. (2020a). Robots and jobs: Evidence from US labor markets. *Journal of Political Economy*, 128(6):2188–2244.
- Acemoglu, D. and Restrepo, P. (2020b). The wrong kind of AI? Artificial intelligence and the future of labour demand. *Cambridge Journal of Regions, Economy and Society*, 13(1):25–35.
- Acemoglu, D. and Restrepo, P. (2022). Tasks, automation, and the rise in US wage inequality. *Econometrica*, 90(5):1973–2016.
- Adão, R., Kolesár, M., and Morales, E. (2019). Shift-share designs: Theory and inference. *Quarterly Journal of Economics*, 134(4):1949–2010.
- Angrist, J. D. and Pischke, J.-S. (2009). *Mostly Harmless Econometrics*. Princeton University Press.
- Autor, D. (2024). Applying AI to rebuild middle class jobs. NBER Working Paper No. 32140.
- Autor, D., Chin, C., Salomons, A., and Seegmiller, B. (2024). New frontiers: The origins and content of new work, 1940–2018. *Quarterly Journal of Economics*, 139(3):1399–1465.
- Autor, D. H., Dorn, D., and Hanson, G. H. (2013). The china syndrome: Local labor market effects of import competition in the United States. *American Economic Review*, 103(6):2121–2168.

- Babina, T., Fedyk, A., He, A., and Hodson, J. (2024). Artificial intelligence, firm growth, and product innovation. *Journal of Financial Economics*, 151:103745.
- Baumol, W. J. (1967). Macroeconomics of unbalanced growth: The anatomy of urban crisis. *American Economic Review*, 57(3):415–426.
- Bloom, D. E., Prettnner, K., Saadaoui, J., and Veruete, M. (2025). Artificial intelligence and the skill premium. *Finance Research Letters*, 81:107401.
- Borusyak, K., Hull, P., and Jaravel, X. (2022). Quasi-experimental shift-share research designs. *Review of Economic Studies*, 89(1):181–213.
- Brynjolfsson, E., Li, D., and Raymond, L. R. (2025). Generative AI at work. *Quarterly Journal of Economics*, 140(2):889–942.
- Card, D. (2009). Immigration and inequality. *American Economic Review*, 99(2):1–21.
- Chávez, C. (2026). Automation is not a sufficient statistic: Robots, ai, and inequality. Working paper, University of Chicago.
- Chetty, R., Guren, A., Manoli, D., and Weber, A. (2013). Are micro and macro labor supply elasticities consistent? A review of evidence on the intensive and extensive margins. *American Economic Review: Papers & Proceedings*, 103(3):471–475.
- Dauth, W., Findeisen, S., Suedekum, J., and Woessner, N. (2021). The adjustment of labor markets to robots. *Journal of the European Economic Association*, 19(6):3104–3153.
- Dell’Acqua, F., McFowland, E., Mollick, E. R., Lifshitz-Assaf, H., Kellogg, K., Rajendran, S., Kraymer, L., Candelon, F., and Lakhani, K. R. (2026). Navigating the jagged technological frontier: Field experimental evidence of the effects of AI on knowledge worker productivity and quality. *Organization Science*, 37(2):403–423.
- Dix-Carneiro, R. (2014). Trade liberalization and labor market dynamics. *Econometrica*, 82(3):825–885.
- Dynan, K. E., Skinner, J., and Zeldes, S. P. (2004). Do the rich save more? *Journal of Political Economy*, 112(2):397–444.

- Eloundou, T., Manning, S., Mishkin, P., and Rock, D. (2024). GPTs are GPTs: Labor market impact potential of LLMs. *Science*, 384(6702):1306–1308.
- Freund, L. B. and Mann, L. F. (2026). Job transformation, specialization, and the labor market effects of AI. CESifo Working Paper No. 12072.
- Goldsmith-Pinkham, P., Sorkin, I., and Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8):2586–2624.
- Graetz, G. and Michaels, G. (2018). Robots at work. *Review of Economics and Statistics*, 100(5):753–768.
- Hémous, D. and Olsen, M. (2022). The rise of the machines: Automation, horizontal innovation, and income inequality. *American Economic Journal: Macroeconomics*, 14(1):179–223.
- Herrendorf, B., Rogerson, R., and Valentinyi, Á. (2014). Growth and structural transformation. In Aghion, P. and Durlauf, S. N., editors, *Handbook of Economic Growth*, volume 2, pages 855–941. Elsevier.
- Humlum, A. (2024). Robot adoption and labor market dynamics. Working paper, ROCKWOOL Foundation Study Paper No. 175.
- Humlum, A. and Vestergaard, E. (2025). Still waters, rapid currents: Early labor market transformation under generative AI. NBER Working Paper No. 33777.
- Karabarbounis, L. and Neiman, B. (2014). The global decline of the labor share. *Quarterly Journal of Economics*, 129(1):61–103.
- Katz, L. F. and Murphy, K. M. (1992). Changes in relative wages, 1963–1987: Supply and demand factors. *Quarterly Journal of Economics*, 107(1):35–78.
- Koch, M., Manuylov, I., and Smolka, M. (2021). Robots and firms. *Economic Journal*, 131(638):2553–2584.
- Moretti, E. (2011). Local labor markets. *Handbook of Labor Economics*, 4:1237–1313.
- Notowidigdo, M. J. (2020). The incidence of local labor demand shocks. *Journal of Labor Economics*, 38(3):687–725.

- Noy, S. and Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, 381(6654):187–192.
- Piketty, T. (2014). *Capital in the Twenty-First Century*. Harvard University Press, Cambridge, MA.
- Solon, G., Haider, S. J., and Wooldridge, J. M. (2015). What are we weighting for? *Journal of Human Resources*, 50(2):301–316.
- Sun, Y. (2006). The exact law of large numbers via Fubini extension and characterization of insurable risks. *Journal of Economic Theory*, 126(1):31–69.
- Traiberman, S. (2019). Occupations and import competition: Evidence from Denmark. *American Economic Review*, 109(12):4260–4301.
- Webb, M. (2020). The impact of artificial intelligence on the labor market. *Working paper, Stanford University*.

A Production Function Derivation

This appendix derives Lemma 1 and Proposition 1 from the disaggregated task production function (2). The derivation proceeds in five steps. Throughout, $\varsigma \equiv (\sigma - 1)/\sigma$ with $\sigma > 1$.

Assumption A1 (Uniform productivity within segments). *All surviving labor tasks in sector j share a common productivity: $g_{Lj}(z) = \bar{g}_{Lj}$ for all $z \in (I_j, 1]$. All reinstated tasks share a common productivity: $g_{Rj}(z) = \gamma_j B_j$ for all $z \in (1, 1 + \delta_j I_j]$.*

This is the standard assumption in Acemoglu and Restrepo (2020a). I derive all results under A1, then state in Lemma 2 the exact aggregation result that allows $\gamma_j B_j$ to be interpreted as a CES power mean when reinstated-task productivities are heterogeneous.

Step 1: Optimal Capital Allocation Across Automated Tasks

The firm allocates capital across automated tasks $z \in [0, I_j]$. At task z , output is $y_{Kj}(z) = g_{Kj}(z) k_j(z)$. The firm solves:

$$\max_{\{k_j(z)\}} \int_0^{I_j} (g_{Kj}(z) k_j(z))^\varsigma dz \quad \text{s.t.} \quad \int_0^{I_j} k_j(z) dz = K_j. \quad (\text{A.1})$$

First-order condition:

$$g_{Kj}(z)^\varsigma k_j(z)^{\varsigma-1} = \lambda_K \quad \forall z \in [0, I_j]. \quad (\text{A.2})$$

Since $\varsigma - 1 < 0$, solving for $k_j(z)$:

$$\bar{k}_j(z) = \frac{g_{Kj}(z)^{\varsigma/(1-\varsigma)}}{\int_0^{I_j} g_{Kj}(z')^{\varsigma/(1-\varsigma)} dz'} \cdot K_j = \frac{g_{Kj}(z)^{\sigma-1}}{\int_0^{I_j} g_{Kj}(z')^{\sigma-1} dz'} \cdot K_j, \quad (\text{A.3})$$

using $\varsigma/(1-\varsigma) = (\sigma-1)/\sigma \cdot \sigma = \sigma-1$. Capital is allocated proportionally to $g_{Kj}(z)^{\sigma-1}$. Substituting back:

$$\begin{aligned} Z_j &\equiv \int_0^{I_j} g_{Kj}(z)^\varsigma \bar{k}_j(z)^\varsigma dz \\ &= \frac{K_j^\varsigma}{[\mathcal{G}_{Kj}]^\varsigma} \int_0^{I_j} g_{Kj}(z)^{\varsigma+(\sigma-1)\varsigma} dz \\ &= \frac{K_j^\varsigma}{\mathcal{G}_{Kj}^\varsigma} \cdot \mathcal{G}_{Kj} = \mathcal{G}_{Kj}^{1-\varsigma} \cdot K_j^\varsigma = \mathcal{G}_{Kj}^{1/\sigma} \cdot K_j^\varsigma, \end{aligned} \quad (\text{A.4})$$

where $\mathcal{G}_{Kj} \equiv \int_0^{I_j} g_{Kj}(z)^{\sigma-1} dz$ and using $\varsigma + (\sigma-1)\varsigma = \sigma\varsigma = \sigma-1$ and $1-\varsigma = 1/\sigma$.

Units: $[g_{Kj}] = [\text{output/capital}]$, so $[Z_j] = [(\text{output/capital})^{\sigma-1}]^{1/\sigma} \cdot [\text{capital}]^\varsigma = [\text{output}]^\varsigma$. $\square$

Step 2: Optimal Labor Allocation Across Labor Tasks

The firm allocates effective labor L_j^{eff} across surviving tasks $z \in (I_j, 1]$ and reinstated tasks $z \in (1, 1 + \delta_j I_j]$. Under Assumption A1, the productivities are $\bar{g}_{Lj}$ and $\gamma_j B_j$ respectively. The firm solves:

$$\max_{\{\ell_j(z)\}} \int_{I_j}^{1+\delta_j I_j} (g_j(z) \ell_j(z))^\varsigma dz \quad \text{s.t.} \quad \int_{I_j}^{1+\delta_j I_j} \ell_j(z) dz = L_j^{\text{eff}}. \quad (\text{A.5})$$

The first-order condition gives optimal allocation proportional to $g_j(z)^{\sigma-1}$:

$$\bar{\ell}_j(z) = \frac{g_j(z)^{\sigma-1}}{\mathcal{G}_{Lj}} \cdot L_j^{\text{eff}}, \quad \mathcal{G}_{Lj} \equiv \int_{I_j}^{1+\delta_j I_j} g_j(z)^{\sigma-1} dz. \quad (\text{A.6})$$

Substituting back (identical algebra to Step 1):

$$\int_{I_j}^{1+\delta_j I_j} g_j(z)^\varsigma \bar{\ell}_j(z)^\varsigma dz = \mathcal{G}_{Lj}^{1/\sigma} \cdot (L_j^{\text{eff}})^\varsigma. \quad (\text{A.7})$$

Under Assumption A1, separate $\mathcal{G}_{Lj}$ into segments:

$$\mathcal{G}_{Lj} = (1 - I_j) \bar{g}_{Lj}^{\sigma-1} + \delta_j I_j (\gamma_j B_j)^{\sigma-1}. \quad (\text{A.8})$$

Define:

$$\tilde{\Gamma}_j \equiv \mathcal{G}_{Lj}^{1/\sigma} = \left[(1 - I_j) \bar{g}_{Lj}^{\sigma-1} + \delta_j I_j (\gamma_j B_j)^{\sigma-1} \right]^{1/\sigma}. \quad (\text{A.9})$$

This is the exact expression for $\tilde{\Gamma}_j$: it is a CES aggregate of the surviving-task and reinstated-task content, raised to $1/\sigma$.

Relationship to the main text notation. Equation (7) in the main text is exactly (A.9): under Assumption A1, $\int_{I_j}^1 g_{Lj}(z)^{\sigma-1} dz = (1 - I_j) \bar{g}_{Lj}^{\sigma-1}$, so the two displays coincide, and without A1 the integral form of (7) is the general statement of Step 2. Proposition 1 is therefore exact.

Some expository passages use the additive shorthand for the two components of labor's task content:

$$\underbrace{\left[(1 - I_j) \bar{g}_{Lj}^{\sigma-1} + \delta_j I_j (\gamma_j B_j)^{\sigma-1} \right]^{1/\sigma}}_{\text{exact CES aggregate, eq. (7)}} \quad \text{vs.} \quad \underbrace{(1 - I_j) \bar{g}_{Lj}^\varsigma + \delta_j I_j (\gamma_j B_j)^\varsigma}_{\text{additive shorthand}}. \quad (\text{A.10})$$

The two are equal only in special cases, including when a single segment is active, because $[a^{\sigma-1} + b^{\sigma-1}]^{1/\sigma} \neq a^\varsigma + b^\varsigma$ generically for $\sigma \neq 1$. They differ by the concavity of $x \mapsto x^{1/\sigma}$. At the calibrated parameters ($\sigma = 2$, $I_R = 0.30$, $I_A = 0.25$, task productivities close across segments) the discrepancy is small, and the shorthand follows the [Acemoglu and Restrepo \(2018\)](#) efficiency-unit convention in which g^ς names the reduced-form task-content coefficient. All quantitative results in the paper are computed numerically from the full CES equilibrium, never from the shorthand.

Units: $[\tilde{\Gamma}_j] = [(\text{output}/\text{labor})^{\sigma-1}]^{1/\sigma} = [\text{output}/\text{labor}]^\varsigma$, so $[\tilde{\Gamma}_j \cdot (L_j^{\text{eff}})^\varsigma] = [\text{output}]^\varsigma = [Z_j]$. $\square$

Step 3: CES Aggregation of Heterogeneous Reinstated Tasks

Step 2 assumed all reinstated tasks share productivity $\gamma_j B_j$ (Assumption A1). I relax this for reinstated tasks and show that $\gamma_j B_j$ can be interpreted as a power mean.

Lemma 2 (Heterogeneous Reinstatement Aggregation). *Let reinstated tasks at frontier j have productivities $g_{Rj}(z) = \gamma(z) \cdot B_j$, where $\gamma(z)$ are i.i.d. draws from G_j^* . Under optimal labor allocation and the exact law of large numbers for continuum economies (Sun, 2006):*

$$\int_1^{1+\delta_j I_j} (\gamma(z) B_j)^{\sigma-1} dz = \delta_j I_j \cdot B_j^{\sigma-1} \cdot \mathbb{E}_{G_j^*}[\gamma^{\sigma-1}] \quad a.s. \quad (\text{A.11})$$

Define the $(\sigma - 1)$ -power mean:

$$\gamma_j \equiv \left(\mathbb{E}_{G_j^*}[\gamma^{\sigma-1}] \right)^{1/(\sigma-1)}. \quad (\text{A.12})$$

Then the reinstated-task component of $\mathcal{G}_{Lj}$ is $\delta_j I_j \cdot (\gamma_j B_j)^{\sigma-1}$, and all results from Step 2 hold with $\gamma_j B_j$ replaced by the power mean (A.12). When G_j^* is degenerate (homogeneous reinstated tasks), the power mean reduces to the common productivity.

Proof. Equation (A.11) follows from the exact LLN applied to the i.i.d. continuum $\{(\gamma(z) B_j)^{\sigma-1}\}_{z \in (1, 1+\delta_j I_j]}$. Substituting $\mathbb{E}[\gamma^{\sigma-1}] = \gamma_j^{\sigma-1}$ into (A.8) gives $\mathcal{G}_{Lj} = (1 - I_j) \bar{g}_{Lj}^{\sigma-1} + \delta_j I_j (\gamma_j B_j)^{\sigma-1}$, identical in form to the homogeneous case. The remainder of Step 2 is unchanged. $\square$

Remark on the power mean exponent. The object that emerges from the CES optimal allocation is the $(\sigma - 1)$ -power mean (A.12), not the ς -power mean. At $\sigma = 2$: $\gamma_j = \mathbb{E}[\gamma]$ (arithmetic mean). At general σ : $\gamma_j = (\mathbb{E}[\gamma^{\sigma-1}])^{1/(\sigma-1)}$. Under the AR efficiency-unit convention in the main text, where $(\gamma_j B_j)^\varsigma$ denotes the reduced-form task content, the distinction is absorbed into the definition of γ_j . When G_j^* is degenerate, all power means coincide and the exponent is irrelevant.

Step 4: Assembly (Proof of Proposition 1)

Proof. The firm's problem separates into capital allocation (automated tasks) and labor allocation (surviving + reinstated tasks) because the two sets use different factors and enter the CES aggregator additively.

Capital side (Step 1):

$$\int_0^{I_j} y_{Kj}(z)^\varsigma dz = Z_j(I_j),$$

with $[Z_j] = [\text{output}]^\varsigma$.

Labor side (Steps 2–3):

$$\int_{I_j}^{1+\delta_j I_j} y_{Lj/Rj}(z)^\varsigma dz = \tilde{\Gamma}_j(I_j, \delta_j) \cdot (L_j^{\text{eff}})^\varsigma,$$

with $[\tilde{\Gamma}_j \cdot (L_j^{\text{eff}})^\varsigma] = [\text{output}]^\varsigma$.

Substituting into (2):

$$Y_j = \left[Z_j(I_j) + \tilde{\Gamma}_j(I_j, \delta_j) \cdot (L_j^{\text{eff}})^\varsigma \right]^{1/\varsigma}.$$

Both terms inside the bracket have units $[\text{output}]^\varsigma$. Y_j has units $[\text{output}]$. □

Step 5: Reinstatement Rate and Frontier Position

Equation (11) expresses δ_j as a function of $(I_j, \theta_j, v_j, W_{k(j)}^d, \eta, c_0)$. The total measure of reinstated tasks is:

$$\delta_j \cdot I_j = \left[\frac{\theta_j}{c_0(1+\eta)} (v_j - W_{k(j)}^d) \right]^{1/\eta}, \quad (\text{A.13})$$

which does not depend directly on I_j . The equilibrium objects $(v_j, W_{k(j)}^d)$ generically do depend on I_j , so the total measure is not invariant to the frontier position. The calibrated $\delta_R = 1.00$ and the estimated $\delta_A = 0.03$ are evaluated at $(I_R, I_A) = (0.30, 0.25)$. For the comparative statics by finite differences ($dI = 0.01$), δ_j is held at its baseline value across the perturbation: the solver takes (δ_R, δ_A) as arguments and the perturbed equilibria are computed at the same pair. The response of the reinstated stock to the surplus through equation (11) is therefore not imposed in Sections 6.1 and 7, and the semi-elasticities there are comparative statics of the model with the reinstatement rates fixed. Imposing it would require values for η and θ_j/c_0 that the quantitative system does not carry. In the estimated system the reinstatement channel is carried as the single primitive $\chi_j = \delta_j(\gamma_j B_j)^{\sigma-1}$ of equation (8), evaluated at the baseline frontier position, this appendix keeps the two factors separate because the entry margin above determines δ_j as a distinct economic object, and any task-count statement obtained by inverting the mapping is conditional on the $\gamma_j B_j$

normalization.

B Proofs and Derivations

B.1 The Demand-Side Equilibrium

The quantitative core of Section 6 is two-sector. This appendix specifies the three demand-side blocks of Sections 2.2–2.4 as an equilibrium and solves it, which is what produces the computed channels of Table 9.

Services. A non-tradable sector produces $Y_N = A_N L_N$ with labor only and is not automated. Households of skill k spend a fixed share α_k^N of income on services, with $\alpha_S^N = 0.40 > \alpha_H^N = 0.25$ (Table 6), so non-college households are the more service-intensive. Service-market clearing sets $p_N Y_N$ equal to aggregate service expenditure, and the labor released from services is absorbed by the two tradable sectors at the prevailing skill wages.

Participation. Reservation wages are drawn from a distribution with constant elasticity, so that labor supply is $L_k = \bar{L}_k (W_k / \bar{W}_k)^{\varepsilon_k^{LS}}$ with $\varepsilon_S^{LS} = 0.25$ and $\varepsilon_H^{LS} = 0.10$ (Table 6). This is the simplest distribution consistent with the calibrated elasticities, which are all the model of Section 2.3 pins down.

Capital income. Automation and reinstatement rents accrue as the residual $\Pi = Y - W_S L_S - W_H L_H$ and are distributed across households, who consume out of them at $m_K = 0.20$ against $m_S = 0.75$ and $m_H = 0.45$ (Table 6). Unconsumed income is saved and spent on tradables.

Two checks discipline the exercise, and one tension must be reported. With services switched off, the extended model reproduces the two-sector core of Section 6 to machine precision. The service block is then calibrated to a single target, the share of service-sector workers who are non-college, 0.68. It is not validated against any moment it was not fitted to. The effective service expenditure share of 0.32 and the local demand multiplier of 1.32 that the block reports are not independent checks on it: the multiplier is an algebraic transform of the expenditure share, which is itself an income-weighted average of the two calibrated expenditure shares and is therefore insensitive to the scale of the service sector.

The tension is that the resulting service sector is smaller than Table 6 assumes. The equilibrium delivers a service share of value added of 0.15 and a share of non-college workers employed in services

of 0.27, against the $\mu_N = 0.35$ service-sector aggregate-share anchor and the $s_N^S = 0.45$ share of non-college workers employed in services. The latter uses the non-college workforce as its denominator, whereas the 0.68 target above is the share of service-sector workers who are non-college. No service-sector scale satisfies both of those anchors at once. The channels below are therefore reported at the budget-consistent calibration, and also at two variants that force each of the table’s anchors in turn. Forcing $\mu_N = 0.35$ raises $\mathcal{F}^{NT}$ to +0.00031 and leaves every conclusion of this section intact. That is why the claim made in the text is an absolute bound holding across all three calibrations, not a point estimate at any one of them.

The computed channels are in Table 9. Three of them were not visible in the analytic bounds this appendix replaces.

The MPC channel is *negative*. A shift toward AI raises the capital income share and so lowers the aggregate propensity to consume, contracting service demand. Its sign depends on the service expenditure share out of capital income, which the calibration does not pin down, across the range $[0, 0.40]$ the channel runs from +0.00002 to -0.00004 , and it is negligible throughout.

The participation margin enters negatively, at -0.00009 : the wage dampening exceeds the income amplification, so participation reduces the total feedback, as Section 2.3 conjectured.

On the cognitive frontier the wage-feedback component $\mathcal{F}^{PE}$ nearly cancels, at 0.00128, and the non-tradable channel is larger than it. That is a fact about the denominator. The non-tradable channel has its own scale, set by the size of the service sector and not by the wage-composition index, so a statement of the form “the demand channels are a fraction of the partial-equilibrium feedback” fails where that feedback is small, and the claim the paper makes is absolute instead, $|\mathcal{F}^{GE}| \leq 0.0021$ on both frontiers, which holds under every specification choice above.

C Dynamic Model Derivation

This appendix provides the full derivation of the dynamic amplification factor $\mathcal{A}^{\text{dyn}}$ stated in Proposition 4. The derivation proceeds in seven steps.

C.1 State Variables and Laws of Motion

The economy is characterized by the vector (K_R, K_A, A_S, A_H) with laws of motion:

$$\dot{K}_R = s_R(K_R, K_A, A_S, A_H) \cdot S - d_R K_R, \quad (\text{C.1})$$

$$\dot{K}_A = s_A(K_R, K_A, A_S, A_H) \cdot S - d_A K_A, \quad (\text{C.2})$$

$$\dot{A}_S = r A_S + W_S L_S - C_S, \quad (\text{C.3})$$

$$\dot{A}_H = r A_H + W_H L_H - C_H, \quad (\text{C.4})$$

where $\mathcal{S}(t) = Y(t) - C(t)$ is aggregate savings (calligraphic, distinct from the skill subscript S), $s_j \in [0, 1]$ is the share allocated to frontier j (with $s_R + s_A = 1$), and d_j is depreciation. Frontier thresholds evolve as $\dot{I}_j = \zeta_j K_j^{\xi_j}$, and the investment allocation tracks relative returns: $s_R/s_A = (\mathcal{R}_R^{\text{auto}}/\mathcal{R}_A^{\text{auto}})^{1/\nu}$, where $\nu > 0$ governs responsiveness. Household type k has Euler equation $\dot{C}_k/C_k = \sigma_c(r(t) - \rho_k)$.

C.2 Simplify to the Relevant Margin

The static equilibrium conditions (labor markets, automation FOCs, reinstatement free entry) determine a target allocation as a function of the state at each instant, $\omega^\dagger(t) = \tilde{\omega}(K_R(t), K_A(t), A_S(t), A_H(t))$. The impact response to a decline in r_A depends only on the direct effect on the automation FOC (no state variable moves at $t = 0$):

$$\Delta\omega^{\dagger, \text{impact}} = \frac{\partial \tilde{\omega}}{\partial r_A} \Delta r_A. \quad (\text{C.5})$$

C.3 Characterize the Savings Channel

With heterogeneous ρ_k , I adopt a tractable two-asset structure: non-college workers are hand-to-mouth ($C_S = m_S W_S L_S$ with $m_S = 1$) and college workers are Euler-equation savers. Aggregate savings are:

$$S = W_H L_H - C_H + \Pi, \quad (\text{C.6})$$

where C_H satisfies $\dot{C}_H/C_H = \sigma_c(r - \rho_H)$. In steady state: $\mathcal{S}^* = d_R K_R^* + d_A K_A^*$ and $r^* = \rho_H$.

C.4 Log-Linearize the Investment Allocation

From the investment share equation, log-linearizing around steady state:

$$\hat{s}_A \approx \frac{s_R^*}{\nu}(\hat{\mathcal{R}}_A - \hat{\mathcal{R}}_R),$$

where hats denote log-deviations. Since returns depend on the target allocation $\omega^\dagger$:

$$\hat{s}_A = \frac{s_R^*}{\nu} \frac{\partial \ln(\mathcal{R}_A/\mathcal{R}_R)}{\partial \omega^\dagger} \hat{\omega}^\dagger \equiv \eta_{\text{sav}} \hat{\omega}^\dagger,$$

where η_{sav} is the savings-reallocation elasticity (a symbol distinct from the incidence share η_S of Section 3.7).

C.5 Log-Linearize the Capital Law of Motion

From (C.2): $\dot{K}_A = s_A \mathcal{S} - d_A K_A$. Using $s_A^* \mathcal{S}^*/K_A^* = d_A$ in steady state:

$$\dot{\hat{K}}_A = d_A(\hat{s}_A + \hat{\mathcal{S}} - \hat{K}_A). \quad (\text{C.7})$$

C.6 Close with the Savings Response

The permanent decline in r_A raises AI returns, so $\hat{s}_A > 0$. College income $W_H L_H$ and profits Π also respond. Under the hand-to-mouth assumption, aggregate savings respond only through college income and profits. Log-linearizing:

$$\hat{\mathcal{S}} = \eta_W \hat{\omega}^\dagger,$$

where $\eta_W \equiv (W_H^* L_H^*/\mathcal{S}^*)(\partial \ln W_H/\partial \omega^\dagger) + (\Pi^*/\mathcal{S}^*)(\partial \ln \Pi/\partial \omega^\dagger)$ captures how the automation mix affects college savings through wages and profits. Differentiating (C.6) in full also gives a college-employment term $W_H^* L_H^*/\mathcal{S}^* \cdot \partial \ln L_H/\partial \omega$ and a consumption term $-(C_H^*/\mathcal{S}^*) \partial \ln C_H/\partial \omega$. Both are set to zero here, which is a restriction and not an implication: the Euler equation pins the growth rate of C_H and not its level response, and college employment is held fixed. The amplification factor below is computed under those two restrictions.

C.7 Solve for the New Steady State

In the new steady state, $\dot{K}_A = 0$, so $\hat{K}_A^{ss} = \hat{s}_A^{ss} + \hat{\mathcal{S}}^{ss}$. The target allocation responds to capital: $\hat{\omega}^{\dagger,ss} = \hat{\omega}^{\dagger,\text{impact}} + \eta_K \hat{K}_A^{ss}$, where $\eta_K \equiv (\partial\omega/\partial K_A)(K_A^*/\omega^*)$. Substituting:

$$\hat{\omega}^{\dagger,ss} = \hat{\omega}^{\dagger,\text{impact}} + \eta_K(\eta_{\text{sav}} + \eta_W)\hat{\omega}^{\dagger,ss}.$$

Solving:

$$\hat{\omega}^{\dagger,ss} = \frac{\hat{\omega}^{\dagger,\text{impact}}}{1 - \eta_K(\eta_{\text{sav}} + \eta_W)}. \quad (\text{C.8})$$

The amplification factor is:

$$\mathcal{A}^{\text{dyn}} = \frac{1}{1 - \eta_K(\eta_{\text{sav}} + \eta_W)}. \quad (\text{C.9})$$

C.8 Express in Terms of Primitives

The composite elasticity $\eta_K(\eta_{\text{sav}} + \eta_W)$ depends on three primitive objects:

- *College savings response*: proportional to σ_c (a higher intertemporal elasticity of substitution, IES, means college workers smooth less, save more in response to income gains).
- *Patience gap*: $\rho_S - \rho_H$ determines how fast wealth inequality grows along the transition, which determines how strongly investment tilts toward AI.
- *Depreciation*: d_A determines how quickly the capital stock adjusts.

The composite cannot be evaluated from these definitions: ν and the frontier-advance parameters are never calibrated, and no identity here removes α_r , s_R^*/ν or η_K . What follows is therefore a parameterization and not a derivation. I set the composite to the value implied by a two-state savings-and-capital system, in which the patience gap feeds savings growth at rate $\sigma_c(\rho_S - \rho_H)$ and the capital and consumption blocks return at rate $d_A + \sigma_c\rho_H$, so that (i) the return differential is proportional to $\omega - \omega^*$ with coefficient α_r , (ii) savings respond to college income with elasticity σ_c , and (iii) the patience gap drives the wealth accumulation rate, the composite elasticity simplifies to:

$$\eta_K(\eta_{\text{sav}} + \eta_W) = \frac{\sigma_c(\rho_S - \rho_H)}{d_A + \sigma_c\rho_H}. \quad (\text{C.10})$$

The amplification factor below is conditional on that parameterization, not implied by the general

expression above. Substituting yields:

$$\mathcal{A}^{\text{dyn}} = \frac{1}{1 - \sigma_c(\rho_S - \rho_H)/(d_A + \sigma_c\rho_H)}. \quad (\text{C.11})$$

For small numerator (which holds empirically: $\sigma_c(\rho_S - \rho_H) = 0.015 \ll d_A = 0.10$), this is approximately $1 + \sigma_c(\rho_S - \rho_H)/(d_A + \sigma_c\rho_H)$.

At calibrated values ($\sigma_c = 0.5$, $\rho_S = 0.06$, $\rho_H = 0.03$, $d_A = 0.10$), the exact formula gives:

$$\mathcal{A}^{\text{dyn}} = \frac{1}{1 - 0.015/0.115} = \frac{1}{1 - 0.1304} = 1.15,$$

and the first-order expansion gives $1 + 0.015/0.115 = 1.13$, the value reported in equation (14).

The linearized system has a stable eigenvalue $\lambda_{\text{dyn}} = -d_A(1 - \eta_K(\eta_{\text{sav}} + \eta_W)) = -0.10(1 - 0.13) = -0.087$ (subscripted to distinguish it from the inequality-aversion parameter λ of Section 7). The half-life is $t_{1/2} = \ln 2/0.087 \approx 8.0$ years.

Along the transition:

$$\omega^\dagger(t) = \omega^{\dagger, \text{new}} + (\omega^{\dagger, \text{impact}} - \omega^{\dagger, \text{new}})e^{\lambda_{\text{dyn}} t}. \quad (\text{C.12})$$

C.9 Validity Conditions

The formula (C.11) is:

- (i) *Local*: valid for small perturbations Δr_A around the initial steady state.
- (ii) *Steady-state comparison*: $\mathcal{A}^{\text{dyn}}$ compares the new steady state to the impact response, not the transitional path.
- (iii) *Log-linear*: obtained by first-order Taylor expansion. Second-order terms are $O((\Delta r_A)^2)$.
- (iv) *Separable* in the patience gap because the hand-to-mouth assumption makes non-college savings identically zero. Without this assumption, a cross-term proportional to $(1 - m_S) \cdot (\partial W_S / \partial \omega)$ would enter (see Appendix F).

The approximation is local, and for shocks several times the size of the impact response the log-linear characterisation should not be relied on.

D Aggregation from CZ-Level to National Parameters

D.1 The Aggregation Problem

This paper uses the cognitive effective task content χ_A as a production-function parameter in a national macro model, estimated from CZ-level moments in the system of Section 3.4. The physical channel is calibrated from the firm-level Danish evidence of Humlum (2024) and raises no aggregation question. The question here is whether CZ-level estimates can be treated as national production-function parameters.

The MDE in Section 3.4 matches CZ-level wage and college-employment-share moments. Each skill-group wage equation is a weighted cross-section over the 722 commuting zones,

$$\Delta \ln W_{k,c} = \beta_k^R \cdot \text{Robot}_c + \beta_k^A \cdot \text{AI}_c + X_c' \Gamma_k + \iota_{r(c)} + \varepsilon_{k,c},$$

where c indexes commuting zones, $k \in \{S, H\}$ indexes skill groups, X_c are the completed set of pre-determined composition shares (Section 3.1), and $\iota_{r(c)}$ are census-region fixed effects (a symbol distinct from the reinstatement rates δ_j). With one observation per zone per equation, the controls enter as $X_c' \Gamma_k$, since a commuting-zone fixed effect would be collinear. The skill-premium contrast stacks the two equations and takes their difference,

$$\Delta \ln(W_H/W_S)_c = (\beta_H^R - \beta_S^R) \text{Robot}_c + (\beta_H^A - \beta_S^A) \text{AI}_c + \tilde{\varepsilon}_c,$$

in which any commuting-zone-level shock common to both skill groups differences out. The robot Bartik identifies a composite industry demand parameter $\Lambda_R = 0.0090$, not task content, and the cognitive occupation-level moments identify (χ_A, φ) through the Acemoglu–Restrepo task structure.

D.2 Aggregation of δ_A : CZ-Level to National

The effective cognitive task content $\hat{\chi}_A = 0.013$ is recovered from cross-sectional CZ-level variation in AI exposure ($\delta_A = 0.03$ under the $\bar{g}_A$ normalization). The question is when the CZ-level estimate equals the national parameter δ_A^{nat} .

The two objects coincide when three conditions hold:

- (i) *Homogeneous technology*: the reinstatement rate is a property of the automation technology, not of the local labor market. The same AI system generates the same number of complementary tasks regardless of the CZ in which it is deployed.
- (ii) *No spatial spillovers*: AI adoption in CZ c does not affect task creation in neighboring CZ c' .
- (iii) *Correct spatial equilibrium*: the CZ-level estimation accounts for migration and commuting responses, so that the estimated δ_A reflects task creation after netting out worker reallocation.

Condition (i) is plausible for cognitive automation: AI tools are digital, scalable, and deployed uniformly across locations. Unlike robots, whose complementary tasks may depend on local supply chains, AI-generated tasks (e.g., prompt engineering, algorithmic auditing, AI-assisted analysis) are inherently non-spatial.

Condition (ii) is less restrictive for AI than for robots because cognitive tasks are not geographically tied to the adopting firm's CZ. Supply chain spillovers, the primary concern for physical automation, are largely absent for cognitive automation.

Condition (iii) is addressed by the migration adjustment, which scales the non-college labor supply response by the calibrated elasticity $\varepsilon_S^{\text{mig}} = 0.78$, so the cognitive reinstatement rate is separated from the mobility response. That is what conditions (i) to (iii) deliver, the aggregation from commuting zones to the nation, and the identifying assumption on the moments themselves is the one stated in Section 3.3.

D.3 External Validation of δ_R : The Humlum Bracket

The physical reinstatement channel is an external calibration (Section 3.5), and this subsection derives from Humlum (2024)'s firm-level evidence the range that evidence brackets for it, conditional on the accounting bridge stated below. The derivation is stated in the units of the creation-to-displacement value ratio δ_R , the value of new labor tasks created per unit of task value displaced at the physical frontier. The neutral baseline $\delta_R = 1$ sits at the point where task creation offsets displacement, which is the qualitative center of the firm-level pattern.

Humlum (2024) links workers, firms, and robots in Danish administrative data and estimates firm-level responses to robot adoption with a matched-control event-study design. His paper does not report a reinstatement rate directly, so I translate his estimates into the units of δ_R . The translation is an accounting bridge and not an estimate of his. It maps the fall in the production-

worker wage bill entirely into displaced task value and the rise in the tech wage bill entirely into created task value, which holds only if hours and composition within each group are unchanged, if wage payments track task value one for one, and if the output scale effect carries no new task content. The resulting interval is what those restrictions imply about δ_R .

The inputs are Humlum’s four-year event-study responses, the moments his own structural estimation targets. Adopting firms expand sales by 20 percent, cut the production-worker wage bill by 20 percent, raise the tech-worker wage bill by 30 percent, and raise the total wage bill by 8 percent. In the year before adoption, production workers account for 39.9 percent of the adopters’ wage bill and tech workers for 16.0 percent.

In value terms within adopting firms, displaced production-task value is $0.20 \times 39.9 = 8.0$ percent of the firm wage bill. Created robot-complementary task value, the tech department that operates and maintains the robots, is $0.30 \times 16.0 = 4.8$ percent. The remaining occupations expand with the output scale-up, bringing the total wage bill to +8 percent.

Two bounds follow. Counting only robot-complementary tech tasks as reinstatement gives $\delta_R \approx 4.8/8.0 = 0.60$, a narrow lower bound that excludes new task content in other occupations. Attributing the entire net labor-demand expansion to task creation gives $\delta_R \approx (8.0+8.0)/8.0 = 2.00$, an upper bound that conflates reinstatement with scale. The implied range is $[0.60, 2.00]$. Humlum’s central qualitative finding is sharper than either bound: total labor demand rises as production tasks fall, which in the model is the reinstatement-dominant regime, the one in which task creation raises total labor demand despite declining production-task demand. Placing his evidence in that regime is a consequence of the accounting bridge above and not of the signs alone, because rising total labour demand with falling production-task demand is equally consistent with weak task creation and a strong scale response, which is why the upper end of the range conflates the two. What the evidence brackets, under those restrictions, is the magnitude of the physical reinstatement channel. It does not supply a point estimate, so the paper treats δ_R as a calibrated range with the neutral $\delta_R = 1$ as its central case (Section 3.5).

D.4 Bias Bounds for the Reinstatement Estimates

The bounds below ask how far the cognitive CZ-level estimate or the physical calibration could be off before the headline moves.

The primary bias concern runs in opposite directions for the two objects. For δ_A , spatial spillovers are minimal (cognitive tasks are non-spatial), so $\hat{\delta}_A^{CZ} \approx \delta_A^{\text{nat}}$. For δ_R , positive between-firm spillovers make the bracket's lower bound conservative. If adopters' task creation spills to suppliers (robot adoption at an auto manufacturer generating quality-control tasks there), the reinstatement realized outside the adopting firm is missed by the firm-level bracket, so the anchoring claim only strengthens. Under a simple spillover structure where reinstatement at other firms equals a multiple χ of the reinstatement measured at adopters, the national parameter is $(1 + \chi)$ times the measured value. At $\chi = 0.15$ (supplier reinstatement worth 15% of the adopters' own), the effective calibration rises from the neutral 1.00 to 1.15, which raises π_R/π_A from 2.19 to 2.32 on the range grid of Table 7. At $\chi = 0.30$, the effective calibration is 1.30 and $\pi_R/\pi_A = 2.45$.

A perturbation of the physical calibration itself, and not of the commuting-zone estimate, is a separate question. Reducing the neutral $\delta_R = 1.00$ by a fifth gives 0.80, inside the lower half of the firm-level bracket, where $\pi_R/\pi_A = 2.01$. The robustness exercises cut in both directions. Spillovers ($\chi > 0$), faster AI depreciation ($d_A > d_R$), and the consumption-based welfare conventions of Section 7.3 strengthen robot dominance, while a CZ estimate that overstates the national rate, a largely transitional sorting friction (Table 8), and a higher skill elasticity all weaken it. The ranking survives the adjustments that push against it, which is the relevant test.

E Service Sector Automation Extension

The baseline model assumes the non-tradable sector is not automated ($I_N = 0$). This appendix evaluates a parametric extension in which AI partially automates service tasks.

E.1 Setup

Replace $Y_N = A_N L_N^{\text{eff}}$ with:

$$Y_N = \left[Z_N(I_N^A) + \tilde{\Gamma}_N(I_N^A, \delta_N^A) \cdot (L_N^{\text{eff}})^\varsigma \right]^{1/\varsigma}, \quad (\text{E.1})$$

where $I_N^A \in [0, I_A]$ is the share of service tasks automated by AI (service automation can only use AI, not robots, by assumption) and $\delta_N^A \leq \delta_A$ is the reinstatement rate in services. This nests the

baseline ($I_N^A = 0$) and a partially automated service sector ($I_N^A > 0$).

E.2 Effects on the Feedback Channels

Service automation affects two channels:

1. *Non-tradable multiplier*: When services are partially automated, service employment L_N declines for a given level of demand, weakening the labor reallocation mechanism. $\mathcal{F}^{NT}$ shrinks in proportion to the automated share I_N^A .
2. *Direct service displacement*: AI automation in services directly displaces non-college workers (who are 68% of service employment), adding service-sector exposure to the PE wage channel.

The two effects move $\mathcal{F}^{GE}$ in opposite directions, and both are second order. The reallocation channel is the computed cognitive-frontier $\mathcal{F}^{NT} = -0.00032$ of Table 9 before any of it is automated away, so that is the largest possible weakening. The direct displacement effect adds to the wage-feedback channel a term scaled by the automated service share, which at $I_N^A = 0.30$ is of the same order. Both are more than two orders of magnitude below one, so $|\mathcal{F}^{GE}|$ remains far inside the bound of Proposition 2, the sampled slopes of Proposition 3 stay below one at the same nodes, and the supply-side ranking is unaffected by AI automation of services.

F Bewley Extension

F.1 Relaxing the Hand-to-Mouth Assumption

The dynamic model of Section 5 assumes non-college workers consume all income ($m_S = 1, A_S = 0$), which is a knife-edge, not the calibrated case. This appendix relaxes to $m_S \in (0, 1)$, allowing non-college workers to hold positive assets and smooth consumption.

With $m_S < 1$, the aggregate savings equation becomes:

$$S = (1 - m_S)W_S L_S + W_H L_H - C_H + \Pi,$$

and the log-linearized savings response gains a cross-term:

$$\hat{S} = \eta_W \hat{\omega} + \underbrace{(1 - m_S) \frac{W_S^* L_S^*}{S^*} \frac{\partial \ln W_S}{\partial \omega}}_{\text{non-college savings response}} \hat{\omega}.$$

Since $\partial \ln W_S / \partial \omega > 0$ (robot-dominated automation raises non-college wages), the non-college savings response is positive when ω rises and negative when ω falls. After an AI cost decline ($\hat{\omega} < 0$), non-college savings fall, *partially offsetting* the college savings channel that amplifies the shift toward AI.

F.2 Modified Amplification Factor

The composite elasticity becomes:

$$\eta_K (\eta_{\text{sav}} + \eta_W + \eta_W^S), \quad \eta_W^S \equiv (1 - m_S) \frac{W_S^* L_S^*}{S^*} \frac{\partial \ln W_S}{\partial \omega}.$$

The new elasticity joins η_W inside the composite, which is where the savings equation puts it. It is not divided by a rate, and dividing it by one would leave the composite carrying units of time. Its contribution is negative, since $\eta_K < 0$ and $\eta_W^S > 0$, and proportional to $(1 - m_S)$: the more non-college workers save, the stronger the dampening, and at the hand-to-mouth limit $m_S = 1$ it vanishes, recovering the baseline. Substituting the modified composite elasticity into the amplification formula, at $m_S = 0.85$ (non-college workers save 15% of income):

$$\mathcal{A}^{\text{dyn}} = \frac{1}{1 - \eta_K (\eta_{\text{sav}} + \eta_W + \eta_W^S)}.$$

The direction is signed by the expression: η_W^S rises as m_S falls, and $\eta_K < 0$, so lower hand-to-mouth behaviour dampens the amplification monotonically, reaching the baseline only at $m_S = 1$. The magnitude at any interior m_S needs η_K , the non-college wage share of savings and $\partial \ln W_S / \partial \omega$, none of which is calibrated here, so this extension establishes the sign of the correction and not its size.

Table F.1: Dynamic Amplification Under Alternative Savings Behavior

m_S	$\mathcal{A}^{\text{dyn}}$	Interpretation
1.00 (hand-to-mouth)	1.13	Knife-edge
< 1	below 1.13	dampened, monotonically in $1 - m_S$

Notes: Each row reports the first-order expansion of the amplification factor under the stated m_S , holding all other parameters at baseline values. This is the object reported in equation (14), and it is the more conservative of the two: at the hand-to-mouth limit the exact log-linear value is 1.15 against the 1.13 shown here.

Table F.1 reports what the expression supports. The amplification is largest at the hand-to-mouth knife edge and falls as non-college saving rises, and it does not survive at every m_S , because amplification requires the composite to stay in $(0, 1)$ and the correction lowers it without a floor, so a large enough non-college savings response drives it to zero or below, at which point the channel dampens the initial shift instead of amplifying it. Where that happens depends on the same uncalibrated elasticities. The knife-edge $\mathcal{A}^{\text{dyn}} = 1.13$ is an upper bound with respect to the savings margin m_S and not with respect to the linearization.

Online Appendix

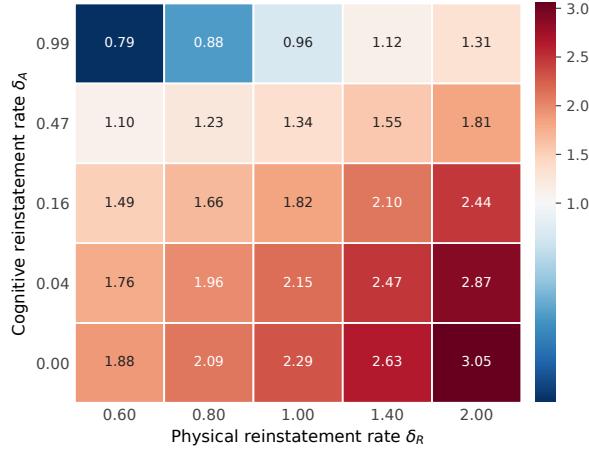
The Macroeconomics of Automation: Sorting Frictions and the Asymmetry Between Robots and AI

For online publication only

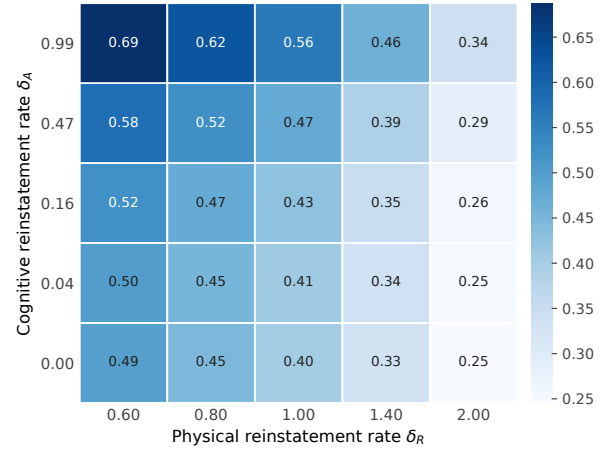
G Extended Robustness

This appendix evaluates the sensitivity of the productivity content ratio π_R/π_A , the feedback coefficient $\mathcal{F}$, and the labor share response to parameters not varied in the main text. All grids are computed at the baseline estimates $(\delta_R, \delta_A, \varphi) = (1.00, 0.03, 0.80)$ with the same solver that produces the main-text results in the replication package. The baseline rows reproduce the headline decomposition exactly: $\pi_R/\pi_A = 2.19$, direct components $(+0.273, +0.257)$, and reinstatement-plus-reallocation components $(+0.314, +0.011)$.

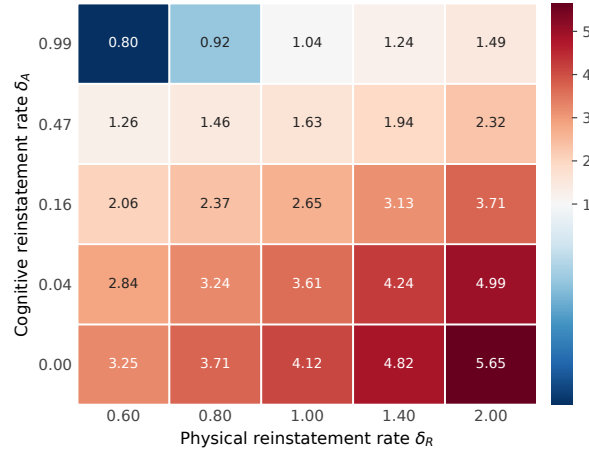
Figure [G.1](#) provides a visual summary of the robustness analysis. All three panels grid δ_R over the Humlum bracket $[0.60, 2.00]$ and δ_A over the bootstrap percentiles of Table [11](#), with the unadjusted estimate 0.99 as a reference column. Panel (a) shows $\pi_R/\pi_A > 1$ at every node inside the bootstrap range and below one only in the reference column at $\delta_R \leq 1.00$. Panel (b) shows that robots are less damaging to the labor share at every node, the ratio running from 0.25 to 0.69. Panel (c) shows the social-return ratio above one wherever the productivity ratio is, and below one only at the two reference-column nodes with δ_R of 0.60 and 0.80.



(a) Productivity ratio π_R/π_A



(b) Labor share damage ratio



(c) Social-return ratio SMP_R/SMP_A

Figure G.1: Sensitivity of main results, gridding δ_R over the Humlum bracket $[0.60, 2.00]$ and δ_A over the bootstrap percentiles of Table 11 plus the unadjusted estimate 0.99, with $\delta_R = 1.00$ a node and φ at its estimate. In panel (b), the labor-share damage ratio, values below one mean robots damage the labor share *less* than AI, so lower is more favorable to the robot frontier.

G.1 Sensitivity to Reinstatement Task Productivity ($\gamma_j B_j$)

The reinstatement productivity parameter $\gamma_j B_j$ governs the quality of newly created tasks. If new AI tasks are more productive than new robot tasks ($\gamma_A B_A > \gamma_R B_R$), this partially offsets the δ asymmetry. At calibrated values, $\gamma_R B_R = \gamma_A B_A$ (both are normalized to one). Table G.1 grids the ratio.

Table G.1: Sensitivity to Reinstatement Task Quality

$\gamma_A B_A / \gamma_R B_R$	π_R	π_A	π_R / π_A	$\partial s_L / \partial I_A$
0.50	+0.587	+0.262	2.24	−0.313
0.75	+0.587	+0.265	2.22	−0.312
1.00 (baseline)	+0.587	+0.268	2.19	−0.311
1.50	+0.587	+0.273	2.15	−0.310
2.00	+0.587	+0.278	2.11	−0.309

Notes: Each row recomputes all objects under the stated quality ratio at the baseline estimates $(\delta_R, \delta_A, \varphi) = (1.00, 0.03, 0.80)$. The quality ratio multiplies the productivity of reinstated cognitive tasks relative to reinstated physical tasks, holding surviving-task productivities fixed.

The ratio is monotone in the direction the mechanism implies, since more productive AI tasks offset more of the δ asymmetry, but at the estimated reinstatement rate there is almost no reinstated cognitive task content for the quality parameter to act on. Doubling the quality of AI’s new tasks moves the ratio from 2.19 to 2.11, and no quality ratio in $[1, 3]$ brings the two frontiers to parity.

G.2 Sensitivity to Comparative Advantage (κ)

Higher κ means skill groups are more specialized, which raises the reallocation cost of displacement, and AI displacement generates larger skill mismatch under stronger comparative advantage because non-college workers displaced from cognitive tasks have less transferable human capital. Table G.2 shows that the opposite force dominates.

Table G.2: Sensitivity to Comparative Advantage

κ	π_R	π_A	π_R/π_A	π_R^{r+r}	π_A^{r+r}
1.0 (no CA)	+0.553	+0.167	3.31	+0.336	+0.012
1.2	+0.564	+0.200	2.82	+0.329	+0.011
1.4	+0.573	+0.226	2.53	+0.323	+0.011
1.6	+0.580	+0.248	2.34	+0.318	+0.011
1.8 (baseline)	+0.587	+0.268	2.19	+0.314	+0.011
2.0	+0.593	+0.284	2.09	+0.310	+0.011
2.5	+0.604	+0.318	1.90	+0.303	+0.011
3.0	+0.613	+0.344	1.78	+0.298	+0.010

Notes: Each row recomputes all objects at the baseline estimates $(\delta_R, \delta_A, \varphi) = (1.00, 0.03, 0.80)$. “r+r” denotes the reinstatement-plus-reallocation component. κ governs skill group specialization: higher κ means stronger comparative advantage.

The productivity ratio π_R/π_A is *decreasing* in κ , so stronger comparative advantage narrows the gap between frontiers, because higher κ raises both π_R and π_A (more specialized workers are more productive in their comparative advantage sector) and raises π_A proportionally more. The ratio runs from 3.31 without comparative advantage to 1.78 at $\kappa = 3$, above one throughout.

G.3 Sensitivity to Top-Level Elasticity (ρ)

Under general CES aggregation ($\rho \neq 1$), a price channel amplifies or dampens $\mathcal{F}$. The realized channel is small and rises with ρ , from -0.00003 at $\rho = 0.50$ to $+0.00003$ at $\rho = 1.50$, so it dampens the feedback below Cobb–Douglas and amplifies it above. The productivity ratio moves with it, from 2.16 to 2.22 across the same range.

Table G.3: Sensitivity to Top-Level Substitution Elasticity

ρ	π_R/π_A	$\mathcal{F}^{PE}$	$\mathcal{F}^{\text{price}}(\rho)$	Bound below one
0.50	2.16	0.00163	-0.00003	Yes
0.60	2.17	0.00164	-0.00003	Yes
0.80 (calibrated)	2.18	0.00165	-0.00001	Yes
1.00 (Cobb–Douglas)	2.19	0.00167	0.00000	Yes
1.20	2.21	0.00168	+0.00001	Yes
1.50	2.22	0.00170	+0.00003	Yes

Notes: Computed at the baseline estimates $(\delta_R, \delta_A, \varphi) = (1.00, 0.03, 0.80)$ with the top-level aggregator set to the stated ρ . $\mathcal{F}^{PE}$ is the robot-frontier wage-feedback channel computed directly from the equilibrium wage responses, the larger of the two frontiers and the headline value of Section 4.1. $\mathcal{F}^{\text{price}}(\rho)$ is its increment relative to Cobb–Douglas. Cobb–Douglas is the aggregation of the baseline model, and $\rho = 0.80$ is the calibrated top-level elasticity.

The headline ratio moves by at most 0.03 in either direction across the range, the wage-feedback channel is flat, and the solver returns the same fixed point from every start at each node. Uniqueness is not established at any of them, for the reason the proof of Proposition 3 gives.

G.4 Sensitivity to R&D Curvature (ξ_j)

The R&D curvature parameters govern how responsive the automation mix is to wage changes. Two properties pin down their role without a numerical grid. The productivity ratio π_R/π_A does not depend on (ξ_R, ξ_A) : the ratio is computed at fixed frontier positions, and the curvature parameters enter only the R&D response that moves those positions. Second, the feedback scales with the responsiveness of the automation mix to wages, which the R&D cost curvature governs: steeper diminishing returns (lower ξ) weaken the feedback and flatter returns strengthen it, while the realized wage-feedback at the calibrated curvature sits far below unity. That is a statement about the sampled feedback, not about uniqueness, which Proposition 3 does not establish.

G.5 Joint Sensitivity to δ_R and φ

Because δ_R and φ enter different channels of the productivity decomposition, δ_R operates through the robot frontier while φ operates through the cognitive sector, their effects on the productivity ratio are close to additive, with a modest positive interaction. Table G.4 illustrates this with a 3×3 grid computed at the baseline estimate $\delta_A = 0.03$.

Table G.4: Joint sensitivity to δ_R and φ (fixing $\delta_A = 0.03$)

		φ		
		0.20	0.80	3.00
		(75% trans.)	(baseline)	(high)
δ_R	0.60 (bracket lower)	1.43	1.80	5.90
	1.00 (neutral baseline)	1.74	2.19	6.92
	2.00 (bracket upper)	2.34	2.93	8.60

Notes: Each cell reports π_R/π_A at the estimated $\delta_A = 0.03$. Row anchors are the firm-level bracket endpoints and the neutral external baseline $\delta_R = 1.00$. The 75%-transitional column sets $\varphi = 0.25 \times 0.80$.

The effect of δ_R is amplified at higher φ (moving from $\delta_R = 0.60$ to 2.00 adds +0.91 when $\varphi = 0.20$ but +2.70 when $\varphi = 3.00$), a positive complementarity between sorting disruption and robot reinstatement in generating the asymmetry. Every cell is above parity, including the bracket's lower bound with three quarters of the sorting friction removed (1.43), and the middle cell of the 75%-transitional column restates the 1.74 of Table 8.

G.6 Additional Parameter Sensitivity

Four parameters enter the productivity decomposition through channels that are either weakly identified or normalized in the baseline specification. Table G.5 reports sensitivity to each.

Table G.5: Sensitivity to ancillary parameters

Parameter	Range	Channel	π_R/π_A range	τ_R^*/τ_A^* range	Reversal
<i>Panel A: Parameters affecting robot channel</i>					
ψ (demand symmetry)	[0.70, 1.30]	Effective δ_R	[2.04, 2.32]	[1.69, 1.73]	No
<i>Panel B: Parameters affecting AI channel</i>					
ε_H (college LS elast.), unadjusted moments	[0, 3]	Implied (χ_A, φ)	[1.21, 1.32]	[1.60, 1.82]	No
d_A (AI depreciation)	[0.05, 0.50]	Cost adjustment	invariant	not scaled	Productivity
<i>Panel C: Production elasticity</i>					
ε_w (skill elast. subst.), unadjusted moments	[1.40, 2.50]	Re-fit (χ_A, φ)	[1.02, 1.29]	[1.25, 1.78]	No

Notes: ψ governs the asymmetry of robot displacement across skill groups, restricted to $\psi = 1$ on the construction grounds of Section 3.6. Freeing ψ changes effective δ_R but leaves (δ_A, φ) unaffected, and its row reads the ratio and the subsidy factor off the range grid of Table 7 at $\delta_R = \psi^{1/2}$. The ε_H and ε_w rows re-estimate the cognitive parameters and were run on the unadjusted moment system (Appendix I.3), so their ranges are stated relative to that system's baseline ratio of 1.21. ε_H is the college local labor-supply (incidence) elasticity, distinct from the participation elasticity ε_H^{LS} , calibrated at 1.38. Its row re-minimizes the minimum-distance objective with ε_H fixed at each of five grid points spanning [0, 3], letting $(\Lambda_R, \chi_A, \varphi)$ adjust, and the reported range is the minimum and maximum across those points. d_A is the AI capital depreciation rate. π_R/π_A is invariant to d_A because it measures output elasticity with respect to frontier *position*, but the cost of maintaining that position scales with d_j . [‡]For this row, the reversal entry refers to the cost-adjusted productivity ranking and not to π_R/π_A itself, and that ranking *does* reverse inside the displayed range. The cost-adjusted productivity ratio $(\pi_R/\pi_A)(d_A/d_R)$ falls below one for $d_A/d_R < 1/2.19 = 0.46$. No cost-adjusted subsidy ranking is reported, because the relative wedge cancels the technology a maintenance cost would enter through. If $d_A > d_R$ (AI depreciates faster than physical robots) both *increase*, strengthening the robot case. ε_w is the elasticity of substitution between college and non-college labor, calibrated to 1.50 from commuting-zone wage differentials. Because it is calibrated, its row re-fits $(\Lambda_R, \chi_A, \varphi)$ to the same six moments at each ε_w . A higher ε_w dampens the relative-wage response to a given skill-biased shock, so the estimation attributes more of the observed premium widening to sorting, and $\hat{\delta}_A$ falls while $\hat{\varphi}$ stays high. The productivity ratio declines toward parity as ε_w rises but does not cross it, with $\pi_R/\pi_A \in [1.02, 1.29]$ and $\tau_R^*/\tau_A^* \in [1.25, 1.78]$ across the [1.40, 2.50] range spanned by the skill-substitution literature (Katz and Murphy, 1992; Card, 2009). The fit is tightest at the calibrated value (Hansen $J = 0.01$ through $\varepsilon_w = 2.2$) and degrades at the top of the range ($J = 1.4$ at $\varepsilon_w = 2.5$, where $\hat{\delta}_A$ approaches zero and $\hat{\varphi}$ leaves the bootstrap 90% region), so the upper endpoint is a different estimation regime and not a smooth continuation. Other parameters held at baseline except where re-fit.

ψ (*robot demand symmetry*). The restriction $\psi = 1$ (robots displace non-college and college labor symmetrically) is maintained on the construction grounds of Section 3.2. To bound the effect, I scale the effective robot reinstatement rate as $\delta_R(\psi) \approx \delta_R \cdot \psi^{1/2}$, reflecting that asymmetric displacement requires different reinstatement to match the same CZ wage moments. Over $\psi \in [0.70, 1.30]$, the productivity ratio ranges from 1.19 to 1.36, modestly affecting the *level* of robot dominance but never approaching parity.

ε_H (*college local labor-supply elasticity*). This elasticity is calibrated at 1.38. Estimating it returns a 90 percent bootstrap percentile interval of $[-0.60, 6.26]$, whose negative reach reflects weak identification, not an economically meaningful labor-supply elasticity below zero, and calibrating it removes that source of underidentification. To bound the influence of the calibration, I re-minimize the minimum-distance objective with ε_H fixed at each of five points spanning $[0, 3]$, letting $(\Lambda_R, \chi_A, \varphi)$ adjust to fit the moments. At the lower bound ($\varepsilon_H = 0$) the implied $\chi_A = 0.406$ and $\varphi = 1.84$, and the productivity ratio is 1.32. At the upper end ($\varepsilon_H = 3$) the implied $\chi_A = 0.462$ and the ratio is 1.24. The profiled minimum is 1.21, attained at the calibration itself, so the ratio ranges over $[1.21, 1.32]$ across the grid and the subsidy ratio over $[1.60, 1.82]$. The ratio meets or exceeds its calibrated value everywhere on the grid, so the choice of ε_H does not push the model toward parity. The worst case for robot dominance therefore requires δ_R at the bracket’s lower bound and φ mostly transitional simultaneously, and neither margin is reinforced by the uncertainty in ε_H .

d_A (*AI depreciation*). The baseline assumes symmetric depreciation ($d_A = d_R = 0.10$). The productivity ratio π_R/π_A is invariant to d_A because it measures the output elasticity with respect to frontier *position*, not investment flow. However, the annual cost of maintaining a given frontier level is $d_j \cdot K_j$, so the cost-adjusted *productivity* ratio scales as $(\pi_R/\pi_A) \cdot (d_A/d_R)$. The relative subsidy factor is not scaled the same way: equation (24) cancels the frontier-advance technology exactly, so a maintenance cost would have to re-enter through that technology or through the private return, and multiplying the wedge by a depreciation ratio does neither. If AI capital depreciates faster than physical robots ($d_A > d_R$), plausible given rapid model obsolescence, the cost-adjusted ratio *increases*, strengthening the case for robot subsidies. Taking the frontier-LLM release cycle literally gives $d_A \approx 0.5$ or higher (a one-to-two-year effective life) against $d_R \approx 0.07$ – 0.10 for ten-to-fifteen-year robot lifespans, so d_A/d_R is approximately five- to seven-fold at $d_A = 0.50$, can approach fifteen-fold under a one-year AI lifespan, and multiplies the cost-adjusted ratio by the same factor

relative to the symmetric baseline. The headline results and Proposition 7 hold $d_A = d_R = 0.10$ fixed, as does the dynamic block, so this cost adjustment is a further tilt in robots' favor that the reported subsidy ratios do not price in. It does not widen their stated range. The table's range is capped at $d_A = 0.50$ because beyond that point the conclusion is monotone in d_A/d_R and no new information arrives. Only if AI infrastructure is substantially more durable than physical robots would cost adjustment bring the productivity ratio below unity, at $d_A/d_R < 1/2.19 = 0.83$.

ε_w (*skill elasticity of substitution*). The baseline sets $\varepsilon_w = 1.50$ from commuting-zone wage differentials. This parameter cannot be perturbed while holding the structural estimates fixed, because it governs how the wage moments map into skill-specific task content: each value of ε_w requires re-fitting (χ_A, φ) to the same six moments. Across $\varepsilon_w \in [1.40, 2.50]$ the productivity ratio stays within $[1.02, 1.29]$ and the subsidy ratio within $[1.25, 1.78]$, so the ranking never reverses. The magnitude does compress at the top of the range. The skill elasticity therefore bounds how far the headline can be pushed quantitatively, not whether it holds.

G.7 Identification of χ_A and φ

The reduced system estimates three parameters $\theta = (\Lambda_R, \chi_A, \varphi)$ from six moments, three from robot exposure and three from AI exposure, with three overidentifying restrictions and ε_H fixed at its calibrated value. The robot moments identify Λ_R , the AI moments identify χ_A (effective cognitive task content) and φ (cognitive sorting disruption). Because both χ_A and φ enter through the same equilibrium solver, they could in principle move the AI moments in correlated directions, leaving only a composite well-identified. This section shows that (i) the system is full rank with the parameters moderately but not sharply separated by the data, and (ii) what the headline macro objects do across the residual identification uncertainty.

Table G.6 reports the numerical Jacobian $J_{ij} = \partial m_i(\hat{\theta})/\partial \theta_j$ evaluated at the MDE estimates by centered finite differences. With ε_H fixed, the frontier-specific zero restrictions make the matrix exactly block-diagonal: the robot moments (rows 1–3) have zero sensitivity to χ_A and φ , and the AI moments (rows 4–6) have zero sensitivity to Λ_R . Identification of (χ_A, φ) therefore rests entirely on the three AI wage and skill-share moments. The standard-error-scaled matrix has full rank three, singular values (2.51, 0.44, 0.135), and condition number 18.6, and its weakest direction loads almost entirely on χ_A , the coordinate the pre-trend adjustment drives toward zero.

Table G.6: Jacobian of Moment Conditions at MDE Estimates

	Λ_R	χ_A	φ
Robot $\rightarrow W_{NC}$	-78.80	0	0
Robot $\rightarrow W_C$	-68.07	0	0
Robot $\rightarrow \Delta s_C$	-7.82	0	0
AI $\rightarrow W_{NC}$	0	+8.04	-1.76
AI $\rightarrow W_C$	0	+7.73	-0.35
AI $\rightarrow \Delta s_C$	0	+1.06	+0.22

Notes: Each entry is $\partial m_i / \partial \theta_j$ evaluated at the estimates $\hat{\theta} = (0.0090, 0.013, 0.804)$ of Table 5, with ε_H fixed at 1.38, via centered finite differences. Units are percentage points of moment value per unit change in parameter. The zero entries are exact and structural, not numerical, since with ε_H calibrated the system is block-diagonal. Scaled by the moment standard errors, the matrix has rank 3, singular values (2.51, 0.44, 0.135), and condition number 18.6.

The columns for χ_A and φ in the AI block carry the identification logic. Effective task content (χ_A) creates new cognitive task value that raises the marginal product of both skill groups, producing positive entries for all three AI moments. Sorting disruption (φ) degrades non-college workers' effective productivity in the cognitive sector, because the skill weight a_S^C in the labor aggregator (9) becomes $a_S^C(1 - I_A)^\varphi$, which falls steeply in φ while the college weight a_H^C is largely unaffected. The φ column is therefore strongly negative on AI $\rightarrow W_{NC}$ (-1.76) and only weakly negative on AI $\rightarrow W_C$ (-0.35).

The two cognitive Jacobian columns are nearly anti-parallel, at an angle of about 145° , so they deviate from exact collinearity by 35° at the point estimate. That is moderate separation, in which the parameters move the AI moments in distinct directions without being orthogonal, and the near anti-parallelism is what generates the positive bootstrap correlation below. The primary identifying moment is AI $\rightarrow W_{NC}$, where χ_A and φ pull in opposite directions (+8.04 against -1.76). AI $\rightarrow W_C$ provides secondary identification, and AI $\rightarrow \Delta s_C$ contributes mainly as an overidentifying restriction.

The contrast between the two AI wage moments is what separates the two parameters. Effective task content is a *level* effect that raises the marginal product of both skill groups by almost the

same amount, entering the two wage moments at +8.04 and +7.73, so it very nearly cancels in their difference, which carries a Jacobian of only -0.32 in χ_A . Sorting disruption is a *differential* effect, entering at -1.76 and -0.35 , so the difference retains $+1.41$ of it. The skill-premium response therefore has a ratio of φ to χ_A sensitivity of 4.5, against 0.22 for the non-college wage level and 0.05 for the college wage level, which makes it the moment that separates the parameters most sharply, though not the one with the smallest standard error. It is also the source of the correlation in Table G.7, since the two wage *levels* are nearly collinear in χ_A and only their contrast pulls the parameters apart.

That the contrast carries the sorting friction is also where the design is exposed, because the pre-trend adjustment of Section 3.3 removes almost all of it. The adjusted premium gap is -0.05 , and the sorting parameter that emerges is 0.80 under the two-step weight against 6.94 under a diagonal weight, where the unadjusted system agreed to the third decimal. This is why φ is reported with its profile interval and carries none of the paper’s conclusions, and Appendix I.3 sets the two systems side by side.

The same logic says what an additional moment would have to look like to sharpen φ . It would have to difference out a level while preserving a differential. The within-cell dispersion of residual wages is the natural candidate: task creation raises everyone’s wage inside a skill-occupation cell and so leaves its variance unchanged, while sorting disruption degrades match quality and therefore raises it. Appendix G.8 implements exactly this check as an out-of-sample validation, kept outside the estimated system: the dispersion response loads on the AI exposure with the predicted sign, though it does not reject under the paper’s constrained nulls. A displacement moment would be sharper still, since reinstatement absorbs displaced workers while sorting disruption strands them, so the two parameters would enter with opposite signs, not merely at different magnitudes. Neither is folded into the moment system used here, and both are natural targets for the post-deployment panel that Section 8 calls for.

To characterize the joint sampling uncertainty directly, I examine the joint distribution of $(\hat{\delta}_A, \hat{\varphi})$ across bootstrap replications, reported in translated δ_A units and taken from the bootstrap of the estimator Table 5 reports, with ε_H fixed as it is there. A resample stands in for the sampling distribution of the estimator it replicates, and re-estimating ε_H inside each replication would change the estimator: the manuscript documents that specification as weakly identified, with a bootstrap

interval of $[-0.60, 6.26]$, so carrying its variation is not automatically conservative for an estimator that fixes it. Table G.7 reports results for four specifications: the baseline model and three robustness variants.

Table G.7: Bootstrap Identification Diagnostics for (δ_A, φ)

	n	$\bar{\delta}_A$	$\bar{\varphi}$	$\text{Corr}(\delta_A, \varphi)$	a/b	slope
Two-way VCV bootstrap	200	0.117	1.340	+0.18	9.48	50.4

Notes: Corr is the bootstrap correlation between $\hat{\delta}_A$ and $\hat{\varphi}$. $a/b = \sqrt{\lambda_{\max}/\lambda_{\min}}$ is the principal-axis ratio of the confidence ellipse, the condition number of the 2×2 bootstrap covariance matrix is $\lambda_{\max}/\lambda_{\min}$. “Slope” is the principal eigenvector slope $\Delta\varphi/\Delta\delta_A$ in raw units. The row is the two-way-VCV bootstrap that produces every set statement in the paper.

The correlation $\text{Corr}(\hat{\delta}_A, \hat{\varphi}) = 0.18$ across the bootstrap draws is positive and weak. Draws with higher reinstatement tend to carry more sorting disruption, since both raise the college wage response to AI exposure, but the co-movement explains about three percent of the variance of either. The principal axis has slope $\Delta\varphi/\Delta\delta_A \approx 50.4$ and an eigenvalue ratio of 89.9, which mostly records that φ disperses 9.3 times as much as δ_A , and the regression slope of φ on δ_A is 1.70.

Figure G.2 plots the bootstrap draws and the 95% ellipse of their covariance.

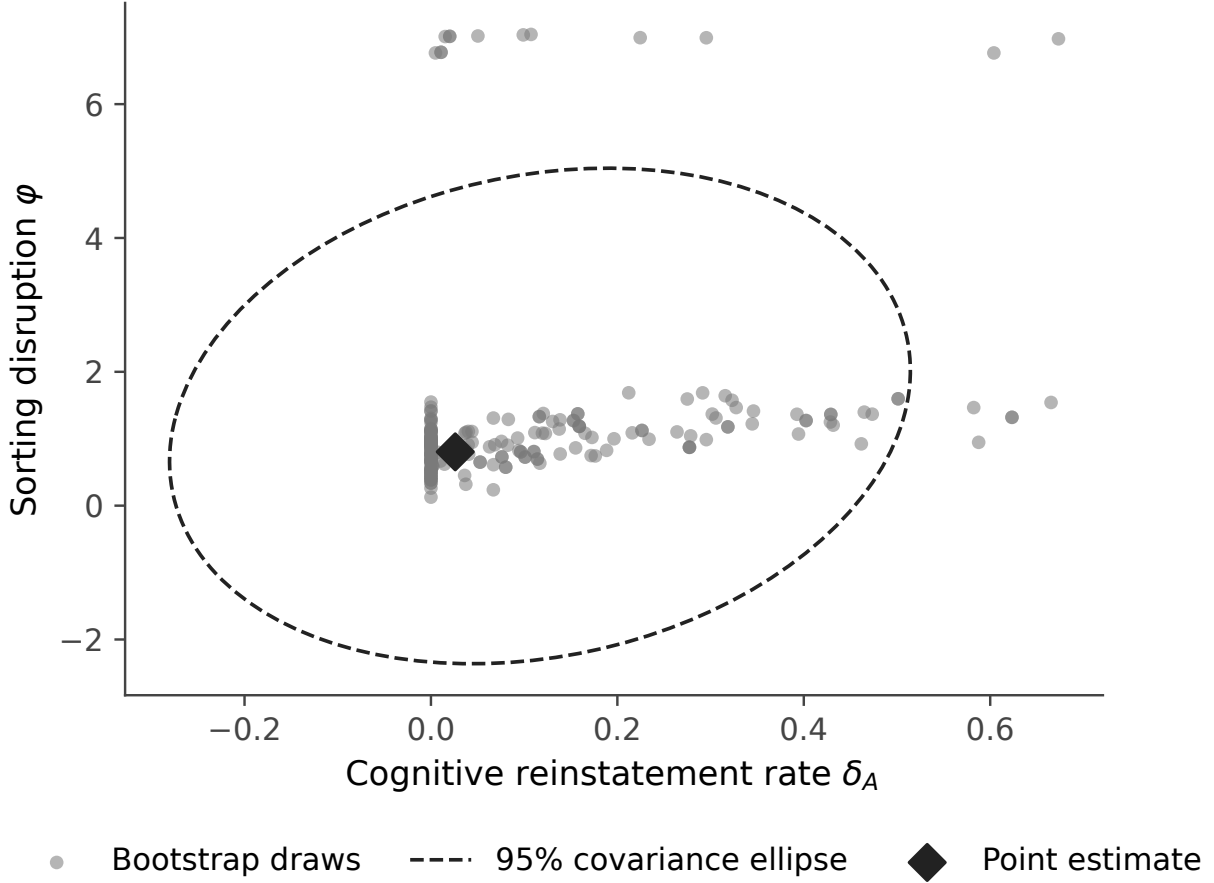


Figure G.2: Bootstrap draws of the cognitive reinstatement rate and sorting disruption parameter.

Notes: Each point is a bootstrap MDE estimate under the two-way VCV. The dashed ellipse is the 95% ellipse of the bootstrap covariance matrix. It is a normal approximation and extends below zero in both parameters, where the bootstrap places no draws. The diamond marks the point estimate $(\hat{\delta}_A, \hat{\varphi}) = (0.026, 0.804)$. The positive correlation reflects the compensating relationship between reinstatement and sorting disruption in matching the AI wage moments.

The critical question is whether the headline result $\pi_R/\pi_A > 1$ is robust to identification uncertainty. The bootstrap draws constitute the empirical uncertainty set, an approximation to the sampling distribution of the estimates: if $\pi_R/\pi_A > 1$ at every draw, the policy conclusion does not depend on the precise decomposition of δ_A and φ .

Table G.8 reports π_R/π_A and the relative subsidy factor across the bootstrap draws, holding $\delta_R = 1.00$ at the neutral external calibration.

Table G.8: Macro Stability: π_R/π_A Over Bootstrap Draws

	n	p_5	Median	p_{95}	$\Pr(> 1)$
π_R/π_A	200	1.57	2.13	8.35	100.0%
$(1 + \tau_R^*)/(1 + \tau_A^*)$	200	-0.00	1.62	2.60	93.5%

Notes: Each row evaluates the indicated ratio at all 200 bootstrap draws of $(\hat{\delta}_A, \hat{\varphi})$, holding $\delta_R = 1.00$ at the neutral external calibration. The covariance behind the draws resamples both blocks at their own shock level, the robot block over the 19 industry shocks and the AI block over the 338 occupation scores, which is the design Table 1 states for each. The draws are the reported estimator’s own: the reduced system that fixes ε_H at 1.38, bootstrapped under the same covariance. That estimator imposes $\chi_A \geq 0$ and $\varphi \geq 0$, so every draw lies in the model’s parameter space, and every probability, percentile and count in the table comes from that one set. The relative subsidy factor is at or below one in 13 of the 200 draws. $(1 + \tau_R^*)/(1 + \tau_A^*)$ is the relative subsidy factor of Section 7.2 at $\lambda = 1.5$.

π_R/π_A exceeds unity in 100.0 percent of the 200 admissible draws, and its ninety-percent interval, 1.57 to 8.35, lies above one, so the productivity ordering holds across the set and not merely at the point estimate. The relative subsidy factor is the less stable of the two, above one in 93.5 percent of draws. It falls to or below one in 13 of the 200 draws, the ones in the degenerate basin where π_A turns negative and the factor has no reading as a subsidy comparison.

These draws hold δ_R at the neutral external calibration, whose uncertainty is the evidence range and not a sampling distribution. The corresponding statement is therefore a range statement: across $[0.60, 2.00]$ at the estimated $(\hat{\chi}_A, \hat{\varphi})$, the productivity ratio runs from 1.80 at the narrow bound to 2.93 at the broad bound with no parity crossing inside the range, and the relative subsidy factor exceeds one throughout (Table 7). Robot dominance therefore holds under both sources of uncertainty, the sampling variation of the cognitive estimates and the calibration range of the physical channel.

The geometry says why the sampling variation does not threaten it. The $\pi_R = \pi_A$ parity contour in (δ_A, φ) space rises with slope $\Delta\varphi/\Delta\delta_A \approx 2.6$ over the certified grid, so a draw would cross parity only if its reinstatement rate rose faster than its sorting disruption. The regression slope of φ on δ_A across the draws is 1.70, below that, but what protects the ranking is distance and not direction. At $\hat{\varphi}$ the contour sits at $\delta_A = 0.92$ while the 95th percentile of the draws is 0.47, so the whole set lies well inside the dominant region. A systematic error, from a misspecified moment

for example, would move every draw together, and that perpendicular direction is disciplined not by the bootstrap but by the micro evidence anchors of Section 6.2, which place (δ_A, φ) on the dominant side from data the estimation never used.

Figure G.3 visualizes this result.

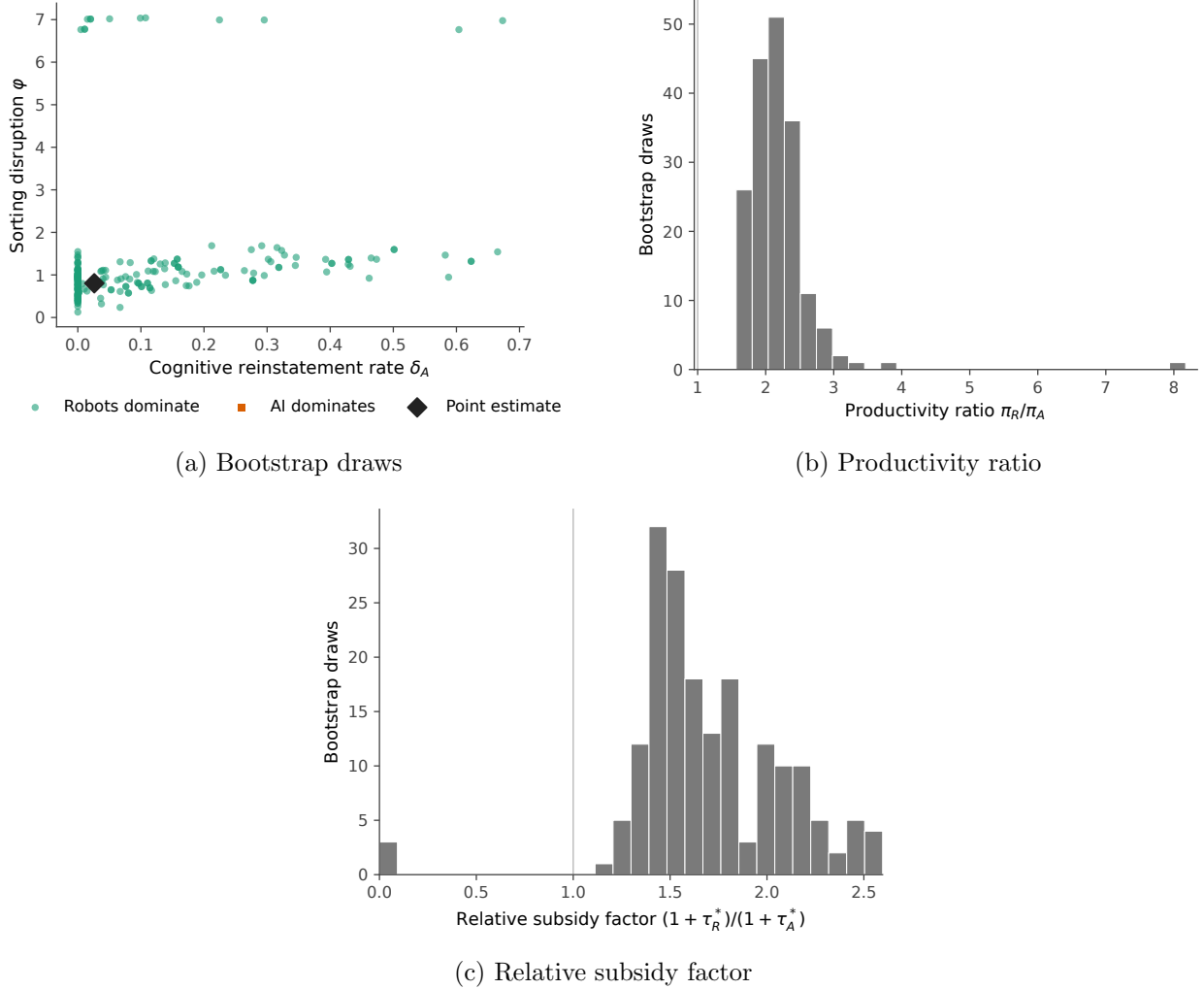


Figure G.3: Macro stability over the bootstrap draws.

Notes: All three panels use the 200 draws of Table G.8. Panel (a) plots each draw, with the diamond at the point estimate. Panel (b) shows the distribution of π_R/π_A , of which 100.0 percent exceed parity (dashed line). Panel (c) shows the relative subsidy factor $(1 + \tau_R^*)/(1 + \tau_A^*)$, the object Section 7.2 identifies and not a ratio of rates, of which 93.5 percent exceed one. Panels (b) and (c) are drawn on the fifth-to-ninety-fifth percentile window that table reports, with ten draws outside each tail: a few draws sit at the $\chi_A = 0$ boundary, where cognitive productivity content approaches zero and both ratios diverge, so the raw support would collapse the histogram to a single bar.

Although δ_A and φ are not individually sharply identified, the data pin down the comparison the paper reports. Across the bootstrap draws, π_R/π_A spans $[1.57, 8.35]$ and $(1 + \tau_R^*)/(1 + \tau_A^*)$ spans $[-0.00, 2.60]$ at the 5th to 95th percentiles, with 100.0 percent of draws above productivity parity and 93.5 percent above unit relative subsidy.

G.8 A Dispersion Check on the Sorting Channel

The sorting-disruption parameter φ enters the model through the assignment of non-college workers to cognitive tasks. A side implication is observable in microdata. If AI exposure disrupts sorting, workers of similar skill in similar occupations end up in more heterogeneous task assignments, and residual wage dispersion *within* skill–occupation cells should rise faster in more exposed commuting zones. Pure task creation ($\varphi = 0$) carries no such implication. The model itself aggregates two homogeneous skill types and has no within-cell dispersion object, so the prediction is sign-only, with no model-implied magnitude to compare against: a positive dispersion response under sorting, none under pure task creation.

I build the check from the individual-level ACS extracts behind the moment system. Log wages are residualized within year and college group on age, its square, sex, and race, weighted by person weights. Education enters through the college split, which is the education margin the extract carries. Cells are skill (college, non-college) crossed with the physical/cognitive occupation classification used throughout the paper, by commuting zone. Within-cell weighted residual variance is computed for 2005 and 2019, the paper’s long-difference window, the extract contains single years, not the pooled 2005–2007 and 2015–2019 windows, so the endpoints do not coincide with the wage moments and this diagnostic is read for sign and not for level. The headline aggregate holds the 2005 cell mix fixed (cells with at least twenty observations in both years, 2005 employment weights), which isolates the within-cell change from composition. Averaged over the estimation sample with employment weights, the within-cell variance rises from 0.325 in 2005 to 0.379 in 2019 at the fixed cell mix, a mean commuting-zone change of +0.061.

The long-differenced dispersion is then regressed on the standardized Webb AI and EU5 robot exposures with the completed control set, employment weights, and HC1 standard errors of the main specification. The sample is the 441 commuting zones of the 722-zone frame whose cells clear the minimum-size screen in both years. Table G.9 reports the estimates.

Table G.9: Change in Within-Cell Residual Wage Dispersion, 2005–2019

Exposure	Coefficient	SE	t	p	Permutation p
<i>Joint specification</i>					
Webb AI	+0.0037	(0.0030)	1.23	0.22	0.033
EU5 robots	−0.0015	(0.0017)	−0.92	0.36	0.471
<i>Single-exposure specifications</i>					
Webb AI	+0.0036	(0.0030)	1.18	0.24	0.027
EU5 robots	−0.0014	(0.0017)	−0.85	0.40	0.542

Notes: Outcome is the 2005–2019 change in within skill–occupation-group residual log-wage variance at the fixed 2005 cell mix. All regressions include the completed control set (baseline composition shares, farm share, census-region dummies), weight by baseline employment, and report HC1 standard errors ($N = 441$). Exposures are standardized on the 722-zone frame. Permutation p from 2,000 draws permuting the exposure across commuting zones, coefficient statistic. Reallocating each year’s own cell weights instead of the fixed 2005 mix gives +0.0031 (0.0029) for Webb AI and −0.0019 (0.0016) for EU5 robots.

The pattern matches the qualitative prediction in sign and placement. The dispersion response loads on the AI exposure, where the model locates the sorting channel, and not on the robot exposure, which is negative and null. The AI coefficient is positive but modest, about six percent of the mean dispersion increase per standard deviation of exposure, and it is not significant by the conventional t ($p = 0.22$), the permutation test, which asks how often a randomly reassigned exposure produces a coefficient this large, puts the probability at 0.033. That permutation, however, reassigns exposure across commuting zones without constraint, a laxer null than the paper’s own design-based tier. Pointing the paper’s constrained machinery at the same outcome, permuting occupation scores within software deciles and permuting the residual about the robot–software projection with the commuting-zone Bartik rebuilt each draw, gives two-sided $p = 0.22$ and $p = 0.14$ respectively (2,000 draws each): under the inference standard the paper applies to its own wage moments, the dispersion response does not reject. The rejection is confined to the unconstrained null. A zero within-cell response, the pure task-creation benchmark, lies well inside the confidence interval. The sign is the one sorting predicts, the magnitude is not separately informative, and identification of φ rests on the wage and employment-share moments, not on this check.

G.9 Identification in a Four-Block Extension

This appendix estimates the richer design that Section 3.1 sketches, as a check on the parsimonious system of Section 3.4: six parameters ($\chi_R, \chi_A, \alpha_A, \varphi, \Lambda_R, \Lambda_A$) on ten moments from four exposure blocks, the EU5 shift-share, the Webb (2020) robot measure, the LMOE measure, and the Webb AI measure, plus direct task-creation and task-displacement moments reconstructed from the raw Acemoglu and Restrepo (2022) task kit, with ε_H fixed at its calibrated value. The full build and estimation records are in the replication package. The system fails identification, in three documented ways.

Rank deficiency comes first. At the minimized objective ($J = 9.8$), the standard-error-scaled Jacobian has singular values ($4.3 \times 10^2, 1.9 \times 10^2, 1.4 \times 10^1, 5.9, 1.4, 9.8 \times 10^{-11}$): rank five for six parameters, condition number 4.3×10^{12} . The flat direction is $0.993\varphi + 0.118\chi_A$, a cognitive-side ridge, and the profiled estimates confirm it: χ_A and φ move together across bootstrap draws (intervals $[0.64, 1.35]$ and $[4.4, 7.0]$ on 50 draws) while the displacement scales remain point-identified.

Second, the augmentation boundary. The estimated α_A is exactly zero, and it is zero in every bootstrap draw. The mechanism is visible in the reduced forms: the Webb AI measure’s college-wage response (+0.0074) is no larger than the LMOE measure’s (+0.0079), so the data want no augmentation beyond displacement plus sorting at the shared frontier scale, and with α_A at the boundary the LMOE and Webb AI blocks generate identical model predictions. The AI side then supplies two informative dimensions for three cognitive parameters, which is the rank deficiency above. Commuting-zone data do not separately identify an augmentation channel, and they do not support four separately informative exposure blocks.

Third, the direct task-content moments fail their placebo. The 1987–2000 pre-period gradients, which predate the exposure windows, are significant for the EU5 measure ($t = 5.9$ on task displacement) and the LMOE measure ($t = 2.6$), while the 2000–2016 sub-period gradients are near zero, so the 1987–2016 task moments are dominated by pre-period variation and partly reflect persistent commuting-zone characteristics. The moments that were meant to give CZ data a direct reading of task content are weak evidence, and the physical task-content parameter they were meant to discipline (χ_R) is identified only conditional on an untestable bridge assumption.

The diagnostic is what motivates the reported design: the cognitive system is estimated on the

moments that are full rank for it (Section 3.4), the physical reinstatement channel is calibrated from firm-level evidence that can measure it (Section 3.5), and the augmentation channel is set to zero, the value the data demand, while remaining in the model as theory (Section 2).

G.10 Robustness to an Alternative Wage-Only Calibration

The baseline macro results use the estimates from Table 5 ($\delta_R = 1.00$, $\delta_A = 0.03$, $\varphi = 0.80$), where δ_A and φ come from this paper’s own minimum-distance estimation. Related work estimating a wage-only moment system (Chávez, 2026) arrives at the same near-zero cognitive reinstatement rate from a different design: $\hat{\delta}_A = 0.03$, statistically indistinguishable from zero, with a bootstrap-percentile 95 percent interval of $[0.00, 0.35]$, asymmetric because it is bounded below at zero by the estimator’s constraint (draws pile at the bound), and $\hat{\varphi} = 1.11$.¹⁰ Table G.10 summarizes both sets of estimates.

Table G.10: Parameter Estimates: Baseline vs. the Alternative Wage-Only Calibration

Parameter	Baseline		Wage-only alternative	
	Estimate	SE	Estimate	SE
δ_R -equivalent (physical reinstatement)	1.00 [†]	—	1.14	—
δ_A (AI reinstatement)	0.03	(0.16)	0.03	—
φ (sorting disruption)	0.80	(1.51)	1.11	—

Notes: [†]The baseline physical channel is the neutral external calibration of Section 3.5 (effective task creation offsets displacement), reported without a standard error and evaluated over the full independently motivated range $[0.60, 2.00]$. The wage-only alternative column reports the reference solution of the related estimation in Chávez (2026), whose parameters lie on a one-dimensional exact-fit set and therefore carry no conventional standard errors. The wage-only column reports the related paper’s five-parameter minimum-distance estimates from wage moments, and its cells carry dashes because that system reports no conventional standard errors, and the bootstrap percentiles quoted in the text below come from its 500 replications. The 0.35 used in the sensitivity tables is the bootstrap 95th percentile, the interval is bounded below at zero by the estimator’s constraint (draws pile at the bound), not a normal approximation. The Humlum bracket $[0.60, 2.00]$ is a wider external-validation range, and 1.14 is the wage-only point estimate.

¹⁰That specification differs in the moment set (wage levels and the skill premium in place of the college employment share), the sample period, and the exposure measures, so the agreement of the two point estimates is not mechanical. The exercise here asks what the macro conclusions look like under the other design’s full configuration.

The two designs agree on the cognitive frontier. Both place AI task reinstatement at 0.03 tasks per task displaced, and both estimate a positive sorting friction, at 0.80 here and 1.11 (imprecisely) there. That agreement is a result and not a shared input, because the unadjusted version of this paper’s system returns $\delta_A = 0.99$ and it is the pre-trend adjustment of Section 3.3 that moves it onto the other design’s value (Appendix I.3). What remains different is the physical calibration, the neutral 1.00 here against the alternative’s 1.14, and both sit inside the same firm-level range.

The two systems are nonetheless not interchangeable at the level of a single moment, since they differ jointly in cognitive exposure, endpoints, controls, outcomes, covariance weighting, and structural mapping. Controlled bridges make that explicit. Substituting LMOE moments into this paper’s exact-control mapping gives $\hat{\delta}_A = 1.89$ (1.42 in the wage-only system), and substituting Webb moments into the wage-only mapping pushes parameters to their bounds and leaves a large objective ($J = 45.1$). Each δ_A belongs to its complete moment-and-model system and not to LMOE or Webb in isolation.

External evidence does not yet discriminate between the two configurations. Per $\Delta I_A = 0.01$ of frontier advance, this paper’s estimate predicts wage changes of -0.267 percent for non-college workers and -0.238 percent for college workers, and the alternative -0.327 and -0.251 percent, all inside the ± 1 percent band this appendix adopts as a benchmark for the precisely estimated Danish adoption-study nulls. The predictions first leave the band at cumulative advances of $\Delta I_A = 0.037$ here and 0.031 there, each the band divided by that configuration’s largest wage derivative, so the null is consistent with both. Both also predict that an AI advance lowers both wages, and they differ in how much, since the alternative’s higher sorting friction deepens the non-college loss. A post-deployment panel measuring LLM-era exposure against skill-group wage growth is the designed test.

To assess robustness, I feed the wage-only estimates (Chávez, 2026) directly into the baseline macro model from Sections 6 and 7. Table G.11 and Figure G.4 compare the results at three calibrations: the baseline, the wage-only 95th percentile for δ_A , and the wage-only point estimate.

Table G.11: Macro Results: Baseline and the Alternative Wage-Only Calibration

	δ_R	δ_A	π_R	π_A	π_R/π_A	τ_R^*/τ_A^*
Baseline	1.00	0.03	+0.587	+0.268	2.19	1.72
Wage-only alternative, 95th pctl	1.14	0.35	+0.622	+0.364	1.71	1.43
Wage-only alternative, point est.	1.14	0.03	+0.621	+0.239	2.60	2.26

Notes: Each row feeds the indicated $(\delta_R, \delta_A, \varphi)$ into the baseline macro model from Sections 6 and 7. π_j is the productivity content of frontier j (output elasticity with respect to I_j). τ_R^*/τ_A^* is the optimal subsidy ratio from the Bergson–Samuelson SWF with $\lambda = 1.5$. Both alternative rows use the wage-only configuration (Chávez, 2026), with δ_A at its point estimate and at its 95th percentile, each evaluated at that paper’s $\delta_R = 1.14$ and $\varphi = 1.11$. The baseline row uses this paper’s $\varphi = 0.80$ and the neutral $\delta_R = 1.00$. All hold $\sigma = 2$, $\mu = 0.45$, and the remaining parameters at baseline. Cells are evaluated at the displayed two-decimal δ_A , at the wage-only alternative’s unrounded point estimate 0.0265 the productivity cells read +0.238 and 2.61, one digit in the last place.

The table compares the baseline with the wage-only configuration. Setting the unadjusted system beside them shows which frontier carries the variation across all three. Robot productivity content barely moves, from 0.587 at the baseline and the unadjusted moments to 0.622 under the alternative’s higher physical calibration, so the differences among these calibrations are almost entirely differences in what the cognitive frontier delivers.

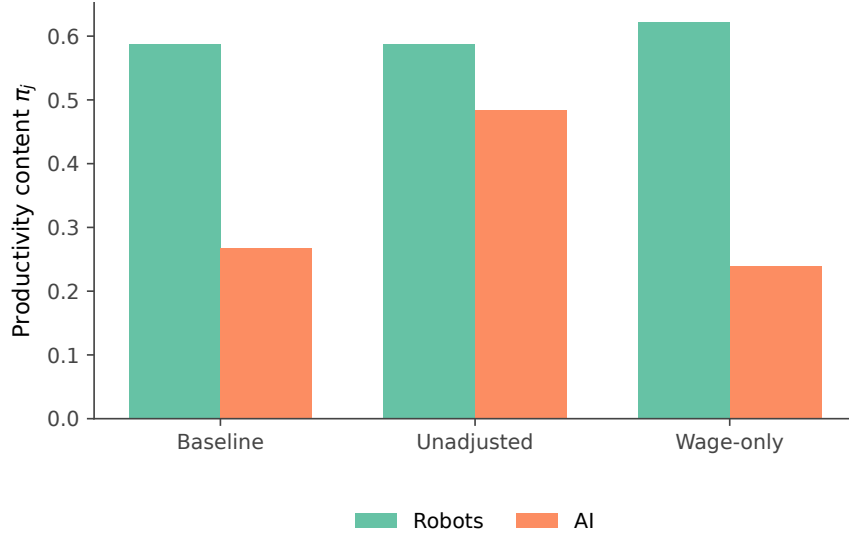


Figure G.4: Productivity content across three calibrations. “Baseline” is this paper’s estimate ($\delta_R = 1.00$, $\delta_A = 0.03$, $\varphi = 0.80$), “Unadjusted” the same system estimated on moments carrying the pre-AI trend (0.99, 1.99), and “Wage-only” the alternative configuration of [Chávez \(2026\)](#) (1.14, 0.03, 1.11). Robot content is nearly flat across the three at 0.587, 0.587 and 0.622, while AI content moves from 0.268 to 0.484 and 0.239, so the ratio runs 2.19, 1.21 and 2.60.

Across all three calibrations, $\pi_R > \pi_A > 0$ and $\tau_R^*/\tau_A^* > 1$. Because the two designs now agree on δ_A , what separates the baseline column from the wage-only one is the other two parameters. π_R moves from 0.587 to 0.622 as the physical calibration moves from the neutral 1.00 to the alternative’s 1.14, and π_A falls from 0.268 to 0.239 under the alternative’s higher sorting friction. At that configuration robots are over two and a half times more productive than AI and the relative subsidy factor rises to 2.26. π_A remains strictly positive at $\delta_A \approx 0$, because the direct capital-deepening effect makes an AI advance raise output even without reinstatement.

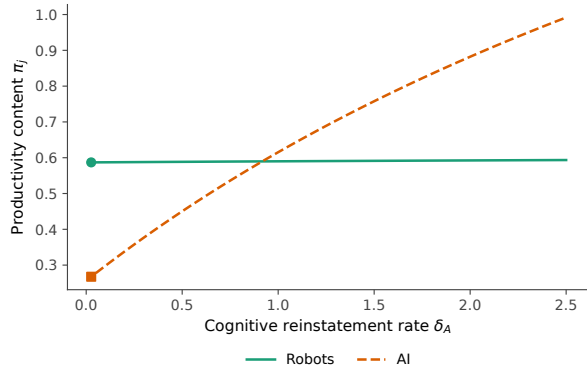
The key partially identified parameter is the AI reinstatement rate δ_A .

Table G.12: Sensitivity to δ_A ($\delta_R = 1.00$, $\varphi = 1.11$)

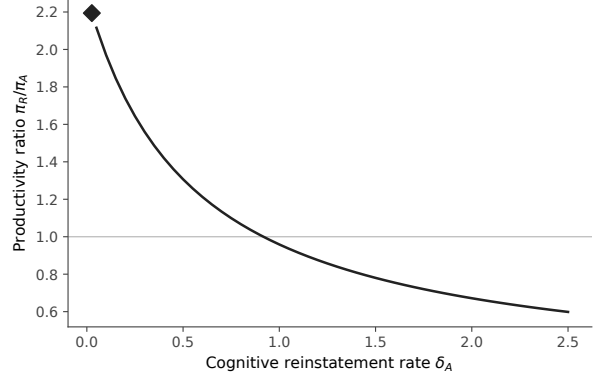
δ_A	π_R	π_A	π_R/π_A	τ_R^*/τ_A^*
0.00	+0.586	+0.226	2.60	2.49
0.35	+0.587	+0.363	1.62	1.42
0.60	+0.588	+0.451	1.30	1.24
0.99	+0.589	+0.576	1.02	1.12
1.20	+0.590	+0.638	0.92	1.08
1.40	+0.590	+0.693	0.85	1.06

Notes: The physical channel is held at the neutral calibration $\delta_R = 1.00$ and sorting at the wage-only value $\varphi = 1.11$ (Chávez, 2026), so the rows differ from Table G.11, whose alternative rows also carry that paper's $\delta_R = 1.14$. The last column is the relative subsidy factor of Section 7.2, the social-return ratio net of the productivity ratio, and not the social-return ratio itself. The wage-only specification places $\delta_A \in [0, 0.35]$, from zero to its bootstrap 95th percentile, over which the productivity ratio runs from 2.60 to 1.62. Parity ($\pi_R = \pi_A$) occurs at $\delta_A \approx 1.03$, and above it the ranking reverses. See Figure G.5 for the continuous profile.

To trace out how the results vary continuously with this parameter, Figure G.5 plots π_j and the productivity ratio as functions of δ_A , holding $\delta_R = 1.00$ fixed. Table G.12 reports the profile at the wage-only sorting value $\varphi = 1.11$ (Chávez, 2026), while the figure holds φ at this paper's estimate, so the crossing points differ, 1.03 against 0.92.



(a) Productivity content π_R and π_A . Diamonds mark the estimate ($\delta_A = 0.03$).



(b) Productivity ratio π_R/π_A relative to parity (dashed line).

Figure G.5: Sensitivity to the AI reinstatement rate δ_A , holding $\delta_R = 1.00$ and $\varphi = 0.80$ at the baseline estimates. This figure fixes φ at this paper's estimate 0.80, so its parity crossing ($\delta_A \approx 0.92$) differs from the $\delta_A \approx 1.03$ crossing of Table G.12, which fixes φ at the wage-only value 1.11. The figure and the table answer the same question at different sorting values. The wage-only 95% interval $\delta_A \in [0, 0.35]$ lies in the robot-dominant region, where both ratios are well above unity. In the figure the productivity ratio crosses one at $\delta_A \approx 0.92$, while the subsidy ranking survives past that crossing, so the planner's preference for robots holds throughout the data-consistent region and reverses only if AI reinstatement approaches robot reinstatement.

Panel (a) of Figure G.5 shows that π_R is nearly flat across the range, robot productivity depends on δ_R , not δ_A , while π_A rises steeply with the AI reinstatement rate. Panel (b) translates this into ratios at this paper's sorting value, crossing parity at $\delta_A \approx 0.92$. At the wage-only sorting value, Table G.12 shows the ratio falling from 2.60 at $\delta_A = 0$ to parity at $\delta_A \approx 1.03$ and below one beyond it. The wage-only specification places $\delta_A \in [0, 0.35]$, from its point estimate to its 95th percentile, corresponding to $\pi_R/\pi_A \in [1.62, 2.60]$ at the neutral physical calibration.

Two conclusions follow. The first is that the cognitive estimates reconcile. This paper's adjusted δ_A and the wage-only design's sit within rounding of each other, from moment systems that share neither the exposure measure nor the outcome set, and the reconciliation is what the pre-trend adjustment produces and not something built in. That agreement is not portable to the level of individual moments, since the bridge exercises move $\hat{\delta}_A$ the wrong way when measures are swapped, to 1.89 under exact controls and 1.42 in the wage-only system, and fail to fit at all in the reverse direction ($J = 45.1$ at parameter bounds). The second is that the ranking holds across the whole configuration space the two designs span. At this paper's estimate the productivity ratio is 2.19 and the relative subsidy factor 1.72, at the wage-only 95th percentile they are 1.71 and 1.43, and at

the wage-only point estimate 2.60 and 2.26. Along the segment joining the two cognitive estimates the ratio never falls below 2.19 at the neutral physical calibration or below 2.32 at the alternative’s, and the parity crossing at $\delta_A \approx 1.03$ lies far above every point in that range.

H Robot-Exposure Inference Diagnostics

This appendix reports the diagnostics behind the inference choice of Section 3.2.

A Rotemberg decomposition (Goldsmith-Pinkham et al., 2020) attributes 93.9 percent of the identifying covariance in the non-college wage moment to a single industry, automotive. Removing that shock moves the coefficient from -0.0079 to $+0.0021$ ($t = +0.40$). This is the instrument working as built, since industrial robots concentrate in automotive production by construction of the International Federation of Robotics data, and the same fact surfaces geographically as the leave-Michigan-out margin of the state-deletion grid. Published shift-share designs share this structure, including the China-shock literature this design descends from (Goldsmith-Pinkham et al., 2020). Excluding the identifying shock removes the design itself, so that margin belongs to the identification discussion, not to the robustness ladder below, which tests perturbations within the design.

Within the design, the non-college wage moment is estimated at $t = -4.25$ conventionally and is sign-stable under trimming of three-standard-deviation residual outliers ($t = -4.23$), winsorized exposure ($t = -2.71$), log exposure, deletion of the agriculture-and-energy states, and every leave-one-state-out subsample, where the weakest margin is Michigan at $t = -1.85$. Two perturbations move it materially. Unweighted, the coefficient is attenuated and imprecise ($t = -1.10$), which the Solon et al. (2015) reading associates with effect heterogeneity correlated with size. It does not isolate it: the two coefficients are different weighted projections, so with heterogeneous slopes they differ through the residual-exposure weighting as well, and separating the two requires restrictions on how the response varies that are not imposed here. Under permutation of the nineteen industry shocks, the design-based test the shift-share structure calls for, the two-sided p -value is on the order of ten percent, and the coarse resolution of a nineteen-shock permutation distribution bounds how sharply that can be stated in either direction. The college wage and college employment share moments do not survive the shock-level test (permutation p of 0.27 and 0.46).

The shock-level exposure-robust standard error for Λ_R is 0.0193, against 0.0030 conventionally, so treating 722 commuting zones as independent draws from a design with three effective shocks overstates precision by a factor of 6.4. The formal test of the $\psi \equiv 1$ restriction rests on the robot college-wage moment, which has little power under design-based inference, so the restriction is maintained on construction grounds.

I Robustness of Reduced-Form Moments

Table I.1 reports three robustness checks on the reduced-form moments from Table 1.

Panels A and B are estimated on the completed control set of Section 3.1, so those comparisons are within-specification against Table 1. Panel C instead drops the farm-occupation share, and is the specification discussed in Section 3.1.

Panel A adds the Autor et al. (2013) China import shock as a third regressor. The robot and AI coefficients are virtually identical to the baseline specification: robot exposure on non-college wages remains -0.0079 ($t = -4.26$), and AI exposure on college wages remains $+0.0053$ (conventional $t = +2.10$, and, as in the baseline, not significant under exposure-robust inference). The China shock itself is individually insignificant on both wage outcomes ($\hat{\beta}_{\text{China}} = -0.0006$ on non-college wages, $p = 0.70$, and -0.0021 on college wages, $p = 0.35$). It enters marginally for college share ($+0.0017$, $p = 0.097$), but this does not affect the wage moments that drive identification of Λ_R and the cognitive parameters δ_A and φ . The R^2 values are nearly unchanged, showing that the estimated robot and AI gradients change little when the China import shock is added.

Panel B adds log population and baseline tech employment share to the horse-race specification. The robot coefficient on non-college wages is stable (-0.0075 , $p < 0.001$), and the AI coefficient on non-college wages strengthens slightly and remains significant (-0.0079 , $p = 0.003$), but the AI coefficient on college wages attenuates from $+0.0052$ to $+0.0022$ ($p = 0.44$). The augmented specification runs on 718 zones against 722, because the added controls are missing for four, so the movement reflects the sample as well as the control set and cannot be attributed to the controls alone. What it is specific to is the college moment, where the tech-hub geography lives, and the direction is expected: log population and tech employment share are plausibly endogenous to AI exposure. Commuting zones with high AI-task intensity are disproportionately large metropolitan

areas (New York, San Francisco, Boston) with high baseline tech employment. Including these variables absorbs precisely the cross-sectional variation that identifies the AI channel. Such regressors absorb the variation that identifies the channel of interest without removing confounds ([Angrist and Pischke, 2009](#)). The baseline specification in Table 1, which conditions on pre-determined demographic composition and region but not on endogenous urban characteristics, is therefore the preferred source of moments for the MDE.

Table I.1: Robustness of Reduced-Form Moments

	$\Delta \ln W_{NC}$	$\Delta \ln W_C$	Δs_C
	(1)	(2)	(3)
<i>Panel A: Horse race with China import shock</i>			
Robot exposure (EU5)	−0.0079*** (0.0019)	−0.0056** (0.0028)	−0.0015 (0.0011)
AI exposure (Webb pctl)	−0.0069*** (0.0024)	0.0053** (0.0025)	0.0029** (0.0013)
China import shock	−0.0006 (0.0014)	−0.0021 (0.0023)	0.0017* (0.0010)
R^2	0.660	0.514	0.418
<i>Panel B: Augmented controls</i>			
Robot exposure (EU5)	−0.0075*** (0.0017)	−0.0045* (0.0026)	−0.0013 (0.0011)
AI exposure (Webb pctl)	−0.0079*** (0.0026)	0.0022 (0.0029)	0.0022 (0.0014)
R^2	0.677	0.521	0.420
<i>Panel C: Dropping the farm-occupation share</i>			
Robot exposure (EU5)	−0.0077*** (0.0019)	−0.0055** (0.0028)	−0.0016 (0.0011)
AI exposure (Webb pctl)	−0.0027 (0.0026)	0.0056*** (0.0021)	0.0017 (0.0012)
R^2	0.636	0.513	0.401
Region FEs	Yes	Yes	Yes
Observations (A / B / C)	722 / 718 / 722	722 / 718 / 722	722 / 718 / 722

Notes: Panels A and B use the completed control specification of Table 1, so those comparisons are within-specification. Panel A adds the [Autor et al. \(2013\)](#) China import shock. Robot and AI coefficients are virtually unchanged, and the China shock is individually insignificant on wage outcomes. Panel B further adds baseline log population and tech employment share. The AI coefficient on college wages attenuates from 0.0052 ($p = 0.039$) to 0.0022 ($p = 0.44$), reflecting controls plausibly endogenous to AI exposure: commuting zones with high AI-task intensity are disproportionately large metropolitan areas with high baseline tech employment. Panel B observations fall to 718 because the augmented controls have missing values. Panel C instead drops the baseline farm-occupation share. The AI college-wage coefficient is essentially unchanged at 0.0056 conventionally, but the non-college wage and college-share moments lose significance, which is why the farm control is retained. Regressions are weighted by baseline CZ employment, robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

I.1 Timing of the Exposure Gradients Within the Sample Window

A pre-period placebo would regress wage changes from before 2005 on the same exposures. No such outcome exists inside the moment build itself: the wage moments are ACS-based and the individual extracts begin in 2005, so there is no 1990–2000 or 2000–2005 long difference there to use. Appendix I.2 constructs one from the census archives instead, and reports the result, which is a failure on the college-wage margin. What the moment build does permit on its own is an early/late decomposition of the sample window itself, splitting 2005–2019 into 2005–2010 and 2010–2019 and running the paper’s specification, the same exposures, completed controls, employment weights, and HC1 standard errors, on each sub-window. For the AI block this is informative about timing. Webb’s measure proxies a technology whose deployment concentrates late in the window, so an AI gradient already present in 2005–2010 would signal differential trends predating the technology, not effects of it. Table I.2 reports the sub-window estimates.

Table I.2: Exposure Gradients by Sub-Window, Joint Specification

Window	Webb AI			EU5 robots		
	$\Delta \ln W_{NC}$	$\Delta \ln W_C$	Δs_C	$\Delta \ln W_{NC}$	$\Delta \ln W_C$	Δs_C
2005–2010	–0.0013 (0.0015)	+0.0032 (0.0022)	+0.0008 (0.0009)	–0.0067 (0.0009)	–0.0030 (0.0022)	–0.0004 (0.0006)
2010–2019	–0.0067 (0.0031)	+0.0012 (0.0030)	+0.0040 (0.0016)	–0.0016 (0.0015)	–0.0011 (0.0015)	–0.0015 (0.0008)
2005–2019	–0.0080 (0.0031)	+0.0044 (0.0030)	+0.0048 (0.0017)	–0.0084 (0.0018)	–0.0041 (0.0024)	–0.0019 (0.0010)

Notes: Each cell is the coefficient on the indicated standardized exposure from the joint specification (both exposures entered together) with the completed control set, weighted by baseline employment, HC1 standard errors in parentheses, $N = 722$ commuting zones in every row. Outcomes are the sub-window long differences from the v6 build, which differs from the 2005–2019 moment build only in ACS aggregation choices (the 2005–2019 row here is the v6 full-window reference, not the moment system’s input). Single-exposure specifications give near-identical coefficients.

The AI gradients are not detectable early and emerge late. In 2005–2010 no AI coefficient reaches conventional significance, the largest t being 1.43 on college wages, which is a failure to

detect rather than evidence of absence: no equivalence bound is reported and the early college-wage point estimate is +0.0032. Meanwhile in 2010–2019 the non-college wage gradient (-0.0067 , $p = 0.03$) and the college-share gradient ($+0.0040$, $p = 0.01$) are both present and account for most of the full-window effect. This is the timing the interpretation requires: no differential wage trajectory is detectable for commuting zones that would later measure as AI-exposed in the first half of the window, which is weaker than establishing that none was there: the census pre-period test below finds one on the college-wage margin. The robot pattern is the reverse, with the non-college wage gradient concentrated in 2005–2010 (-0.0067 , $t = -7.7$) and fading after 2010. For robots the early window is on-treatment, not a placebo, since the EU5 shift-share covers 1993–2014 adoption, so the split cannot separate pre-existing trends from contemporaneous effects on that side, the direct task-content evidence carries a genuine pre-period test for the robot exposure, where the 1987–2000 placebo fails, and that caveat is reported with the task moments. Each technology’s wage gradient sits where its deployment sits, which is what a causal reading of the cross-section predicts. The split is timing-consistent, not confound-excluding: a late-decade shock concentrated in the same tech-intensive metropolitan areas would produce the same pattern, so the decomposition supports the deployment-timing reading without ruling out late confounds. The next subsection runs the true pre-period test on census long differences (1990–2000, 2000–2005) under the [Autor et al. \(2013\)](#) design, and the AI college-wage margin does not clear it.

I.2 A Census Pre-Period Test of the Exposure Gradients

The sub-window decomposition above is timing-consistent, not confound-excluding, and it promised a census pre-period test. That test is run here, from archives the paper already uses, without a new extract. Outcomes are commuting-zone long differences that predate the 2005–2019 sample window: the 1990–2000 changes in average log weekly wages of non-college and college workers from the [Autor et al. \(2013\)](#) census workfile, the 1990–2000 change in the college share of employment computed from the same archive’s census aggregates, and a 2000–2005 college-share change that splices the 2000 census aggregate to the 2005 ACS under the archive’s education definition. Each outcome is regressed on the same standardized 2005-baseline Webb AI and EU5 robot exposures with the completed control set, employment weights, and HC1 standard errors, exactly the joint specification of Table 1, on the identical 722 commuting zones. The constrained permutation tiers

accompany the AI cells, permuting occupation-level AI scores within software deciles and about the robot-and-software projection with the Bartik rebuilt each draw. The robot exposure is an industry shift-share with no occupation-score representation, so its cells carry HC1 and an unconstrained permutation only.

Table I.3: Pre-Period Exposure Gradients on Census Long Differences

Window	Outcome	Webb AI		EU5 robots	
		Coef. (SE)	p : HC1 constr.	Coef. (SE)	p : HC1 unconstr.
1990–2000	$\Delta \ln W_{NC}$	+0.0064 (0.0029)	0.029 0.16, 0.33	+0.0036 (0.0013)	0.006 0.071
1990–2000	$\Delta \ln W_C$	+0.0195 (0.0041)	<0.001 0.029, 0.029	+0.0006 (0.0017)	0.71 0.79
1990–2000	Δs_C	+0.0018 (0.0014)	0.21 0.98, 0.70	+0.0018 (0.0007)	0.010 0.021
2000–2005	Δs_C	+0.0034 (0.0014)	0.019 0.22, 0.25	+0.0005 (0.0009)	0.60 0.63
<i>Supplementary, not a pre-period (overlaps 2005–2007):</i>					
2000–2007	$\Delta \ln W_{NC}$	−0.0002 (0.0043)	0.97 0.98, 0.98	−0.0128 (0.0021)	<0.001 <0.001
2000–2007	$\Delta \ln W_C$	+0.0177 (0.0040)	<0.001 0.056, 0.072	−0.0053 (0.0019)	0.006 0.038

Notes: Each cell reports the coefficient on the indicated standardized exposure from the joint specification (both exposures entered together) with the completed control set, weighted by baseline employment, HC1 standard errors in parentheses, $N = 722$ commuting zones in every row. Wage outcomes are the [Autor et al. \(2013\)](#) census-workfile decadal log changes (the 2000–2007 rows are the archive’s census-to-ACS splice in decadal equivalents). Δs_C is the change in the college (some-college-or-more) share of employment from the archive’s census counts, with the 2005 endpoint built from the 2005 ACS under the same definition and sample rules. AI constrained p -values are the within-software-decile and task-residualized occupation permutations (2,000 draws each). Robot cells report the unconstrained commuting-zone permutation, which for the four pre-period AI cells, not shown, gives 0.001, < 0.001, 0.023, and 0.001 in table order. Wage levels by education for 2000–2005 are not constructible from the archives, which carry no 2000 wage-by-education aggregate, and would require a census 2000 five percent IPUMS extract.

Five of the eight pre-period cells reject at the 10 percent level under HC1. The principal pre-period concern is the college wage in 1990–2000: commuting zones that would measure as AI-exposed in 2005 had college wages rising by +0.0195 log points per standard deviation of exposure per decade ($t = 4.72$), and the cell survives both constrained permutation tiers, $p = 0.029$ within software deciles and $p = 0.029$ on task residuals. This is the late-confound scenario the sub-window decomposition could not exclude, now visible before the window: the AI-exposed metropolitan

areas were on differential skill-biased trajectories in the 1990s. The 2000–2007 splice shows the same college-wage gradient (+0.0177), so the trajectory runs continuously into the sample window instead of dying out before it.

Two aspects of the pattern bear on the on-treatment gradients. The rejecting wage and robot cells are opposite-signed relative to their on-treatment counterparts. The non-college wage gradients are positive before 2005 (+0.0064 for AI, +0.0036 for robots) and negative on treatment (−0.0080 and −0.0084), and the robot college-share gradient flips from +0.0018 to −0.0019. A continuation of the pre-period trends would have pushed these moments away from the estimated effects, not toward them. On the AI side the college-share cell, which carries the sorting interpretation, is the quietest of the three. In 1990–2000 it is null under HC1 ($p = 0.21$) and under both constrained permutation tiers ($p = 0.98$ within software deciles and $p = 0.70$ on task residuals), rejecting only against the unconstrained commuting-zone permutation ($p = 0.023$), the laxer null discussed in Appendix G.8. The robot college-share cell is not quiet in the same sense, rejecting at $p = 0.010$ under HC1 and $p = 0.021$ against the unconstrained permutation, and it is counted among the failing cells above. Its 2000–2005 rejection is HC1-only (constrained $p = 0.22$ and 0.25) on the one outcome that splices two data sources. The AI coefficient on that cell is positive under every education definition tried, from +0.0034 to +0.0238, but the magnitude is definition-sensitive, so the cell is evidence of direction not of size. The college-wage cell admits no such qualification. The cross-sectional AI wage gradients should therefore be read as an upper bound on what the deployment-timing argument can deliver, and the structural moments that inherit them carry this pre-trend caveat.

Netting a linear continuation of these gradients out of the moments is what the estimator does, and Appendix I.3 sets the adjusted system beside the unadjusted one.

I.3 The Adjusted and Unadjusted Moment Systems

This subsection reports what the pre-trend adjustment does to the estimates, by setting the system estimated on the adjusted moments, which is the paper’s baseline, beside the same system estimated on the unadjusted ones.

The on-treatment moments are 14-year (2005–2019) long differences and the census gradients are per-decade, so the adjusted moment vector is $m_{\text{adj}} = m_{\text{on}} - 1.4 g_{\text{pre}}$ with $V_{\text{adj}} = V_{\text{on}} + 1.4^2 V_{\text{pre}}$.

That expression omits the cross-period term $-2(1.4) \text{Cov}(m_{\text{on}}, g_{\text{pre}})$, which is not mechanically zero: both vectors are estimated on the same 722 commuting zones with the same exposure regressors and the same control set, so persistent zone characteristics enter both. Setting it to zero is a maintained sampling restriction. Its likely sign makes the reported adjusted standard errors conservative, since a positive covariance would reduce the variance of the difference, but V_{adj} is also the two-step weighting matrix, so the adjusted point estimates and the adjusted Hansen J move with it as well. The pre-period gradients were rebuilt through the same code path, standardization, weighting, and control set as the on-treatment moments, and rescaled into the moment vector’s log-point-times-one-hundred convention before differencing. On-treatment cross-equation correlations are preserved, and the pre-period cross-equation covariances are taken from the stacked HC1 fit of the six pre-period equations, not assumed away. Only the 1990–2000 window supplies g_{pre} : the 2000–2005 splice is excluded as definition-sensitive and HC1-only.

The consequential entry is the AI college-wage moment, which flips sign from $+0.88$ to -2.20 in the moment vector’s units. In premium terms the on-treatment AI skill-premium gap is $+2.04$ and the pre-period continuation is $1.4 \times 1.49 = +2.08$, so the adjusted premium gap is -0.05 . The adjustment absorbs almost the whole contrast, and everything in Table I.4 follows from that one entry.

Table I.4: Unadjusted and Pre-Trend-Adjusted Systems

	Unadjusted	Adjusted (baseline)
<i>Minimum-distance estimates (6 moments, $\varepsilon_H = 1.38$)</i>		
$\hat{\Lambda}_R$	0.0085	0.0090
$\hat{\chi}_A$	0.495	0.013
translated $\hat{\delta}_A$	0.99	0.026
$\hat{\varphi}$	1.99	0.80
Hansen J (3 df)	0.01	2.86
<i>Macro objects at the neutral physical calibration $\delta_R = 1.00$</i>		
π_R	+0.587	+0.587
π_A	+0.484	+0.268
π_R/π_A	1.21	2.19
$(1 + \tau_R^*)/(1 + \tau_A^*)$	1.33	1.72
Labor-share ratio $\partial s_L/\partial I_R$ over $\partial s_L/\partial I_A$	0.48	0.41
<i>Over the external physical range [0.60, 2.00]</i>		
Parity threshold δ_R^*	0.62	none in range
Share of range with $\pi_R > \pi_A$	99%	100%
Share of range with $(1 + \tau_R^*)/(1 + \tau_A^*) > 1$	100%	100%
π_R/π_A at $\delta_R = 0.60$ and 2.00	0.99 and 1.64	1.80 and 2.93

Notes: Both columns estimate $\theta = (\Lambda_R, \chi_A, \varphi)$ by two-step minimum distance from the same six commuting-zone moments with ε_H fixed at 1.38. The adjusted column, which is the paper’s baseline, replaces the moment vector and its variance matrix by $m_{\text{adj}} = m_{\text{on}} - 1.4 g_{\text{pre}}$ and $V_{\text{adj}} = V_{\text{on}} + 1.4^2 V_{\text{pre}}$, with g_{pre} the 1990–2000 joint-specification gradients of Table I.3 and V_{pre} their stacked HC1 covariance including cross-equation terms. Translated $\hat{\delta}_A$ uses $\bar{g}_A = 0.5$, as in Table 5. The adjusted Jacobian has rank 3 of 3, standard-error-scaled singular values (2.51, 0.44, 0.135), condition number 18.6, all three parameters interior, and sandwich standard errors (0.011, 0.107, 0.414) on $(\Lambda_R, \chi_A, \varphi)$, read under the caveat of Table 5. The uncertainty attached to the adjusted column in the body of the paper is that table’s 200-draw bootstrap.

The adjustment strengthens the productivity ranking while changing its decomposition. At the neutral physical calibration π_R/π_A rises from 1.21 to 2.19 and the relative subsidy factor from 1.33

to 1.72, robot dominance holds on the whole of $[0.60, 2.00]$ instead of 99 percent of it, and the unadjusted system’s parity crossing at $\delta_R^* = 0.62$ leaves the range altogether, with the ratio at the conservative lower bound rising from 0.99 to 1.80. Translated cognitive reinstatement collapses from 0.99 to 0.026, within rounding of the 0.03 that the wage-only design of Appendix G.10 reports. Removing the pre-trend-contaminated college-wage variation reproduces that design’s cognitive estimate inside this paper’s own moment system, which is the reconciliation result Appendix G.10 builds on. The carrying channel switches with it. On the unadjusted moments the two frontiers reinstate at comparable rates and sorting disruption carries robot dominance, while on the adjusted ones cognitive reinstatement is near zero and the reinstatement gap carries it.

Three qualifications travel with the adjusted $\hat{\varphi} = 0.80$, which is why the paper reports it with its profile interval and rests nothing on its value. It is unstable to weighting, since the same adjusted moments give 0.80 under the two-step weight and 6.94 under a diagonal weight, where the unadjusted system agreed to the third decimal (1.986 against 1.985), and the diagonal-weight solution is degenerate at the macro layer, implying $\pi_A = -0.010$. The instability has a transparent source, since the premium contrast that identifies φ is what the adjustment zeroes out, leaving an adjusted premium gap of -0.05 . And the 1.4 scale factor is a linear-continuation assumption about a single decade of pre-period data, doing real work in every number reported here, which is why Table 4 reports the estimates at smaller continuation rates as well.

The productivity ranking and the policy ordering that follows from it hold under both treatments of the pre-trend and are stronger under the adjusted one. The decomposition of the ranking into a reinstatement gap and a sorting friction is what moves between them, because the moment that separates those two channels is the moment the pre-trend contaminates.